Science Monk · Presents

KCET Physics

Previous Year Question Papers

2016 – 2026 · Eleven Years · 660 Questions

The important thing
is not to stop questioning

Albert Einstein

every question retyped *every diagram redrawn*



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Science Monk

"Appa Deepo Bhava"
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About this book. Each paper has 60 multiple-choice questions, one mark each, to be attempted in 80 minutes, with no negative marking. Questions keep their original numbers and option labels, so they match the official KEA answer keys for the version named above. Every question has been retyped and every diagram redrawn from the English version of the paper.

Prepared by Sumeet Birajadar for Science Monk online classes.

KCET 2026
Physics Question Paper

Day 1 · First Session · Version A1 · 60 Questions · 60 Marks · 80 Minutes

1. Match the physical quantities given in List-I with dimensions expressed in terms of mass (M), length (L), time (T) and electric current (A) given in List-II.

List-I		*List-II*	
(a)	Torque	(i)	$[M^{-1}L^{-2}T^{4}A^{2}]$
(b)	Gravitational constant	(ii)	$[M^{1}L^{2}T^{-1}]$
(c)	Capacitance	(iii)	$[M^{-1}L^{3}T^{-2}]$
(d)	Planck's constant	(iv)	$[M^{1}L^{2}T^{-2}]$

Codes:

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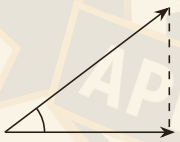
(1) a–iv, b–ii, c–iii, d–i **(2)** a–iv, b–iii, c–i, d–ii
(3) a–iv, b–i, c–iii, d–ii **(4)** a–ii, b–i, c–iii, d–iv

2. A car covers the first half of the distance between two places at 40 km/h and another half at 50 km/h. The average speed of the car is

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(1) 45.00 km/h **(2)** 44.44 km/h
(3) 43.14 km/h **(4)** 42.04 km/h

3. Two bodies are projected with the same velocity. If one is projected at an angle of 30° and the other at 45° to the horizontal, then the ratio of maximum heights attained is



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(1) 3:1 **(2)** 1:2 **(3)** 4:1 **(4)** 1:3

4. The velocity of a particle moving along x-axis is given as $V = x^2 - 5x + 4$ (in m/s) where x denotes the x-coordinate of the particle in metres. The magnitude of the acceleration of the particle when the velocity of the particle is zero is

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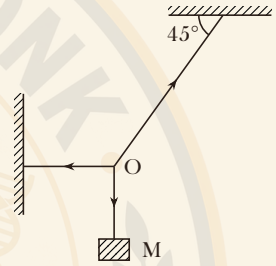
(1) 2 m/s^2 **(2)** 3 m/s^2 **(3)** Zero **(4)** 1 m/s^2

5. A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 6 m/s^2. What would be his weight in kg? (g = 10 m/s^2)

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(1) Zero **(2)** 48 kg **(3)** 120 kg **(4)** 128 kg

6. A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string.



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(1) $\sqrt{2}\,Mg$ **(2)** $\sqrt{3}\,Mg$ **(3)** Mg **(4)** $3\,Mg$

7. Two bodies with kinetic energies in the ratio of 3:1 are moving with equal linear momentum. The ratio of their masses is

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(1) 1:4 **(2)** 1:3 **(3)** 1:2 **(4)** 1:1

8. A horizontal force of 5 N is applied on a stationary body of mass 5 kg, which is initially at rest on a frictionless table. The change in kinetic energy of the body in 10 s is

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(1) 25 J **(2)** Zero **(3)** 125 J **(4)** 250 J

9. The angular momentum of a moving body remains constant, if

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(1) net external force is applied
(2) net pressure is applied
(3) net external torque is applied
(4) net external torque is not applied

10. If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?

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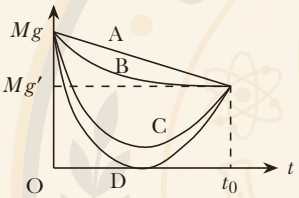
(1) 6 h **(2)** 18 h **(3)** 24 h **(4)** 30 h

11. Imagine a new planet having the same density as that of the earth, but it is two times bigger than the earth in size. If the acceleration due to gravity on the surface of the earth is g and that on the surface of the new planet is g', then

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(1) $g' = \frac{g}{4}$ **(2)** $g' = 8g$ **(3)** $g' = 2g$ **(4)** $g' = 4g$

12. Suppose the acceleration due to gravity at the earth's surface is g m/s^2 and at the surface of moon it is g' m/s^2. An M kg passenger goes from the earth to the moon in a spaceship moving with a constant velocity (Neglect all other objects in the sky). Which curve best represents the weight (net gravitational force) as a function of time?



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(1) A **(2)** B **(3)** C **(4)** D

13. Instrument fitted in the carburetor of the automobile to provide the correct mixture of air and fuel necessary for combustion works on

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(1) Pascal's Law
(2) Bernoulli's Principle
(3) Newton's Law of Cooling
(4) Archimedes' Principle

14. There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the ratio of extensions produced in the wires by applying same load will be

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(1) 1:1 **(2)** 1:2 **(3)** 2:1 **(4)** 4:1

15. In a capillary tube experiment, a vertical 30 cm long capillary tube is dipped in water, water rises upto a height of 10 cm due to capillarity. If this experiment is conducted in a freely falling elevator, then the length of the water column becomes

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(1) 10 cm **(2)** 20 cm **(3)** 30 cm **(4)** Zero

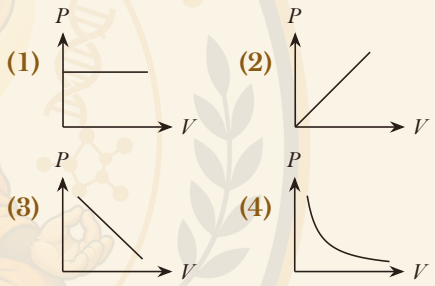
16. In thermodynamic processes, which of the following statements is not true?

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(1) In an isothermal process, the temperature remains constant
(2) In an isobaric process, the volume remains constant
(3) In an adiabatic process, the system is insulated from the surroundings
(4) In an adiabatic process, PV^{γ} = a constant

17. The graph of pressure P and volume V of 1 mole of an ideal gas at constant temperature is

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18. A mass of 1 kg is executing SHM. Its displacement is given by $x = 6.0\cos(100t + \pi/4)$ cm. What is the maximum kinetic energy?

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(1) 3 J **(2)** 6 J **(3)** 9 J **(4)** 18 J

19. A source of frequency ν gives 6 beats/second when sounded with a source of frequency 200 Hz. The second harmonic of frequency 2ν of the source gives 8 beats/second when sounded with a source of frequency 420 Hz. The value of ν is

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(1) 205 Hz **(2)** 206 Hz **(3)** 195 Hz **(4)** 210 Hz

20. Following are statements of a few processes taking place in nature.

I. Free expansion of a gas
II. The combustion of a mixture of petrol and air ignited by a spark
III. The leaking of gas from the kitchen cylinder
IV. The transfer of heat from one heated part of a liquid to the other colder part

Which amongst these processes are irreversible in nature?

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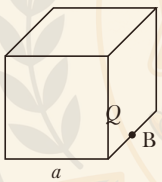
(1) I and II **(2)** III and IV
(3) II, III and IV **(4)** I, II, III and IV

21. An electron falls through a distance of 1.5 cm in a uniform electric field of magnitude 2.0×10^4 N/C from rest. The time taken to cover this distance in second is _____ ($e = 1.6 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg)

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(1) 2.9×10^{-9} **(2)** 2.9×10^{9}
(3) 4×10^{-6} **(4)** 4×10^{6}

22. What will be the total electric flux through the faces of the cube as given in the figure with side of length 'a' if a charge Q is placed at B, midpoint of an edge of the cube (see figure)?

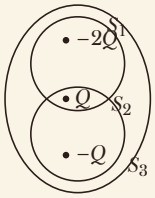


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(1) $\frac{Q}{8\varepsilon_0}$ **(2)** $\frac{Q}{3\varepsilon_0}$ **(3)** $\frac{Q}{4\varepsilon_0}$ **(4)** $\frac{Q}{2\varepsilon_0}$

23. Consider three point charges $-2Q$, Q and $-Q$ and three surfaces S_1, S_2 and S_3 as shown in the figure. Match the entries of List-I with that of List-II.

List-I	*List-II*
(a) Net flux through S_1	(i) $-2Q/\varepsilon_0$
(b) Net flux through S_2	(ii) $-Q/\varepsilon_0$
(c) Net flux through S_3	(iii) Zero



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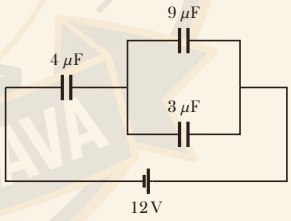
(1) a–ii, b–i, c–iii **(2)** a–iii, b–ii, c–i
(3) a–i, b–ii, c–iii **(4)** a–ii, b–iii, c–i

24. A parallel plate capacitor has a uniform electric field 'E' in the space between the plates. If the distance between the plates is 'd' and area of each plate is 'A', the energy stored in the capacitor is

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(1) $\frac{1}{2}\epsilon_0 E^2$ **(2)** $\epsilon_0 EAd$
(3) $\frac{1}{2}\epsilon_0 E^2 Ad$ **(4)** $\frac{E^2 Ad}{\epsilon_0}$

25. In the circuit shown in the figure, the potential difference across the 4 μF capacitor is



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(1) 3 V **(2)** 4 V **(3)** 9 V **(4)** 12 V

26. An electric dipole of dipole moment $\vec{P}$ is placed in the uniform electric field $\vec{E}$. Then which of the following statements are correct?
Statement I: The torque on the dipole is $\vec{P} \times \vec{E}$
Statement II: The potential energy of the dipole is $-\vec{P} \cdot \vec{E}$
Statement III: The net force on the dipole is non zero

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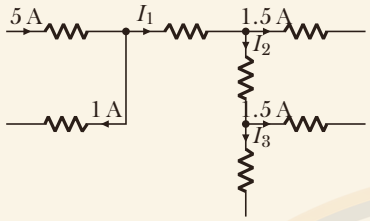
(1) I, II and III **(2)** I and II only
(3) II and III only **(4)** I and III only

27. A 200 J of work is done in moving a charge 5 C from a point A where the potential is −20 V to another point B where potential is V volt. The value of V at B is

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(1) 10 V **(2)** 20 V **(3)** 40 V **(4)** 60 V

28. In the figure, the values of currents I_1, I_2 and I_3 respectively are



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(1) 6 A, 1.5 A and 1 A
(2) 4 A, 2.5 A and 2 A
(3) 4 A, 2.5 A and 1 A
(4) 6 A, 4.5 A and 1.5 A

29. The number of electrons moving per second through the filament of a lamp of 60 W operating at 120 V is nearly ($e = 1.6 \times 10^{-19}$ C) _____

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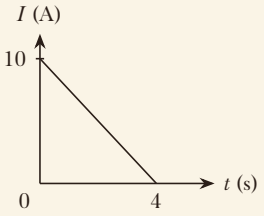
(1) 6.2×10^{18} **(2)** 6.2×10^{19}
(3) 3.1×10^{18} **(4)** 3.1×10^{19}

30. Given below are two statements:
Statement I: The resistivity of a conductor is independent of its temperature
Statement II: The resistivity of a semiconductor decreases with increase in temperature
Select the correct option.

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(1) Both Statement I and Statement II are false
(2) Both Statement I and Statement II are true
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

31. Current flowing through a wire decreases linearly from 10 A to zero in 4 s as shown in the graph. Find the total charge flowing through the wire in the given time interval.



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(1) 40 C **(2)** 20 C **(3)** 10 C **(4)** 80 C

32. In a conducting region, 10^{19} electrons and 10^{19} protons move to the left, while 10^{19} α-particles move to the right per second. The resulting electric current is ($e = 1.6 \times 10^{-19}$ C) _____

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(1) 3.2 A towards left **(2)** 3.2 A towards right
(3) 1.6 A towards left **(4)** 1.6 A towards right

33. A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this charge. Then

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(1) A observes both electric and magnetic fields
(2) B observes both electric and magnetic fields
(3) A observes only magnetic field
(4) B observes only electric field

34. A proton, an electron and an α-particle enter at right angles to a uniform magnetic field with the same velocity. If R_p, R_e and R_α are the radii of circular paths of these particles, then

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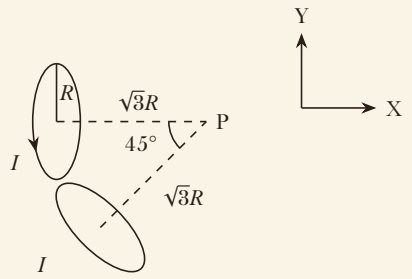
(1) $R_\alpha = R_p = R_e$ **(2)** $R_\alpha > R_p > R_e$
(3) $R_\alpha < R_p < R_e$ **(4)** $R_\alpha > R_p = R_e$

35. Biot–Savart law indicates that an electron moving with a velocity $\vec{V}$ produces a magnetic field $\vec{B}$ around it such that

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(1) $\vec{B}$ is parallel to $\vec{V}$
(2) $\vec{B}$ is perpendicular to $\vec{V}$
(3) $\vec{B}$ is anti-parallel to $\vec{V}$
(4) $\vec{B}$ is inclined to $\vec{V}$ by 45°

36. Two identical circular current loops carrying equal currents are placed with their axes inclined at 45° to each other as shown in the figure. The resultant magnetic field at P is



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(1) $\frac{\mu_0 I}{16\sqrt{2}R}\left[(\sqrt{2}+1)\hat{i}+\hat{j}\right]$

(2) $\frac{\mu_0 I}{16\sqrt{2}R}\left[\sqrt{2}\,\hat{i}+\hat{j}\right]$

(3) $\frac{\mu_0 I}{16R}\left[(\sqrt{2}+1)\hat{i}+\hat{j}\right]$

(4) $\frac{\mu_0 I}{16R}\left[\sqrt{2}\,\hat{i}+\hat{j}\right]$

37. If a paramagnetic bar is brought near a bar magnet, then it is

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(1) Attracted by both the poles of the bar magnet
(2) Repelled by both the poles of the bar magnet
(3) Attracted by the South-pole and repelled by the North-pole of the bar magnet
(4) Attracted by the North-pole and repelled by the South-pole of the bar magnet

38. Pick out the WRONG statements about magnetic substances (χ = magnetic susceptibility) (μ_r = relative permeability).

I. Substances with $-1 \le \chi < 0$ are diamagnetic
II. Substances with $\chi \gg 1$ are paramagnetic
III. Substances with $\chi \ll 1$ are ferromagnetic
IV. Substances with $\mu_r \gg 1$ are ferromagnetic

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(1) I and II **(2)** III and IV
(3) II and III **(4)** II and IV

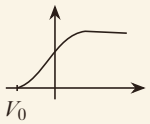
39. Work function of the metal is

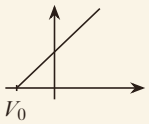
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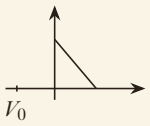
(1) Maximum possible energy acquired by an electron
(2) Equal for all metals
(3) Minimum energy required by an electron to just eject from metal surface
(4) Maximum energy which is given to electron to move out of metal surface

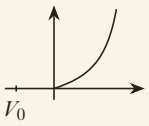
40. Variation of photoelectric current with anode potential is shown below. Choose the correct option (V_0 = stopping potential). (Vertical axis: photoelectric current; horizontal axis: anode potential.)

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(1) 

(2) 

(3) 

(4) 

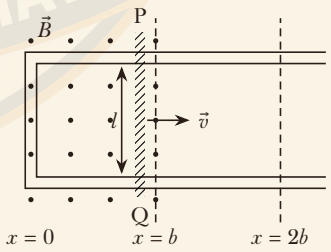
41. In Faraday–Henry's experiment, a coil is connected to a galvanometer. For the deflection of pointer in the galvanometer, which of the following statement/s is/are WRONG?
The pointer in the galvanometer deflects –

(a) When the bar magnet is moved towards the stationary coil along its axis
(b) When the bar magnet is moved away from the stationary coil along its axis
(c) When the coil is moved towards the stationary bar magnet along its axis
(d) When the coil and the magnet are moved without relative motion between them

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(1) a and b **(2)** b and c
(3) a, b and c **(4)** Only d

42. In the figure shown, the conductor PQ of length l is moved from $x = 0$ to $x = b$ and then up to $x = 2b$ with a constant velocity $\vec{v}$. A uniform magnetic field $\vec{B}$ is perpendicular to the plane of the paper and extends from $x = 0$ to $x = b$ and it is zero from $x > b$. The magnitude of emf induced in the conductor is



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(1) Blx ; $0 \le x < b$ **(2)** Zero ; $0 \le x < b$
(3) Blv ; $0 \le x \le b$ **(4)** Blv ; $b \le x < 2b$

43. In a circuit containing a pure resistor connected to an AC source,

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(1) Voltage leads the current by 90°
(2) Current leads the voltage by 90°
(3) Voltage and current are in same phase with each other
(4) Current leads the voltage by 180°

44. A light bulb rated 100 W is connected to an AC source of 220 V, 50 Hz. The rms current through the bulb is

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(1) 0.454 A **(2)** 0.545 A
(3) 2.20 A **(4)** 0.22 A

45. A small town with a demand of 900 kW of electric power at 220 V is situated 20 km away from an electric power generating station. The two-wire line has resistance per unit length of $5 \times 10^{-4}\ \Omega\ m^{-1}$. The town gets power from the line through 45000 V to 220 V step-down transformer at a substation in the town. The line power loss in the form of heat is

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(1) 4 kW **(2)** 8 kW **(3)** 40 kW **(4)** 80 kW

46. Match the following Maxwell's equations: (The symbols used here have their usual meanings)

List-I
(a) Gauss' law for electrostatics (b) Gauss' law for magnetism
(c) Faraday's law (d) Ampere–Maxwell law

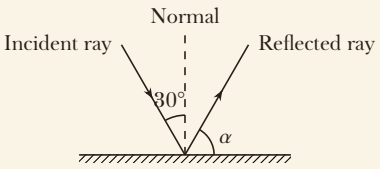
List-II
(i) $\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\varepsilon_0}$
(ii) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left[i_c + \varepsilon_0 \frac{d\phi_E}{dt} \right]$
(iii) $\oint \vec{B} \cdot d\vec{A} = 0$
(iv) $\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$

Codes:

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(1) a–i, b–iii, c–iv, d–ii **(2)** a–ii, b–iii, c–i, d–iv
(3) a–i, b–ii, c–iii, d–iv **(4)** a–ii, b–iii, c–iv, d–i

47. With reference to the figure shown below, match the following:



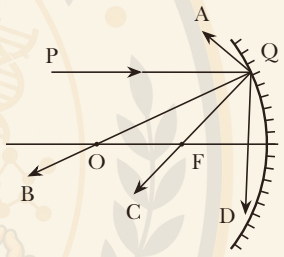
List-I	*List-II*
(a) Angle of reflection	(i) 60°
(b) Value of α	(ii) 120°
(c) Angle of deviation	(iii) 30°

Codes:

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(1) a–i, b–ii, c–iii **(2)** a–ii, b–i, c–iii
(3) a–iii, b–i, c–ii **(4)** a–iii, b–ii, c–i

48. The direction of a ray of light incident on a concave mirror is shown by PQ, while direction in which the ray would travel after reflection is shown by four rays marked as A, B, C and D as shown in the figure. Which of the four rays correctly shows the direction of the reflected ray? (PQ is a paraxial ray.)



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(1) D **(2)** C **(3)** B **(4)** A

49. The incorrect statement about refractive index for a pair of media is

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(1) It depends upon nature of the first medium
(2) It depends upon nature of the second medium
(3) It depends upon wavelength of light
(4) It depends upon angle of incidence

50. The critical angle for a monochromatic light going from medium A to medium B is θ. If the speed of light in medium A is V, then the speed of light in medium B is

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(1) $V(1 - \cos\theta)$ **(2)** $\frac{V}{\cos\theta}$
(3) $\frac{V}{\sin\theta}$ **(4)** $V(1 - \sin\theta)$

51. What range of electromagnetic spectrum is considered as light?

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(1) 1 mm to 700 nm **(2)** 400 nm to 1 nm
(3) 400 nm to 700 nm **(4)** 1 nm to 10^{-3} nm

52. In Young's double slit experiment, how many maxima can be seen on a screen (including central maxima) if $d = \frac{5\lambda}{2}$ (where λ is wavelength of light and d is distance between the two slits)?

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(1) 5 **(2)** 4 **(3)** 7 **(4)** 1

53. The radius of first orbit in hydrogen atom is 5.3×10^{-11} m. The kinetic energy E_K, potential energy E_P and total energy E_T of electron in first orbit are

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(1) $E_K = -13.6$ eV, $E_P = 27.2$ eV, $E_T = 13.6$ eV
(2) $E_K = 13.6$ eV, $E_P = -27.2$ eV, $E_T = -13.6$ eV
(3) $E_K = -27.2$ eV, $E_P = -13.6$ eV, $E_T = 13.6$ eV
(4) $E_K = 13.6$ eV, $E_P = -6.8$ eV, $E_T = -13.6$ eV

54. Bohr's second postulate implies quantisation of

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(1) Charge of an electron
(2) Energy of an electron
(3) Angular momentum of electron
(4) Radiated energy by an electron

55. An electron transition takes place from excited state to ground state in hydrogen atom, then

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(1) Its kinetic energy increases but potential energy and total energy decrease
(2) Its kinetic energy, potential energy and total energy decrease
(3) Kinetic energy decreases, potential energy increases but total energy remains same
(4) Kinetic energy and total energy decrease but potential energy increases

56. An n-type and p-type semiconductor can be obtained by respectively doping pure silicon with

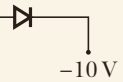
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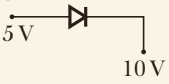
(1) Arsenic and Phosphorous respectively
(2) Indium and Aluminium respectively
(3) Phosphorous and Indium respectively
(4) Aluminium and Boron respectively

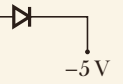
57. In which of the following figures, diode is reverse biased?

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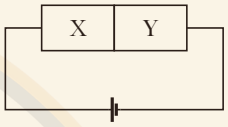
(1) 

(2) 

(3) 

(4) 

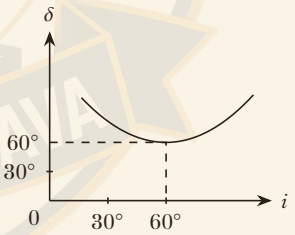
58. A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to a battery as shown in the figure. Which of the following statements is correct?



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(1) X is p-type, Y is n-type and the junction is forward biased
(2) X is n-type, Y is p-type and the junction is forward biased
(3) X is p-type, Y is n-type and the junction is reverse biased
(4) X is n-type, Y is p-type and the junction is reverse biased

59. From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is

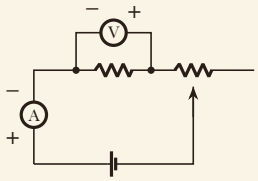


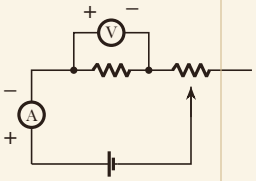
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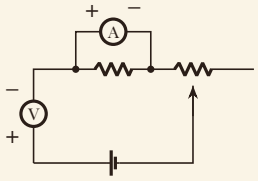
(1) $\frac{\sqrt{3}}{2}$ **(2)** $\frac{3}{2}$ **(3)** $\sqrt{3}$ **(4)** $\sqrt{2}$

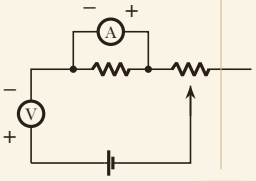
60. Which of the following circuits is correct for verification of Ohm's law?

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(1) 

(2) 

(3) 

(4) 

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KCET 2025
Physics Question Paper

Day 1 · First Session · Version A1 · 60 Questions · 60 Marks · 80 Minutes

1. Which of the following expressions can be deduced on the basis of dimensional analysis? (All symbols have their usual meanings)

(1) $s = ut + \frac{1}{2}at^2$ **(2)** $x = A\cos\omega t$

(3) $N = N_0 e^{-\lambda t}$ **(4)** $F = 6\pi\eta r v$

2. Two stones begin to fall from rest from the same height, with the second stone starting to fall 'Δt' seconds after the first falls from rest. The distance of separation between the two stones becomes 'H', 't_0' seconds after the first stone starts its motion. Then t_0 is equal to

(1) $\frac{H}{g\Delta t}$ **(2)** $\frac{H}{\Delta t} + \frac{\Delta t}{2g}$

(3) $\frac{H}{g\Delta t} - \frac{\Delta t}{2}$ **(4)** $\frac{H}{g\Delta t} + \frac{\Delta t}{2}$

3. In the projectile motion of a particle on a level ground, which of the following remains constant with reference to time and position?

(1) Vertical component of the velocity of the projectile

(2) Average velocity between any two points on the path

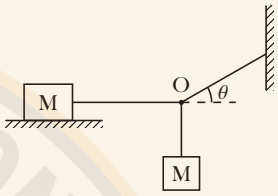
(3) Horizontal component of velocity

(4) Angle between the instantaneous velocity with the horizontal

4. A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The speed of the particle is $2\,\text{m s}^{-1}$. When the particle attains the lowest 'y' co-ordinate, the acceleration of the particle is (in m s^{-2})

(1) $0.4\,\hat{i}$ **(2)** $0.4\,\hat{j}$ **(3)** $0.8\,\hat{i}$ **(4)** $0.8\,\hat{j}$

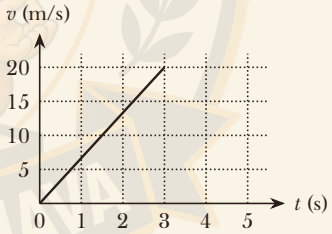
5. A wooden block of mass M lies on a rough floor. Another wooden block of the same mass is hanging from the point O through strings as shown in the figure. To achieve equilibrium, the co-efficient of static friction between the block on the floor with the floor itself is



(1) $\mu = \cos\theta$ **(2)** $\mu = \cot\theta$

(3) $\mu = \sin\theta$ **(4)** $\mu = \tan\theta$

6. A block of certain mass is placed on a rough floor. The coefficients of static and kinetic friction between the block and the floor are 0.4 and 0.25 respectively. A constant horizontal force $F = 20$ N acts on it so that the velocity of the block varies with time according to the following graph. The mass of the block is nearly (Take $g \simeq 10\,\text{m s}^{-2}$)



(1) 2.2 kg **(2)** 4.4 kg **(3)** 1.2 kg **(4)** 1.0 kg

7. A body of mass 0.25 kg travels along a straight line from $x = 0$ to $x = 2$ m with a speed $v = kx^{3/2}$ where $k = 2$ SI units. The work done by the net force during this displacement is

(1) 4 J **(2)** 8 J **(3)** 16 J **(4)** 32 J

8. During an elastic collision between two bodies, which of the following statements are correct?

I. The initial kinetic energy is equal to the final kinetic energy of the system.

II. The linear momentum is conserved.

III. The kinetic energy during Δt (the collision time) is not conserved.

(1) I and II only **(2)** II and III only

(3) I and III only **(4)** I, II and III

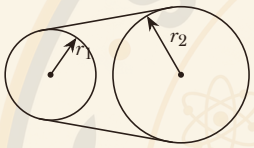
9. Three particles of mass 1 kg, 2 kg and 3 kg are placed at the vertices A, B and C respectively of an equilateral triangle ABC of side 1 m. The centre of mass of the system from vertex A (located at origin) is

(1) (0, 0) **(2)** $\left(\frac{7}{12}, \frac{3\sqrt{3}}{12}\right)$

(3) $\left(\frac{9}{12}, \frac{3\sqrt{3}}{12}\right)$ **(4)** $\left(\frac{7}{12}, \frac{6+3\sqrt{3}}{12}\right)$

10. Two fly wheels are connected by a non-slipping belt as shown in the figure. $I_1 = 4\,\text{kg m}^2$, $r_1 = 20\,\text{cm}$, $I_2 = 20\,\text{kg m}^2$ and $r_2 = 30\,\text{cm}$. A torque of 10 N m is applied on the smaller wheel. Then match the entries of column I with appropriate entries of column II.

I	Quantities	II	Value (SI)
(a)	Angular acceleration of smaller wheel	(i)	$\frac{5}{3}$
(b)	Torque on the larger wheel	(ii)	$\frac{100}{3}$
(c)	Angular acceleration of larger wheel	(iii)	$\frac{5}{2}$



(1) a–iii, b–ii, c–i **(2)** a–ii, b–iii, c–i

(3) a–iii, b–i, c–ii **(4)** a–ii, b–i, c–iii

11. If r_p, v_p, L_p and r_a, v_a, L_a are radii, velocities and angular momenta of a planet at perihelion and aphelion of its elliptical orbit around the Sun respectively, then

(1) $r_p < r_a,\ v_p < v_a,\ L_a < L_p$

(2) $r_p > r_a,\ v_p > v_a,\ L_a > L_p$

(3) $r_p < r_a,\ v_p > v_a,\ L_a = L_p$

(4) $r_p > r_a,\ v_p < v_a,\ L_a = L_p$

12. The total energy of a satellite in a circular orbit at a distance $(R + h)$ from the center of the Earth varies as [R is the radius of the Earth and h is the height of the orbit from Earth's surface]

(1) $\frac{1}{R+h}$ **(2)** $-\frac{1}{R+h}$

(3) $\frac{1}{(R+h)^2}$ **(4)** $-\frac{1}{(R+h)^2}$

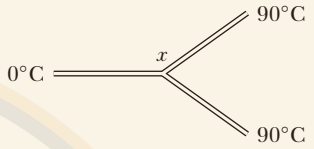
13. Two wires A and B are made of same material. Their diameters are in the ratio of 1:2 and lengths are in the ratio of 1:3. If they are stretched by the same force, then increase in their lengths will be in the ratio of

(1) 4:3 **(2)** 3:4 **(3)** 2:3 **(4)** 3:2

14. A horizontal pipe carries water in a streamlined flow. At a point along the pipe, where the cross-sectional area is $10\,\text{cm}^2$, the velocity of water is $1\,\text{m s}^{-1}$ and the pressure is 2000 Pa. What is the pressure of water at another point where the cross-sectional area is $5\,\text{cm}^2$? [Density of water = $1000\,\text{kg m}^{-3}$]

(1) 200 Pa **(2)** 300 Pa **(3)** 400 Pa **(4)** 500 Pa

15. Three metal rods of the same material and identical in all respects are joined as shown in the figure. The temperatures at the ends of these rods are maintained as indicated. Assuming no heat energy loss occurs through the curved surfaces of the rods, the temperature at the junction x is



(1) 45°C **(2)** 60°C **(3)** 30°C **(4)** 20°C

16. A gas is taken from state A to state B along two different paths 1 and 2. The heat absorbed and work done by the system along these two paths are Q_1 and Q_2 and W_1 and W_2 respectively. Then

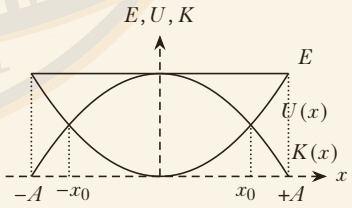
(1) $Q_1 = Q_2$ **(2)** $W_1 = W_2$

(3) $Q_1 - W_1 = Q_2 - W_2$ **(4)** $Q_1 + W_1 = Q_2 + W_2$

17. At 27°C temperature, the mean kinetic energy of the atoms of an ideal gas is E_1. If the temperature is increased to 327°C, then the mean kinetic energy of the atoms will be

(1) $\frac{E_1}{2}$ **(2)** $\frac{E_1}{\sqrt{2}}$ **(3)** $\sqrt{2}\,E_1$ **(4)** $2E_1$

18. The variations of kinetic energy $K(x)$, potential energy $U(x)$ and total energy as a function of displacement of a particle in SHM is as shown in the figure. The value of $|x_0|$ is



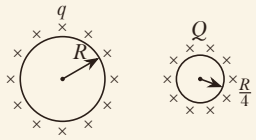
(1) $\frac{A}{2}$ **(2)** $2A$ **(3)** $\frac{A}{\sqrt{2}}$ **(4)** $\sqrt{2}\,A$

19. The angle between the particle velocity and wave velocity in a transverse wave is [except when the particle passes through the mean position]

(1) Zero radian **(2)** $\frac{\pi}{4}$ radian

(3) $\frac{\pi}{2}$ radian **(4)** π radian

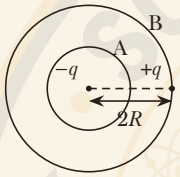
20. A metallic sphere of radius R carrying a charge q is kept at certain distance from another metallic sphere of radius $R/4$ carrying a charge Q. What is the electric flux at any point inside the metallic sphere of radius R due to the sphere of radius $R/4$?



(1) $\frac{Q}{\varepsilon_0}$ **(2)** $\frac{Q}{\varepsilon_0} - \frac{q}{\varepsilon_0}$

(3) Zero **(4)** $\frac{q}{\varepsilon_0} - \frac{Q}{\varepsilon_0}$

21. You are given a dipole of charge $+q$ and $-q$ separated by a distance $2R$. A sphere 'A' of radius 'R' passes through the centre of the dipole as shown below and another sphere 'B' of radius '$2R$' passes through the charge $+q$. Then the electric flux through the sphere A is

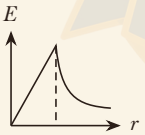


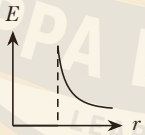
(1) $-q/\varepsilon_0$ **(2)** q/ε_0 **(3)** Zero **(4)** $2q/\varepsilon_0$

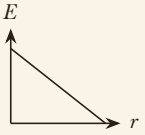
22. A potential at a point A is -3 V and that at another point B is 5 V. What is the work done in carrying a charge of 5 mC from B to A?

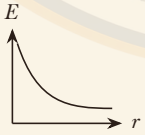
(1) -40 J **(2)** -0.04 J **(3)** -0.4 J **(4)** -4 J

23. Charges are uniformly spread on the surface of a conducting sphere. The electric field from the centre of sphere to a point outside the sphere varies with distance r from the centre as

(1) 

(2) 

(3) 

(4) 

24. Match Column-I with Column-II related to an electric dipole of dipole moment $\vec{p}$ that is placed in a uniform electric field $\vec{E}$.

Column-I (angle between $\vec{p}$ and $\vec{E}$)	*Column-II* (P.E.)
(a) 180°	(i) $-pE$
(b) 120°	(ii) pE
(c) 90°	(iii) $\frac{1}{2}pE$
	(iv) Zero

(1) a–ii, b–iii, c–iv **(2)** a–i, b–ii, c–iii

(3) a–ii, b–iii, c–i **(4)** a–ii, b–i, c–iv

25. Which of the following statements is not true?

(1) Electric field is always perpendicular to an equipotential surface.

(2) Work done to move a charge on an equipotential surface is not zero

(3) Equipotential surfaces are the surfaces where the potential is constant

(4) Equipotential surfaces for a uniform electric field are parallel and equidistant from each other.

26. Which of the following is a correct statement?

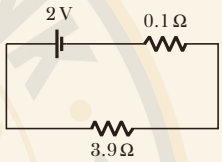
(1) Gauss's law is true for any closed surface

(2) Gauss's law is true for any open surface

(3) Gauss's law is not applicable when charges are not symmetrically distributed over a closed surface.

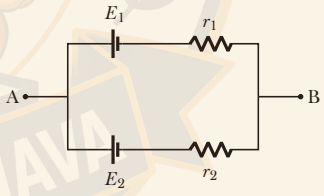
(4) Gauss's law does not hold good for a charge situated outside the Gaussian surface.

27. In the following circuit, the terminal voltage across the cell is



(1) 0.52 V **(2)** 1.68 V **(3)** 1.95 V **(4)** 2.71 V

28. Two cells of emfs E_1 and E_2 and internal resistances r_1 and r_2 ($E_2 > E_1$ and $r_2 > r_1$) respectively, are connected in parallel as shown in figure. The equivalent emf of the combination is E_{eq}. Then



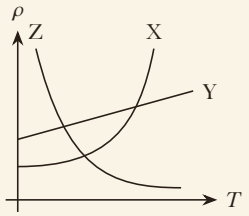
(1) $E_1 < E_{eq} < E_2$ and E_{eq} is nearer E_1

(2) $E_1 < E_{eq} < E_2$ and E_{eq} is nearer E_2

(3) $E_{eq} > E_2$

(4) $E_{eq} < E_1$

29. The variations of resistivity ρ with absolute temperature T for three different materials X, Y and Z are shown in the graph below. Identify the materials X, Y and Z.



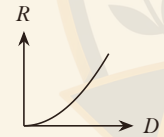
(1) X – copper, Y – nichrome, Z – semiconductor
(2) X – copper, Y – semiconductor, Z – nichrome
(3) X – semiconductor, Y – nichrome, Z – copper
(4) X – nichrome, Y – copper, Z – semiconductor

30. Given, a current carrying wire of non-uniform cross-section, which of the following is constant throughout the length of wire?
(1) Current, electric field and drift speed
(2) Drift speed
(3) Current and drift speed
(4) Current only

31. The graph between variation of resistance of a metal wire as a function of its diameter keeping other parameters like length and temperature constant is

(1) 

(2) 

(3) 

(4) 

32. Two thin long parallel wires separated by a distance 'r' from each other in vacuum carry a current of I ampere in opposite directions. Then, they will
(1) Attract each other with a force per unit length of $\frac{\mu_0 I^2}{2\pi r^2}$
(2) Attract each other with a force per unit length of $\frac{\mu_0 I^2}{2\pi r}$
(3) Repel each other with a force per unit length of $\frac{\mu_0 I^2}{2\pi r}$
(4) Repel each other with a force per unit length of $\frac{\mu_0 I^2}{2\pi r^2}$

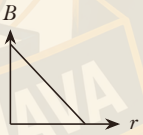
33. A solenoid is 1 m long and 4 cm in diameter. It has five layers of windings of 1000 turns each and carries a current of 7 A. The magnetic field at the centre of the solenoid is
(1) 439.6 T **(2)** 0.4396×10^{-5} T
(3) 4.396×10^{-2} T **(4)** 43.96×10^{-2} T

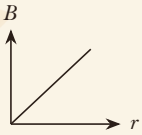
34. Two similar galvanometers are converted into an ammeter and a milliammeter. The shunt resistance of ammeter as compared to the shunt resistance of milliammeter will be
(1) Equal **(2)** Zero **(3)** More **(4)** Less

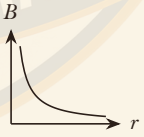
35. Which of the following statements is true in respect of diamagnetic substances?
(1) Susceptibility is small and negative
(2) They are feebly attracted by magnets
(3) Permeability is greater than 1000
(4) Susceptibility decreases with temperature.

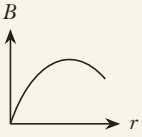
36. Identify the correct statement
(1) The magnetic field inside a solenoid is non-uniform
(2) A current carrying conductor produces an electric field around it.
(3) A straight current carrying conductor has circular magnetic field lines around it.
(4) The direction of magnetic field due to a current element is given by Fleming's Left Hand Rule

37. Which of the following graphs represents the variation of magnetic field B with perpendicular distance 'r' from an infinitely long, straight conductor carrying current?

(1) 

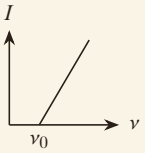
(2) 

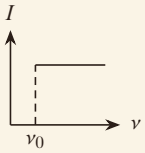
(3) 

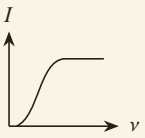
(4) 

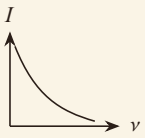
38. If we consider an electron and a photon of same de-Broglie wavelength, then they will have same
(1) Momentum
(2) Angular momentum
(3) Energy
(4) Velocity

39. The anode voltage of a photocell is kept fixed. The frequency of the light falling on the cathode is gradually increased. Then the correct graph which shows the variation of photo current I with the frequency ν of incident light is

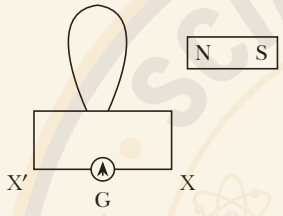
(1) 

(2) 

(3) 

(4) 

40. When a bar magnet is pushed towards the coil, along its axis, as shown in the figure, the galvanometer pointer deflects towards X. When this magnet is pulled away from the coil, the galvanometer pointer



(1) deflects towards X (2) deflects towards X′
(3) does not deflect (4) oscillates

41. A square loop of side 2 m lies in the Y-Z plane in a region having a magnetic field $\vec{B} = (5\hat{i} + 3\hat{j} - 4\hat{k})$ T. The magnitude of magnetic flux through the square loop is

(1) 10 Wb (2) 20 Wb (3) 12 Wb (4) 16 Wb

42. In domestic electric mains supply, the voltage and the current are

(1) AC voltage and AC current
(2) AC voltage and DC current
(3) DC voltage and DC current
(4) DC voltage and AC current

43. A sinusoidal voltage produced by an AC generator at any instant t is given by an equation $V = 311 \sin 314t$. The rms value of voltage and frequency are respectively

(1) 200 V, 100 Hz (2) 200 V, 50 Hz
(3) 220 V, 100 Hz (4) 220 V, 50 Hz

44. A series LCR circuit containing an AC source of 100 V has an inductor and a capacitor of reactances 24 Ω and 16 Ω respectively. If a resistance of 6 Ω is connected in series, then the potential difference across the series combination of inductor and capacitor only is

(1) 40 V (2) 80 V (3) 400 V (4) 8 V

45. Match the following types of waves with their wavelength ranges

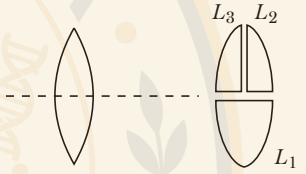
Waves		*Wavelength ranges*	
i.	Microwave	(a)	700 nm to 400 nm
ii.	Visible light	(b)	1 nm to 10^{-3} nm
iii.	Ultraviolet	(c)	0.1 m to 1 mm
iv.	X-rays	(d)	400 nm to 1 nm

(1) i–a, ii–d, iii–b, iv–c (2) i–c, ii–a, iii–d, iv–b
(3) i–d, ii–b, iii–c, iv–a (4) i–b, ii–c, iii–a, iv–d

46. A ray of light passes from vacuum into a medium of refractive index n. If the angle of incidence is twice the angle of refraction, then the angle of incidence in terms of refractive index is

(1) $\cos^{-1}\left(\frac{n}{2}\right)$ (2) $\sin^{-1}\left(\frac{n}{2}\right)$
(3) $2\cos^{-1}\left(\frac{n}{2}\right)$ (4) $2\sin^{-1}\left(\frac{n}{2}\right)$

47. A convex lens has power P. It is cut into two halves along its principal axis. Further one piece (out of two halves) is cut into two halves perpendicular to the principal axis as shown in figure. Choose the incorrect option for the reported lens pieces.

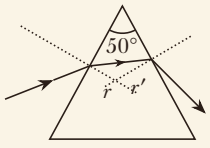


(1) Power of L_1 is $\frac{P}{2}$ (2) Power of L_2 is $\frac{P}{2}$
(3) Power of L_3 is $\frac{P}{2}$ (4) Power of L_1 is P

48. The image formed by an objective lens of a compound microscope is

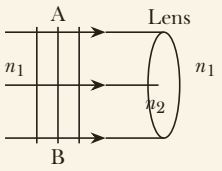
(1) Virtual and diminished
(2) Real and diminished
(3) Real and enlarged
(4) Virtual and enlarged

49. If r and r' denote the angles inside the prism having angle of prism 50° considering that during interval of time from $t = 0$ to $t = t$, r varies with time as $r = 10° + t^2$. During this time r' will vary with time as



(1) $40° - t^2$ (2) $40° + t^2$
(3) $50° - t^2$ (4) $50° + t^2$

50. If AB is incident plane wave front then refracted wave front is ($n_2 > n_1$)



(1) 

(2) 

(3) 

(4) 

51. The total energy carried by the light wave when it travels from a rarer to a non-reflecting and non-absorbing medium

(1) decreases
(2) remains same
(3) increases
(4) either increases or decreases depending upon angle of incidence

52. If the radius of first Bohr orbit is r, then the radius of the second Bohr orbit will be

(1) $2r$ **(2)** $8r$ **(3)** $4r$ **(4)** $2\sqrt{2}\,r$

53. Match the following types of nuclei with examples shown

Column-I		*Column-II*	
(A)	Isotopes	(i)	${}_3Li^7$, ${}_4Be^7$
(B)	Isobars	(ii)	${}_8O^{18}$, ${}_9F^{19}$
(C)	Isotones	(iii)	${}_1H^1$, ${}_1H^2$

(1) A–iii, B–i, C–ii **(2)** A–ii, B–iii, C–i
(3) A–i, B–iii, C–ii **(4)** A–iii, B–ii, C–i

54. Which of the following statements is incorrect with reference to 'Nuclear force'?

(1) Potential energy is minimum, if the separation between the nucleons is 0.8 fm
(2) Nuclear force becomes attractive for nucleon distances larger than 0.8 fm
(3) Nuclear force becomes repulsive for nucleon distances less than 0.8 fm
(4) Nuclear force is always attractive

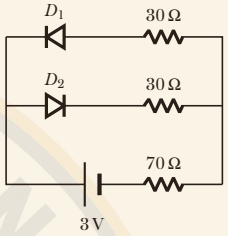
55. The range of electrical conductivity (σ) and resistivity (ρ) for metals, among the following, is

(1) $\rho \rightarrow 10^{-2}$–$10^{-8}\ \Omega$m; $\sigma \rightarrow 10^{2}$–$10^{8}\ \text{S m}^{-1}$
(2) $\rho \rightarrow 10^{-5}$–$10^{6}\ \Omega$m; $\sigma \rightarrow 10^{5}$–$10^{-6}\ \text{S m}^{-1}$
(3) $\rho \rightarrow 10^{11}$–$10^{19}\ \Omega$m; $\sigma \rightarrow 10^{-11}$–$10^{-19}\ \text{S m}^{-1}$
(4) $\rho \rightarrow 10^{2}$–$10^{8}\ \Omega$m; $\sigma \rightarrow 10^{-2}$–$10^{-8}\ \text{S m}^{-1}$

56. Which of the following statements is correct for an n-type semiconductor?

(1) The donor energy level lies just below the bottom of the conduction band
(2) The donor energy level lies closely above the top of the valence band
(3) The donor energy level lies at the half way mark of forbidden energy gap
(4) The donor energy level does not exist

57. The circuit shown in the figure contains two ideal diodes D_1 and D_2. If a cell of emf 3 V and negligible internal resistance is connected as shown, then the current through 70 Ω resistance (in ampere) is



(1) 0 **(2)** 0.01 **(3)** 0.02 **(4)** 0.03

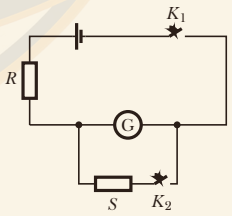
58. In determining the refractive index of a glass slab using a travelling microscope, the following readings are tabulated.

(a) Reading of travelling microscope for ink mark = 5.123 cm
(b) Reading of travelling microscope for ink mark through glass slab = 6.123 cm
(c) Reading of travelling microscope for chalk dust on glass slab = 8.123 cm

From the data, the refractive index of a glass slab is

(1) 1.390 **(2)** 1.500 **(3)** 1.601 **(4)** 1.399

59. In an experiment to determine the figure of merit of a galvanometer by half deflection method, a student constructed the following circuit.



He unplugged a resistance of 5200 Ω in R. When K_1 is closed and K_2 is open, the deflection observed in the galvanometer is 26 div. When K_2 is also closed and a resistance of 90 Ω is removed in S, the deflection becomes 13 div. The resistance of galvanometer is nearly

(1) 116.0 Ω **(2)** 45.0 Ω
(3) 103.0 Ω **(4)** 91.6 Ω

60. While determining the coefficient of viscosity of the given liquid, a spherical steel ball sinks by a distance $h = 0.9\,\text{m}$. The radius of the ball is $r = \sqrt{3}\times10^{-3}\,\text{m}$. The time taken by the ball to sink in three trials are tabulated as follows.

Trial No.	Time taken by the ball to fall by h (s)
1	2.75
2	2.65
3	2.70

The difference between the densities of the steel ball and the liquid is $7000\,\text{kg m}^{-3}$. If $g = 10.0\,\text{m s}^{-2}$, then the coefficient of viscosity of the given liquid at room temperature is

(1) 0.28 Pa s **(2)** 0.14 Pa s

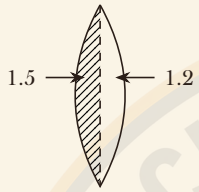
(3) 0.14×10^{-3} Pa s **(4)** 14 Pa s

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KCET 2024
Physics Question Paper

Day 2 · First Session · Version A-2 · 60 Questions · 60 Marks · 80 Minutes

1. An equiconvex lens of radius of curvature 14 cm is made up of two different materials. Left half and right half of vertical portion is made up of material of refractive index 1.5 and 1.2 respectively as shown in the figure. If a point object is placed at a distance of 40 cm, calculate the image distance.

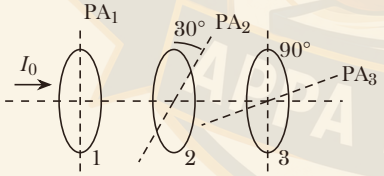


(A) 25 cm **(B)** 50 cm **(C)** 35 cm **(D)** 40 cm

2. A galaxy is moving away from the Earth so that a spectral line at 600 nm is observed at 601 nm. Then the speed of the galaxy with respect to the Earth is

(A) 500 km s^{-1} **(B)** 50 km s^{-1}
(C) 200 km s^{-1} **(D)** 20 km s^{-1}

3. Three polaroid sheets are co-axially placed as indicated in the diagram. Pass axes of the polaroids 2 and 3 make 30° and 90° with pass axis of polaroid sheet 1. If I_0 is the intensity of the incident unpolarised light entering sheet 1, the intensity of the emergent light through sheet 3 is



(A) zero **(B)** $\frac{3I_0}{32}$ **(C)** $\frac{3I_0}{8}$ **(D)** $\frac{3I_0}{16}$

4. In Young's double slit experiment, an electron beam is used to produce interference fringes of width β_1. Now the electron beam is replaced by a beam of protons with the same experimental set-up and same speed. The fringe width obtained is β_2. The correct relation between β_1 and β_2 is

(A) $\beta_1 = \beta_2$
(B) No fringes are formed
(C) $\beta_1 < \beta_2$
(D) $\beta_1 > \beta_2$

5. Light of energy E falls normally on a metal of work function $\frac{E}{3}$. The kinetic energies (K) of the photo electrons are

(A) $K = \frac{2E}{3}$ **(B)** $K = \frac{E}{3}$
(C) $0 \le K \le \frac{2E}{3}$ **(D)** $0 \le K \le \frac{E}{3}$

6. The photoelectric work function for photo metal is 2.4 eV. Among the four wavelengths, the wavelength of light for which photo-emission does not take place is

(A) 200 nm **(B)** 300 nm
(C) 700 nm **(D)** 400 nm

7. In alpha particle scattering experiment, if v is the initial velocity of the particle, then the distance of closest approach is d. If the velocity is doubled, then the distance of closest approach becomes

(A) $4d$ **(B)** $2d$ **(C)** $\frac{d}{2}$ **(D)** $\frac{d}{4}$

8. The ratio of area of first excited state to ground state of orbit of hydrogen atom is

(A) 1:16 **(B)** 1:4 **(C)** 4:1 **(D)** 16:1

9. The ratio of volume of Al^{27} nucleus to its surface area is (Given $R_0 = 1.2 \times 10^{-15}$ m)

(A) 2.1×10^{-15} m **(B)** 1.3×10^{-15} m
(C) 0.22×10^{-15} m **(D)** 1.2×10^{-15} m

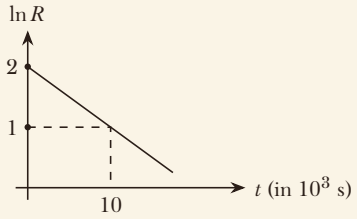
10. Consider the nuclear fission reaction

$$^{1}_{0}n + ^{235}_{92}U \rightarrow ^{144}_{56}Ba + ^{89}_{36}Kr + 3\,^{1}_{0}n.$$

Assuming all the kinetic energy is carried away by the fast neutrons only and total binding energies of $^{235}_{92}U$, $^{144}_{56}Ba$ and $^{89}_{36}Kr$ to be 1800 MeV, 1200 MeV and 780 MeV respectively, the average kinetic energy carried by each fast neutron is (in MeV)

(A) 200 **(B)** 180 **(C)** 67 **(D)** 60

11. The natural logarithm of the activity R of a radioactive sample varies with time t as shown. At $t = 0$, there are N_0 undecayed nuclei. Then N_0 is equal to [Take $e^2 = 7.5$]

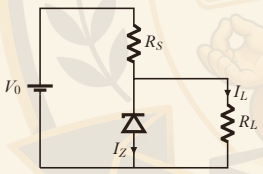


(A) 7,500 **(B)** 3,500
(C) 75,000 **(D)** 1,50,000

12. Depletion region in an unbiased semiconductor diode is a region consisting of
(A) both free electrons and holes
(B) neither free electrons nor holes
(C) only free electrons
(D) only holes

13. The upper level of valence band and lower level of conduction band overlap in the case of
(A) silicon **(B)** copper
(C) carbon **(D)** germanium

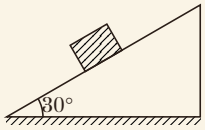
14. In the diagram shown, the Zener diode has a reverse breakdown voltage of V_Z. The current through the load resistance R_L is I_L. The current through the Zener diode is



(A) $\frac{V_0 - V_Z}{R_S}$ **(B)** $\frac{V_0 - V_Z}{R_L}$
(C) $\frac{V_Z}{R_L}$ **(D)** $\left(\frac{V_0 - V_Z}{R_S}\right) - I_L$

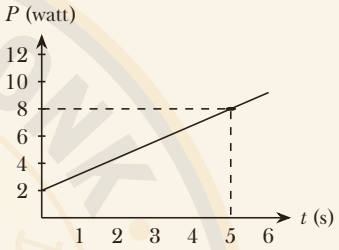
15. A p-n junction diode is connected to a battery of emf 5.7 V in series with a resistance 5 kΩ such that it is forward biased. If the barrier potential of the diode is 0.7 V, neglecting the diode resistance, the current in the circuit is
(A) 1.14 mA **(B)** 1 mA
(C) 1 A **(D)** 1.14 A

16. A block of certain mass is placed on a rough inclined plane. The angle between the plane and the horizontal is 30°. The coefficients of static and kinetic frictions between the block and the inclined plane are 0.6 and 0.5 respectively. Then the magnitude of the acceleration of the block is [Take $g = 10\,m\,s^{-2}$]



(A) $2\,m\,s^{-2}$ **(B)** zero
(C) $0.196\,m\,s^{-2}$ **(D)** $0.67\,m\,s^{-2}$

17. A particle of mass 500 g is at rest. It is free to move along a straight line. The power delivered to the particle varies with time according to the following graph:



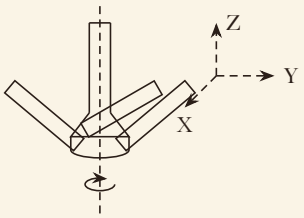
The momentum of the particle at $t = 5$ s is
(A) $2\sqrt{5}$ N s **(B)** $5\sqrt{2}$ N s
(C) 5 N s **(D)** 5.5 N s

18. Dimensional formula for activity of a radioactive substance is
(A) $[M^0L^1T^{-1}]$ **(B)** $[M^0L^{-1}T^0]$
(C) $[M^0L^0T^{-1}]$ **(D)** $[M^{-1}L^0T^0]$

19. An athlete runs along a circular track of diameter 80 m. The distance travelled and the magnitude of displacement of the athlete when he covers $\frac{3}{4}$th of the circle is (in m)
(A) 60π, $40\sqrt{2}$ **(B)** 40π, $60\sqrt{2}$
(C) 120π, $80\sqrt{2}$ **(D)** 80π, $120\sqrt{2}$

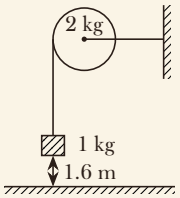
20. Among the given pair of vectors, the resultant of two vectors can never be 3 units. The vectors are
(A) 1 unit and 2 units **(B)** 2 units and 5 units
(C) 3 units and 6 units **(D)** 4 units and 8 units

21. A ceiling fan is rotating around a fixed axle as shown. The direction of angular velocity is along _____.



(A) $+\hat{j}$ **(B)** $-\hat{j}$ **(C)** $+\hat{k}$ **(D)** $-\hat{k}$

22. A body of mass 1 kg is suspended by a weightless string which passes over a frictionless pulley of mass 2 kg as shown in the figure. The mass is released from a height of 1.6 m from the ground. With what velocity does it strike the ground?



(A) $16\ \text{m s}^{-1}$ **(B)** $8\ \text{m s}^{-1}$
(C) $4\sqrt{2}\ \text{m s}^{-1}$ **(D)** $4\ \text{m s}^{-1}$

23. What is the value of acceleration due to gravity at a height equal to half the radius of the Earth, from its surface?

(A) $4.4\ \text{m s}^{-2}$ **(B)** $6.5\ \text{m s}^{-2}$
(C) zero **(D)** $9.8\ \text{m s}^{-2}$

24. A thick metal wire of density ρ and length 'L' is hung from a rigid support. The increase in length of the wire due to its own weight is (Y = Young's modulus of the material of the wire)

(A) $\frac{\rho g L}{Y}$ **(B)** $\frac{1}{2}\frac{\rho g L^2}{Y}$
(C) $\frac{\rho g L^2}{Y}$ **(D)** $\frac{1}{4Y}\rho g L^2$

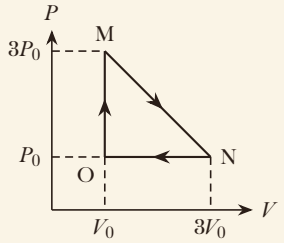
25. Water flows through a horizontal pipe of varying cross-section at a rate of $0.314\ \text{m}^3\text{s}^{-1}$. The velocity of water at a point where the radius of the pipe is 10 cm is

(A) $0.1\ \text{m s}^{-1}$ **(B)** $1\ \text{m s}^{-1}$
(C) $10\ \text{m s}^{-1}$ **(D)** $100\ \text{m s}^{-1}$

26. A solid cube of mass m at a temperature θ_0 is heated at a constant rate. It becomes liquid at temperature θ_1 and vapour at temperature θ_2. Let s_1 and s_2 be specific heats in its solid and liquid states respectively. If L_f and L_v are latent heats of fusion and vaporisation respectively, then the minimum heat energy supplied to the cube until it vaporises is

(A) $ms_1(\theta_1 - \theta_0) + ms_2(\theta_2 - \theta_1)$
(B) $mL_f + ms_2(\theta_2 - \theta_1) + mL_v$
(C) $ms_1(\theta_1 - \theta_0) + mL_f + ms_2(\theta_2 - \theta_1) + mL_v$
(D) $ms_1(\theta_1 - \theta_0) + mL_f + ms_2(\theta_2 - \theta_0) + mL_v$

27. One mole of an ideal monoatomic gas is taken round the cyclic process MNOM. The work done by the gas is



(A) $4.5\ P_0V_0$ **(B)** $4\ P_0V_0$
(C) $9\ P_0V_0$ **(D)** $2\ P_0V_0$

28. The ratio of molar specific heats of oxygen is

(A) 1.4 **(B)** 1.67 **(C)** 1.33 **(D)** 1.28

29. For a particle executing simple harmonic motion (SHM), at its mean position

(A) velocity is zero and acceleration is maximum
(B) velocity is maximum and acceleration is zero
(C) both velocity and acceleration are maximum
(D) both velocity and acceleration are zero

30. A motor-cyclist moving towards a huge cliff with a speed of $18\ \text{km h}^{-1}$, blows a horn of source frequency 325 Hz. If the speed of the sound in air is $330\ \text{m s}^{-1}$, the number of beats heard by him is

(A) 5 **(B)** 4 **(C)** 10 **(D)** 7

31. A body has a charge of $-3.2\ \mu$C. The number of excess electrons it has is

(A) 5.12×10^{25} **(B)** 5×10^{12}
(C) 2×10^{13} **(D)** 5.12×10^{13}

32. A point charge A of $+10\ \mu$C and another point charge B of $+20\ \mu$C are kept 1 m apart in free space. The electrostatic force on A due to B is $\vec{F}_1$ and the electrostatic force on B due to A is $\vec{F}_2$. Then

(A) $\vec{F}_1 = -2\vec{F}_2$ **(B)** $\vec{F}_1 = -\vec{F}_2$
(C) $2\vec{F}_1 = -\vec{F}_2$ **(D)** $\vec{F}_1 = \vec{F}_2$

33. A uniform electric field $E = 3 \times 10^5\ \text{N C}^{-1}$ is acting along the positive Y-axis. The electric flux through a rectangle of area 10 cm × 30 cm whose plane is parallel to the Z-X plane is

(A) $12 \times 10^3\ \text{V m}$ **(B)** $9 \times 10^3\ \text{V m}$
(C) $15 \times 10^3\ \text{V m}$ **(D)** $18 \times 10^3\ \text{V m}$

34. The total electric flux through a closed spherical surface of radius 'r' enclosing an electric dipole of dipole moment $2aq$ is (Given ε_0 = permittivity of free space)

(A) zero **(B)** $\frac{q}{\varepsilon_0}$ **(C)** $\frac{2q}{\varepsilon_0}$ **(D)** $\frac{8\pi r^2 q}{\varepsilon_0}$

35. Under electrostatic condition of a charged conductor, which among the following statements is true?

(A) The electric field on the surface of a charged conductor is $\frac{\sigma}{2\varepsilon_0}$, where σ is the surface charge density.

(B) The electric potential inside a charged conductor is always zero.

(C) Any excess charge resides on the surface of the conductor.

(D) The net electric field is tangential to the surface of the conductor.

36. A cube of side 1 cm contains 100 molecules each having an induced dipole moment of 0.2×10^{-6} C m in an external electric field of 4 N C^{-1}. The electric susceptibility of the material is _____ $C^2N^{-1}m^{-2}$.

(A) 50 **(B)** 5 **(C)** 0.5 **(D)** 0.05

37. A capacitor of capacitance 5 μF is charged by a battery of emf 10 V. At an instant of time, the potential difference across the capacitor is 4 V and the time rate of change of potential difference across the capacitor is 0.6 V s^{-1}. Then the time rate at which energy is stored in the capacitor at that instant is

(A) 12 μW **(B)** 3 μW

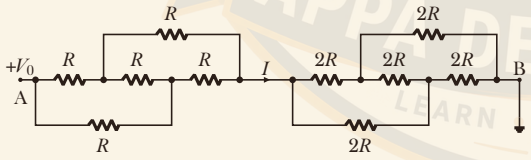
(C) zero **(D)** 30 μW

38. $\vec{E}$ is the electric field inside a conductor whose material has conductivity σ and resistivity ρ. The current density inside the conductor is $\vec{j}$. The correct form of Ohm's law is

(A) $\vec{E} = \sigma\vec{j}$ **(B)** $\vec{j} = \rho\vec{E}$

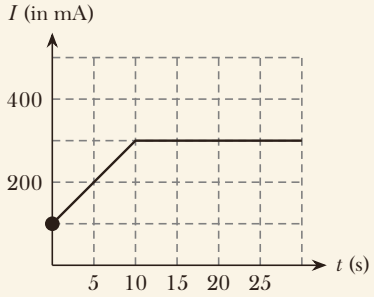
(C) $\vec{E} = \rho\vec{j}$ **(D)** $\vec{E}\cdot\vec{j} = \rho$

39. In the circuit shown, the end A is at potential V_0 and end B is grounded. The electric current I indicated in the circuit is



(A) $\frac{V_0}{R}$ **(B)** $\frac{2V_0}{R}$ **(C)** $\frac{3V_0}{R}$ **(D)** $\frac{V_0}{3R}$

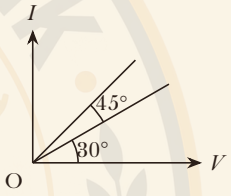
40. The electric current flowing through a given conductor varies with time as shown in the graph below. The number of free electrons which flow through a given cross-section of the conductor in the time interval $0 \le t \le 20$ s is



(A) 3.125×10^{19} **(B)** 1.6×10^{19}

(C) 6.25×10^{18} **(D)** 1.625×10^{18}

41. The I–V graph for a conductor at two different temperatures 100°C and 400°C is as shown in the figure. The temperature coefficient of resistance of the conductor is about (in per degree Celsius)



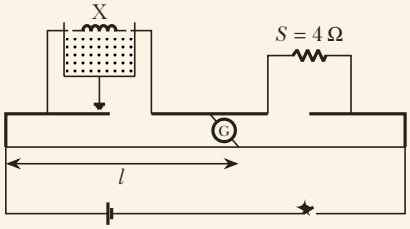
(A) 3×10^{-3} **(B)** 6×10^{-3}

(C) 9×10^{-3} **(D)** 12×10^{-3}

42. An electric bulb of 60 W, 120 V is to be connected to 220 V source. What resistance should be connected in series with the bulb, so that the bulb glows properly?

(A) 50 Ω **(B)** 100 Ω **(C)** 200 Ω **(D)** 288 Ω

43. In an experiment to determine the temperature coefficient of resistance of a conductor, a coil of wire X is immersed in a liquid. It is heated by an external agent. A meter bridge set up is used to determine resistance of the coil X at different temperatures. The balancing points measured at temperatures $t_1 = 0$°C and $t_2 = 100$°C are 50 cm and 60 cm respectively. If the standard resistance taken out is $S = 4\ \Omega$ in both trials, the temperature coefficient of the coil is



(A) 0.05°C^{-1} **(B)** 0.02°C^{-1}

(C) 0.005°C^{-1} **(D)** 2.0°C^{-1}

44. A moving electron produces
(A) only electric field
(B) both electric and magnetic field
(C) only magnetic field
(D) neither electric nor magnetic field

45. A coil having 9 turns carrying a current produces magnetic field B_1 at the centre. Now the coil is rewounded into 3 turns carrying same current. Then the magnetic field at the centre B_2 = _____.
(A) $\frac{B_1}{9}$ **(B)** $9B_1$ **(C)** $3B_1$ **(D)** $\frac{B_1}{3}$

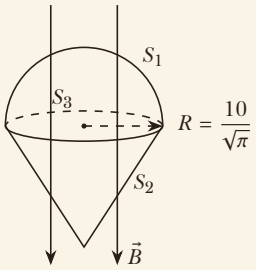
46. A particle of specific charge $\frac{q}{m} = \pi\,\text{C kg}^{-1}$ is projected from the origin towards positive x-axis with the velocity $10\,\text{m s}^{-1}$ in a uniform magnetic field $\vec{B} = -2\hat{k}\,\text{T}$. The velocity $\vec{v}$ of particle after time $t = \frac{1}{12}$ s will be (in m s^{-1})
(A) $5(\hat{i} + \hat{j})$ **(B)** $5(\hat{i} + \sqrt{3}\,\hat{j})$
(C) $5(\sqrt{3}\,\hat{i} - \hat{j})$ **(D)** $5(\sqrt{3}\,\hat{i} + \hat{j})$

47. The magnetic field at the centre of a circular coil of radius R carrying current I is 64 times the magnetic field at a distance x on its axis from the centre of the coil. Then the value of x is
(A) $\frac{R}{4}\sqrt{15}$ **(B)** $R\sqrt{3}$ **(C)** $\frac{R}{4}$ **(D)** $R\sqrt{15}$

48. Magnetic hysterisis is exhibited by _____ magnetic materials.
(A) only para
(B) only dia
(C) only ferro
(D) both para and ferro

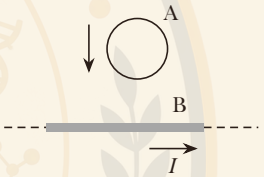
49. Magnetic susceptibility of Mg at 300 K is 1.2×10^{-5}. What is its susceptibility at 200 K?
(A) 18×10^{-5} **(B)** 180×10^{-5}
(C) 1.8×10^{-5} **(D)** 0.18×10^{-5}

50. A uniform magnetic field of strength $B = 2\,\text{mT}$ exists vertically downwards. These magnetic field lines pass through a closed surface as shown in the figure. The closed surface consists of a hemisphere S_1, a right circular cone S_2 and a circular surface S_3. The magnetic flux through S_1 and S_2 are respectively



(A) $\Phi_{S_1} = -20\,\mu\text{Wb}$, $\Phi_{S_2} = +20\,\mu\text{Wb}$
(B) $\Phi_{S_1} = +20\,\mu\text{Wb}$, $\Phi_{S_2} = -20\,\mu\text{Wb}$
(C) $\Phi_{S_1} = -40\,\mu\text{Wb}$, $\Phi_{S_2} = +40\,\mu\text{Wb}$
(D) $\Phi_{S_1} = +40\,\mu\text{Wb}$, $\Phi_{S_2} = -40\,\mu\text{Wb}$

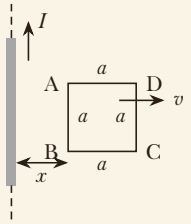
51. In the figure, a conducting ring of certain resistance is falling towards a current carrying straight long conductor. The ring and conductor are in the same plane. Then the

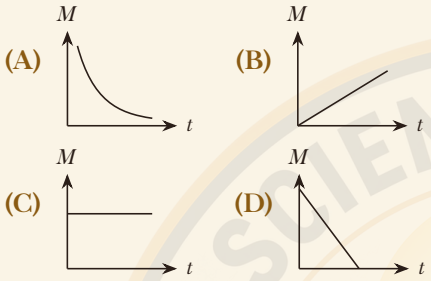


(A) induced electric current is zero
(B) induced electric current is anticlockwise
(C) induced electric current is clockwise
(D) ring will come to rest

52. An induced current of 2 A flows through a coil. The resistance of the coil is 10 Ω. What is the change in magnetic flux associated with the coil in 1 ms?
(A) 0.2×10^{-2} Wb **(B)** 2×10^{-2} Wb
(C) 22×10^{-2} Wb **(D)** 0.22×10^{-2} Wb

53. A square loop of side length 'a' is moving away from an infinitely long current carrying conductor at a constant speed 'v' as shown. Let 'x' be the instantaneous distance between the long conductor and side AB. The mutual inductance (M) of the square loop – long conductor pair changes with time (t) according to which of the following graphs?

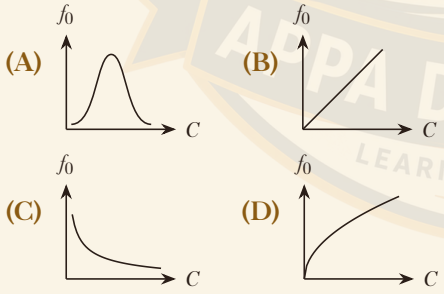




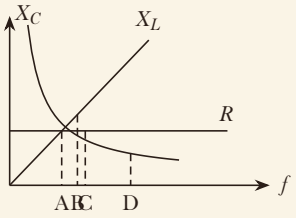
54. Which of the following combinations should be selected for better tuning of an LCR circuit used for communication?

(A) $R = 20\ \Omega$, $L = 1.5$ H, $C = 35\ \mu$F
(B) $R = 25\ \Omega$, $L = 2.5$ H, $C = 45\ \mu$F
(C) $R = 25\ \Omega$, $L = 1.5$ H, $C = 45\ \mu$F
(D) $R = 15\ \Omega$, $L = 3.5$ H, $C = 30\ \mu$F

55. In an LCR series circuit, the value of only capacitance C is varied. The resulting variation of resonance frequency (f_0) as a function of C can be represented as



56. The figure shows variation of R, X_L and X_C with frequency 'f' in a series LCR circuit. Then for what frequency point is the circuit capacitive?



(A) B **(B)** D **(C)** A **(D)** C

57. Electromagnetic waves are incident normally on a perfectly reflecting surface having surface area A. If I is the intensity of the incident electromagnetic radiation and c is the speed of light in vacuum, the force exerted by the electromagnetic wave on the reflecting surface is

(A) $\frac{2IA}{c}$ **(B)** $\frac{IA}{c}$ **(C)** $\frac{IA}{2c}$ **(D)** $\frac{I}{2Ac}$

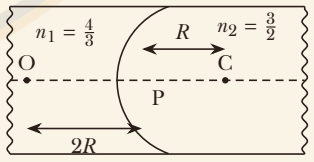
58. The final image formed by an astronomical telescope is

(A) real, erect and diminished
(B) virtual, inverted and diminished
(C) real, inverted and magnified
(D) virtual, inverted and magnified

59. If the angle of minimum deviation is equal to angle of a prism for an equilateral prism, then the speed of light inside the prism is _____.

(A) $3 \times 10^8\ \text{m s}^{-1}$ **(B)** $2\sqrt{3} \times 10^8\ \text{m s}^{-1}$
(C) $\sqrt{3} \times 10^8\ \text{m s}^{-1}$ **(D)** $\frac{\sqrt{3}}{2} \times 10^8\ \text{m s}^{-1}$

60. A luminous point object O is placed at a distance $2R$ from the spherical boundary separating two transparent media of refractive indices n_1 and n_2 as shown, where R is the radius of curvature of the spherical surface. If $n_1 = \frac{4}{3}$, $n_2 = \frac{3}{2}$ and $R = 10$ cm, the image is obtained at a distance from P equal to



(A) 30 cm in the rarer medium
(B) 30 cm in the denser medium
(C) 18 cm in the rarer medium
(D) 18 cm in the denser medium

KCET 2023
Physics Question Paper

Day 2 · First Session · Version A-3 · 60 Questions · 60 Marks · 80 Minutes

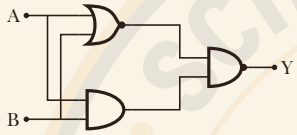
1. When a p-n junction diode is in forward bias, which type of charge carriers flows in the connecting wire?

(A) Ions **(B)** Protons
(C) Holes **(D)** Free electrons

2. A full-wave rectifier with diodes D_1 and D_2 is used to rectify 50 Hz alternating voltage. The diode D_1 conducts _____ times in one second.

(A) 25 **(B)** 75 **(C)** 50 **(D)** 100

3. The truth table for the given circuit is



(A)

A	B	Y
1	1	1
1	0	0
0	1	1
0	0	1

(B)

A	B	Y
1	1	1
1	0	1
0	1	1
0	0	1

(C)

A	B	Y
1	1	1
1	0	1
0	1	1
0	0	0

(D)

A	B	Y
1	1	1
1	0	1
0	1	0
0	0	1

4. The energy gap of an LED is 2.4 eV. When the LED is switched 'ON', the momentum of the emitted photons is

(A) 2.56×10^{-27} kg m s^{-1}
(B) 1.28×10^{-11} kg m s^{-1}
(C) 0.64×10^{-27} kg m s^{-1}
(D) 1.28×10^{-27} kg m s^{-1}

5. In the following equation representing β^- decay, the number of neutrons in the nucleus X is

$$^{210}_{83}\text{Bi} \longrightarrow \text{X} + e^{-1} + \bar{\nu}$$

(A) 127 **(B)** 125 **(C)** 84 **(D)** 126

6. A nucleus with mass number 220 initially at rest emits an alpha particle. If the Q value of reaction is 5.5 MeV, calculate the value of kinetic energy of alpha particle.

(A) 5.4 MeV **(B)** 7.4 MeV
(C) 4.5 MeV **(D)** 6.5 MeV

7. A radioactive sample has half-life of 3 years. The time required for the activity of the sample to reduce to $\frac{1}{5}$th of its initial value is about

(A) 7 years **(B)** 15 years
(C) 5 years **(D)** 10 years

8. A body of mass 10 kg is kept on a horizontal surface. The coefficient of kinetic friction between the body and the surface is 0.5. A horizontal force of 60 N is applied on the body. The resulting acceleration of the body is about

(A) 5 m s^{-2} **(B)** 6 m s^{-2}
(C) zero **(D)** 1 m s^{-2}

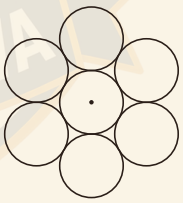
9. A ball of mass 0.2 kg is thrown vertically down from a height of 10 m. It collides with the floor and loses 50% of its energy and then rises back to the same height. The value of its initial velocity is

(A) 14 m s^{-1} **(B)** 196 m s^{-1}
(C) 20 m s^{-1} **(D)** zero

10. The moment of inertia of a rigid body about an axis

(A) does not depend on its shape.
(B) depends on the position of axis of rotation.
(C) does not depend on its size.
(D) does not depend on its mass.

11. Seven identical discs are arranged in a planar pattern, so as to touch each other as shown in the figure. Each disc has mass 'm' radius R. What is the moment of inertia of system of six discs about an axis passing through the centre of central disc and normal to plane of all discs?



(A) $100\,mR^2$ **(B)** $55\,\frac{mR^2}{2}$
(C) $85\,\frac{mR^2}{2}$ **(D)** $27\,mR^2$

12. The true length of a wire is 3.678 cm. When the length of this wire is measured using instrument A, the length of the wire is 3.5 cm. When the length of the wire is measured using instrument B, it is found to have length 3.38 cm. Then the

(A) measurement with A is more accurate while measurement with B is more precise.

(B) measurement with B is more accurate and precise.

(C) measurement with A is more precise while measurement with B is more accurate.

(D) measurement with A is more accurate and precise.

13. A body is moving along a straight line with initial velocity v_0. Its acceleration a is constant. After t seconds, its velocity becomes v. The average velocity of the body over the given time interval is

(A) $\bar{v} = \dfrac{v^2 + v_0^2}{2at}$ **(B)** $\bar{v} = \dfrac{v^2 + v_0^2}{at}$

(C) $\bar{v} = \dfrac{v^2 - v_0^2}{2at}$ **(D)** $\bar{v} = \dfrac{v^2 - v_0^2}{at}$

14. A particle is in uniform circular motion. Related to one complete revolution of the particle, which among the statements is incorrect?

(A) Displacement of the particle is zero.

(B) Average speed of the particle is zero.

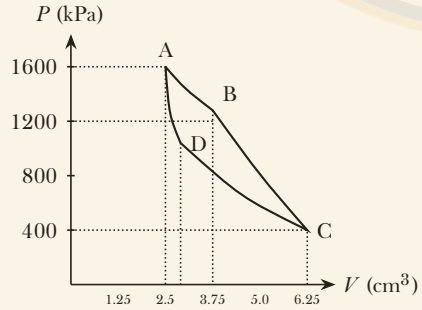
(C) Average velocity of the particle is zero.

(D) Average acceleration of the particle is zero.

15. 100 g of ice at 0°C is mixed with 100 g of water at 100°C. The final temperature of the mixture is [Take $L_f = 3.36 \times 10^5\,\text{J kg}^{-1}$ and $S_w = 4.2 \times 10^3\,\text{J kg}^{-1}\text{K}^{-1}$]

(A) 10°C **(B)** 50°C **(C)** 1°C **(D)** 40°C

16. The P-V diagram of a Carnot's engine is shown in the graph below. The engine uses 1 mole of an ideal gas as working substance. From the graph, the area enclosed by the P-V diagram is [The heat supplied to the gas is 8000 J]



(A) 2000 J **(B)** 3000 J **(C)** 1000 J **(D)** 1200 J

17. When a planet revolves around the Sun, in general, for the planet

(A) linear momentum and aerial velocity are constant.

(B) kinetic and potential energy of the planet are constant.

(C) angular momentum about the Sun and aerial velocity of the planet are constant.

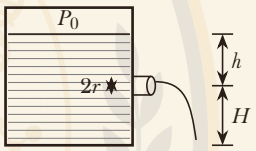
(D) linear momentum and linear velocity are constant.

18. A stretched wire of a material whose Young's modulus $Y = 2 \times 10^{11}\,\text{N m}^{-2}$ has Poisson's ratio 0.25. Its lateral strain $\varepsilon_l = 10^{-3}$. The elastic energy density of the wire is

(A) $1 \times 10^5\,\text{J m}^{-3}$ **(B)** $4 \times 10^5\,\text{J m}^{-3}$

(C) $8 \times 10^5\,\text{J m}^{-3}$ **(D)** $16 \times 10^5\,\text{J m}^{-3}$

19. A closed water tank has cross-sectional area A. It has a small hole at a depth of h from the free surface of water. The radius of the hole is r so that $r \ll \sqrt{A/\pi}$. If P_0 is the pressure inside the tank above water level, and P_a is the atmospheric pressure, the rate of flow of the water coming out of the hole is [ρ is the density of water]



(A) $\pi r^2 \sqrt{2gh + \dfrac{2(P_0 - P_a)}{\rho}}$

(B) $\pi r^2 \sqrt{2gH}$

(C) $\pi r^2 \sqrt{gh + \dfrac{2(P_0 - P_a)}{\rho}}$

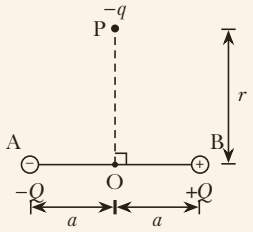
(D) $\pi r^2 \sqrt{2gh}$

20. A positively charged glass rod is brought near uncharged metal sphere, which is mounted on an insulated stand. If the glass rod is removed, the net charge on the metal sphere is

(A) Zero **(B)** 1.6×10^{-19} C

(C) Positive charge **(D)** Negative charge

21. In the situation shown in the diagram, magnitude of $q \ll |Q|$ and $r \gg a$. The net force on the free charge $-q$ and net torque on it about O at the instant shown are respectively [$p = 2aQ$ is the dipole moment]



(A) $\frac{1}{4\pi\varepsilon_0}\frac{pq}{r^2}\hat{k},\ \frac{1}{4\pi\varepsilon_0}\frac{pq}{r^3}\hat{i}$

(B) $-\frac{1}{4\pi\varepsilon_0}\frac{pq}{r^2}\hat{k},\ -\frac{1}{4\pi\varepsilon_0}\frac{pq}{r^3}\hat{i}$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{pq}{r^3}\hat{i},\ +\frac{1}{4\pi\varepsilon_0}\frac{pq}{r^2}\hat{k}$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{pq}{r^3}\hat{i},\ -\frac{1}{4\pi\varepsilon_0}\frac{pq}{r^2}\hat{k}$

22. Pressure of ideal gas at constant volume is proportional to _____.

(A) average potential energy of the molecules

(B) total energy of the gas

(C) average kinetic energy of the molecules

(D) force between the molecules

23. A block of mass m is connected to a light spring of force constant k. The system is placed inside a damping medium of damping constant b. The instantaneous values of displacement, acceleration and energy of the block are x, a and E respectively. The initial amplitude of oscillation is A and ω' is the angular frequency of oscillations. The incorrect expression related to the damped oscillations is

(A) $\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$

(B) $E = \frac{1}{2}kA^2e^{-bt/m}$

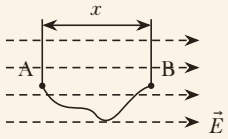
(C) $m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$

(D) $x = Ae^{-\frac{b}{m}t}\cos(\omega' t + \phi)$

24. The speed of sound in an ideal gas at a given temperature T is v. The rms speed of gas molecules at that temperature is v_{rms}. The ratio of the velocities v and v_{rms} for helium and oxygen gases are X and X' respectively. Then $\frac{X}{X'}$ is equal to

(A) $\frac{5}{\sqrt{21}}$ **(B)** $\sqrt{\frac{5}{21}}$ **(C)** $\frac{21}{5}$ **(D)** $\frac{21}{\sqrt{5}}$

25. A uniform electric field vector $\vec{E}$ exists along horizontal direction as shown. The electric potential at A is V_A. A small point charge q is slowly taken from A to B along the curved path as shown. The potential energy of the charge when it is at point B is



(A) $q[V_A + Ex]$ **(B)** $q[Ex - V_A]$

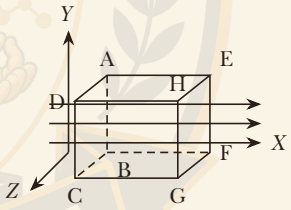
(C) qEx **(D)** $q[V_A - Ex]$

26. A parallel plate capacitor of capacitance C_1 with a dielectric slab in between its plates is connected to a battery. It has a potential difference V_1 across its plates. When the dielectric slab is removed, keeping the capacitor connected to the battery, the new capacitance and potential difference are C_2 and V_2 respectively. Then,

(A) $V_1 > V_2,\ C_1 > C_2$ **(B)** $V_1 < V_2,\ C_1 > C_2$

(C) $V_1 = V_2,\ C_1 > C_2$ **(D)** $V_1 = V_2,\ C_1 < C_2$

27. A cubical Gaussian surface has side of length $a = 10$ cm. Electric field lines are parallel to x-axis as shown. The magnitudes of electric fields through surfaces ABCD and EFGH are 6 kN C^{-1} and 9 kN C^{-1} respectively. Then the total charge enclosed by the cube is [Take $\varepsilon_0 = 9\times10^{-12}$ F m^{-1}]



(A) 1.35 nC **(B)** −1.35 nC

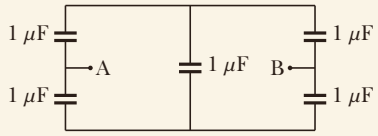
(C) 0.27 nC **(D)** −0.27 nC

28. Electric field at a distance 'r' from an infinitely long uniformly charged straight conductor, having linear charge density λ is E_1. Another uniformly charged conductor having same linear charge density λ is bent into a semicircle of radius 'r'. The electric field at its centre is E_2. Then

(A) $E_2 = \frac{E_1}{r}$ **(B)** $E_1 = E_2$

(C) $E_1 = \pi r E_2$ **(D)** $E_2 = \pi r E_1$

29. Five capacitors each of value 1 μF are connected as shown in the figure. The equivalent capacitance between A and B is



(A) 1 μF **(B)** 2 μF **(C)** 5 μF **(D)** 3 μF

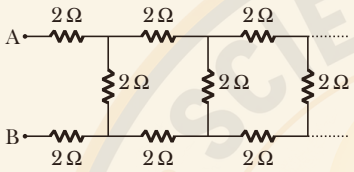
30. For a given electric current the drift velocity of conduction electrons in a copper wire is v_d and their mobility is μ. When the current is increased at constant temperature

(A) v_d remains the same, μ increases
(B) v_d decreases, μ remains the same
(C) v_d remains the same, μ decreases
(D) v_d increases, μ remains the same

31. Ten identical cells each emf 2 V and internal resistance 1 Ω are connected in series with two cells wrongly connected. A resistor of 10 Ω is connected to the combination. What is the current through the resistor?

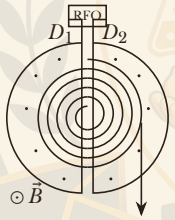
(A) 2.4 A **(B)** 0.6 A **(C)** 1.2 A **(D)** 1.8 A

32. The equivalent resistance between the points A and B in the following circuit is



(A) 5.5 Ω **(B)** 0.05 Ω **(C)** 5 Ω **(D)** 0.5 Ω

33. A charged particle is subjected to acceleration in a cyclotron as shown. The charged particle undergoes increase in its speed



(A) Only inside D_2
(B) Inside D_1, D_2 and the gaps
(C) Only inside D_1
(D) Only in the gap between D_1 and D_2

34. The resistance of a carbon resistor is 4.7 kΩ ± 5%. The colour of the third band is

(A) red **(B)** violet **(C)** orange **(D)** gold

35. The four bands of a colour coded resistor are of the colours gray, red, gold and gold. The value of the resistance of the resistor is

(A) 82 Ω ± 10% **(B)** 8.2 Ω ± 5%
(C) 82 Ω ± 5% **(D)** 5.2 Ω ± 5%

36. A wire of resistance R is connected across a cell of emf ε and internal resistance r. The current through the circuit is I. In time t, the work done by the battery to establish the current I is

(A) $\frac{\varepsilon^2 t}{R}$ **(B)** IRt **(C)** I^2Rt **(D)** εIt

37. The Curie temperatures of Cobalt and iron are 1400 K and 1000 K respectively. At T = 1600 K, the ratio of magnetic susceptibility of Cobalt to that of iron is

(A) 3 **(B)** $\frac{7}{5}$ **(C)** $\frac{5}{7}$ **(D)** $\frac{1}{3}$

38. The torque acting on a magnetic dipole placed in uniform magnetic field is zero, when the angle between the dipole axis and the magnetic field is _____.

(A) 45° **(B)** 60° **(C)** 90° **(D)** zero

39. The horizontal component of Earth's magnetic field at a place is 3×10^{-5} T. If the dip at that place is 45°, the resultant magnetic field at that place is

(A) $\frac{3}{\sqrt{2}} \times 10^{-5}$ T **(B)** $\frac{3}{2}\sqrt{3} \times 10^{-5}$ T
(C) $3\sqrt{2} \times 10^{-5}$ T **(D)** 3×10^{-5} T

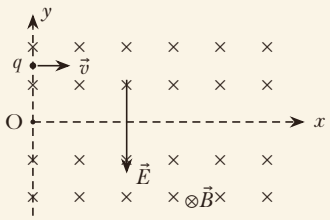
40. A proton and an alpha-particle moving with the same velocity enter a uniform magnetic field with their velocities perpendicular to the magnetic field. The ratio of radii of their circular paths is

(A) 1:4 **(B)** 4:1 **(C)** 1:2 **(D)** 2:1

41. A moving coil galvanometer is converted into an ammeter of range 0 to 5 mA. The galvanometer resistance is 90 Ω and the shunt resistance has a value of 10 Ω. If there are 50 divisions in the galvanometer-turned-ammeter on either sides of zero, its current sensitivity is

(A) 1×10^5 A/div **(B)** 2×10^4 A/div
(C) 1×10^5 div/A **(D)** 2×10^4 div/A

42. A positively charged particle of mass m is passed through a velocity selector. It moves horizontally rightward without deviation along the line $y = \frac{2mv}{qB}$ with a speed v. The electric field is vertically downwards and magnetic field is into the plane of the paper. Now, the electric field is switched off at $t = 0$. The angular momentum of the charged particle about origin O at $t = \frac{\pi m}{qB}$ is



(A) $\frac{2mE^2}{qB^3}$ **(B)** zero **(C)** $\frac{mE^3}{qB^2}$ **(D)** $\frac{mE^2}{qB^3}$

43. In series LCR circuit at resonance, the phase difference between voltage and current is

(A) π **(B)** $\frac{\pi}{4}$ **(C)** $\frac{\pi}{2}$ **(D)** zero

44. An ideal transformer has a turns ratio of 10. When the primary is connected to 220 V, 50 Hz ac source, the power output is

(A) $\frac{1}{10}$th the power input
(B) equal to power input
(C) zero
(D) 10 times the power input

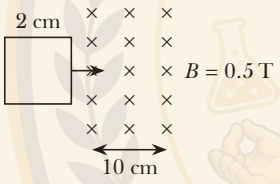
45. The current in a coil changes from 2 A to 5 A in 0.3 s. The magnitude of emf induced in the coil is 1.0 V. The value of self-inductance of the coil is

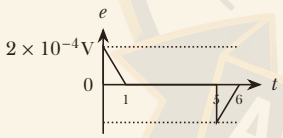
(A) 100 mH **(B)** 0.1 mH
(C) 10 mH **(D)** 1.0 mH

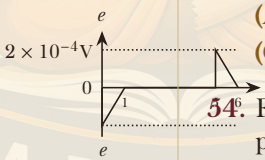
46. A metallic rod of length 1 m held along east-west direction is allowed to fall down freely. Given horizontal component of earth's magnetic field $B_H = 3\times10^{-5}$ T. The emf induced in the rod at an instant $t = 2$ s after it is released is (Take $g = 10\ m\,s^{-2}$)

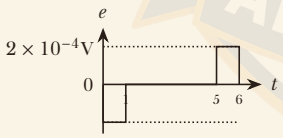
(A) 3×10^{-3} V **(B)** 3×10^{-4} V
(C) 6×10^{-3} V **(D)** 6×10^{-4} V

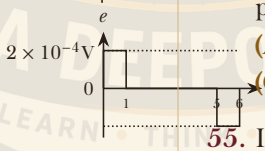
47. A square loop of side 2 cm enters a magnetic field with a constant speed of $2\ cm\,s^{-1}$ as shown. The front edge enters the field at $t = 0$ s. Which of the following graph correctly depicts the induced emf in the loop? (Take clockwise direction positive)



(A)


(B)


(C)


(D)


48. For a given pair of transparent media, the critical angle for which colour is maximum?

(A) Red **(B)** Blue **(C)** Violet **(D)** Green

49. An equiconvex lens made of glass of refractive index $\frac{3}{2}$ has focal length f in air. It is completely immersed in water of refractive index $\frac{4}{3}$. The percentage change in the focal length is

(A) 300% decrease **(B)** 400% decrease
(C) 300% increase **(D)** 400% increase

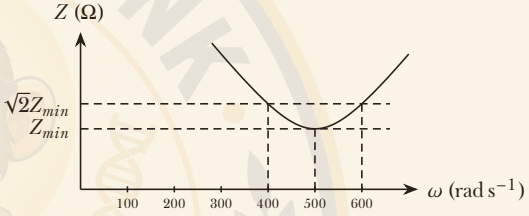
50. A point object is moving at a constant speed of $1\ m\,s^{-1}$ along the principal axis of a convex lens of focal length 10 cm. The speed of the image is also $1\ m\,s^{-1}$, when the object is at _____ cm from the optic centre of the lens.

(A) 15 **(B)** 20 **(C)** 5 **(D)** 10

51. When light propagates through a given homogeneous medium, the velocities of

(A) primary wavefronts are lesser than those of secondary wavelets.
(B) primary wavefronts are greater than or equal to those of secondary wavelets.
(C) primary wavefront and wavelets are equal.
(D) primary wavefront are larger than those of secondary wavelets.

52. Total impedance of a series LCR circuit varies with angular frequency of the AC source connected to it as shown in the graph. The quality factor Q of the series LCR circuit is



(A) 2.5 **(B)** 5 **(C)** 1 **(D)** 0.4

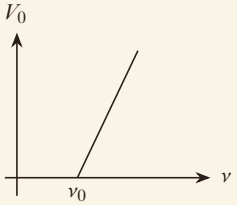
53. The ratio of the magnitudes of electric field to the magnetic field of an electromagnetic wave is of the order of

(A) $10^5\ m\,s^{-1}$ **(B)** $10^{-5}\ m\,s^{-1}$
(C) $10^8\ m\,s^{-1}$ **(D)** $10^{-8}\ m\,s^{-1}$

54. For a point object, which of the following always produces virtual image in air?

(A) Plano-convex lens **(B)** Convex mirror
(C) Biconvex lens **(D)** Concave mirror

55. In an experiment to study photo-electric effect the observed variation of stopping potential with frequency of incident radiation is as shown in the figure. The slope and y-intercept are respectively (V_0: stopping potential; ν: frequency of incident radiation)



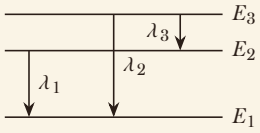
(A) $\frac{h\nu}{e}$, ν_0 **(B)** $\frac{h\nu}{e}$, $-\frac{h}{e}$
(C) $h\nu$, $-h\nu_0$ **(D)** $\frac{h}{e}$, $-\frac{h\nu_0}{e}$

56. In the Rutherford's alpha scattering experiment, as the impact parameter increases, the scattering angle of the alpha particle

(A) is always 90° **(B)** decreases

(C) increases **(D)** remains the same

57. Three energy levels of hydrogen atom and the corresponding wavelength of the emitted radiation due to different electron transition are as shown. Then



(A) $\lambda_1 = \dfrac{\lambda_2\lambda_3}{\lambda_2+\lambda_3}$ **(B)** $\lambda_2 = \lambda_1 + \lambda_3$

(C) $\lambda_2 = \dfrac{\lambda_1\lambda_3}{\lambda_1+\lambda_3}$ **(D)** $\lambda_3 = \dfrac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

58. An unpolarised light of intensity I is passed through two polaroids kept one after the other with their planes parallel to each other. The intensity of light emerging from second polaroid is $\dfrac{I}{4}$. The angle between the pass axes of the polaroids is

(A) 0° **(B)** 60° **(C)** 30° **(D)** 45°

59. In the Young's double slit experiment, the intensity of light passing through each of the two double slits is $2 \times 10^{-2}\,\text{W m}^{-2}$. The screen-slit distance is very large in comparison with slit-slit distance. The fringe width is β. The distance between the central maximum and a point P on the screen is $x = \dfrac{\beta}{3}$. Then the total light intensity at that point is

(A) $4 \times 10^{-2}\,\text{W m}^{-2}$ **(B)** $2 \times 10^{-2}\,\text{W m}^{-2}$

(C) $16 \times 10^{-2}\,\text{W m}^{-2}$ **(D)** $8 \times 10^{-2}\,\text{W m}^{-2}$

60. A 60 W source emits monochromatic light of wavelength 662.5 nm. The number of photons emitted per second is

(A) 2×10^{20} **(B)** 5×10^{26}

(C) 2×10^{29} **(D)** 5×10^{17}

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KCET 2022
Physics Question Paper

Version D-1 · 60 Questions · 60 Marks · 80 Minutes

1. In case of Fraunhoffer diffraction at a single slit the diffraction pattern on the screen is correct for which of the following statements?

(A) Central dark band having uniform brightness on either side.

(B) Central dark band having alternate dark and bright bands of decreasing intensity on either side.

(C) Central bright band having dark bands on either side.

(D) Central bright band having alternate dark and bright bands of decreasing intensity on either side.

2. When a Compact Disc (CD) is illuminated by small source of white light coloured bands are observed. This is due to

(A) Interference **(B)** Scattering

(C) Reflection **(D)** Diffraction

3. Consider a glass slab which is silvered at one side and the other side is transparent. Given the refractive index of the glass slab to be 1.5. If a ray of light is incident at an angle of 45° on the transparent side, the deviation of the ray of light from its initial path, when it comes out of the slab is

(A) 120° **(B)** 90° **(C)** 45° **(D)** 180°

4. Focal length of a convex lens will be maximum for

(A) Green light **(B)** Blue light

(C) Red light **(D)** Yellow light

5. For light diverging from a finite point source

(A) the wave front is parabolic.

(B) the wave front is cylindrical

(C) the intensity at the wave front does not depend on the distance.

(D) the intensity decreases in proportion to the distance squared.

6. The fringe width for red colour as compared to that for violet colour is approximately

(A) 4 times **(B)** 3 times

(C) 8 times **(D)** Double

7. In accordance with the Bohr's model, the quantum number that characterises the Earth's revolution around the Sun in an orbit of radius 1.5×10^{11} m with orbital speed 3×10^4 m s^{-1} is [Given mass of Earth = 6×10^{24} kg]

(A) 8.57×10^{64} **(B)** 5.98×10^{86}

(C) 2.57×10^{74} **(D)** 2.57×10^{38}

8. An electron is revolving in its Bohr orbit having Bohr radius of 0.529 Å, then the radius of third orbit is

(A) 4.761 Å **(B)** 4234 nm

(C) 5125 nm **(D)** 4496 Å

9. Binding energy of a Nitrogen nucleus $\left[{}^{14}_{7}\mathrm{N}\right]$, given $m\left[{}^{14}_{7}\mathrm{N}\right] = 14.00307$ u

(A) 206.5 MeV **(B)** 104.7 MeV

(C) 78 MeV **(D)** 85 MeV

10. In a photo electric experiment, if both the intensity and frequency of the incident light are doubled, then the saturation photo electric current

(A) is doubled

(B) remains constant

(C) becomes four times

(D) is halved

11. The kinetic energy of the photoelectrons increases by 0.52 eV when the wavelength of incident light is changed from 500 nm to another wavelength which is approximately

(A) 1250 nm **(B)** 700 nm

(C) 1000 nm **(D)** 400 nm

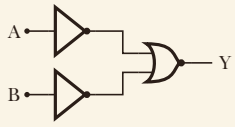
12. The de-Broglie wavelength of a particle of kinetic energy 'K' is λ; the wavelength of the particle, if its kinetic energy is $\frac{K}{4}$ is

(A) $\frac{\lambda}{2}$ **(B)** λ **(C)** 4λ **(D)** 2λ

13. The radius of hydrogen atom in the ground state is 0.53 Å. After collision with an electron, it is found to have a radius of 2.12 Å, the principal quantum number 'n' of the final state of the atom is

(A) $n = 3$ **(B)** $n = 1$ **(C)** $n = 4$ **(D)** $n = 2$

14. Which logic gate is represented by the following combination of logic gates?



(A) AND **(B)** OR
(C) NOR **(D)** NAND

15. A metallic rod of mass per unit length $0.5\,\text{kg m}^{-1}$ is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. A magnetic field of strength 0.25 T is acting on it in the vertical direction. When a current 'I' is flowing through it, the rod is not allowed to slide down. The quantity of current required to keep the rod stationary is

(A) 14.76 A **(B)** 7.14 A
(C) 11.32 A **(D)** 5.98 A

16. A nuclear reactor delivers a power of 10^9 W, the amount of fuel consumed by the reactor in one hour is

(A) 0.72 g **(B)** 0.04 g **(C)** 0.96 g **(D)** 0.08 g

17. Which of the following radiations is deflected by electric field?

(A) γ-rays **(B)** X-rays
(C) α-particles **(D)** Neutrons

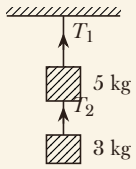
18. The resistivity of a semiconductor at room temperature is in between

(A) 10^6 to $10^8\ \Omega$ cm **(B)** 10^{-2} to $10^{-5}\ \Omega$ cm
(C) 10^{10} to $10^{12}\ \Omega$ cm **(D)** 10^{-3} to $10^6\ \Omega$ cm

19. The forbidden energy gap for 'Ge' crystal at '0' K is

(A) 2.57 eV **(B)** 0.071 eV
(C) 6.57 eV **(D)** 0.71 eV

20. Two masses of 5 kg and 3 kg are suspended with the help of massless inextensible strings as shown in fig. When whole system is going upwards with acceleration $2\,\text{m/s}^2$, the value of T_1 is (use $g = 9.8\,\text{m/s}^2$)



(A) 23.6 N **(B)** 94.4 N **(C)** 59 N **(D)** 35.4 N

21. The Vernier scale of a travelling microscope has 50 divisions which coincides with 49 main scale divisions. If each main scale division is 0.5 mm, then the least count of the microscope is

(A) 0.01 mm **(B)** 0.01 cm
(C) 0.5 cm **(D)** 0.5 mm

22. The displacement 'x' (in metre) of a particle of mass 'm' (in kg) moving in one dimension under the action of a force, is related to time 't' (in sec.) by, $t = \sqrt{x} + 3$. The displacement of the particle when its velocity is zero, will be

(A) 6 m **(B)** 4 m **(C)** 2 m **(D)** 0 m

23. Two objects are projected at an angle $\theta°$ and $(90 - \theta)°$, to the horizontal with the same speed. The ratio of their maximum vertical heights is

(A) $1 : \tan\theta$ **(B)** 1 : 1
(C) $\tan^2\theta : 1$ **(D)** $\tan\theta : 1$

24. A car is moving in a circular horizontal track of radius 10 m with a constant speed of $10\,\text{m s}^{-1}$. A bob is suspended from the roof of the car by a light wire of length 1.0 m. The angle made by the wire with the vertical is (in radian)

(A) 0 **(B)** $\frac{\pi}{6}$ **(C)** $\frac{\pi}{3}$ **(D)** $\frac{\pi}{4}$

25. A tiny spherical oil drop carrying a net charge q is balanced in still air, with a vertical uniform electric field of strength $\frac{81}{7}\pi \times 10^5\,\text{V/m}$. When the field is switched off, the drop is observed to fall with terminal velocity $2 \times 10^{-3}\,\text{m s}^{-1}$. Here $g = 9.8\,\text{m/s}^2$, viscosity of air is $1.8 \times 10^{-5}\,\text{N s/m}^2$ and the density of oil is $900\,\text{kg m}^{-3}$. The magnitude of 'q' is

(A) 1.6×10^{-19} C **(B)** 0.8×10^{-19} C
(C) 3.2×10^{-19} C **(D)** 8×10^{-19} C

26. "Heat cannot be itself flow from a body at lower temperature to a body at higher temperature". This statement corresponds to

(A) Conservation of mass
(B) Second law of Thermodynamics
(C) First law of thermodynamics
(D) Conservation of momentum

27. A smooth chain of length 2 m is kept on a table such that its length of 60 cm hangs freely from the edge of the table. The total mass of the chain is 4 kg. The work done in pulling the entire chain on the table is, (Take $g = 10\,\text{m/s}^2$)

(A) 3.6 J **(B)** 12.9 J **(C)** 2.0 J **(D)** 6.3 J

28. The angular speed of a motor wheel is increased from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration of the motor wheel is

(A) $6\pi\ \text{rad/s}^2$ **(B)** $2\pi\ \text{rad/s}^2$
(C) $8\pi\ \text{rad/s}^2$ **(D)** $4\pi\ \text{rad/s}^2$

29. The centre of mass of an extended body on the surface of the earth and its centre of gravity

(A) can never be at the same point.

(B) are always at the same point for any size of the body.

(C) centre of mass coincides with the centre of gravity of a body if the size of the body is negligible as compared to the size (or radius) of the earth.

(D) are always at the same point only for spherical bodies.

30. A metallic rod breaks when strain produced is 0.2%. The Young's modulus of the material of the rod is 7×10^9 N/m^2. The area of cross section to support a load of 10^4 N is

(A) 7.1×10^{-4} m^2 **(B)** 7.1×10^{-8} m^2

(C) 7.1×10^{-2} m^2 **(D)** 7.1×10^{-6} m^2

31. A charged particle of mass 'm' and charge 'q' is released from rest in an uniform electric field $\vec{E}$. Neglecting the effect of gravity, the kinetic energy of the charged particle after 't' second is

(A) $\frac{Eqm}{t}$ **(B)** $\frac{2E^2t^2}{mq}$

(C) $\frac{E^2q^2t^2}{2m}$ **(D)** $\frac{Eq^2m}{2t^2}$

32. The electric field and the potential of an electric dipole vary with distance r as

(A) $\frac{1}{r^2}$ and $\frac{1}{r^3}$ **(B)** $\frac{1}{r}$ and $\frac{1}{r^2}$

(C) $\frac{1}{r^3}$ and $\frac{1}{r^2}$ **(D)** $\frac{1}{r^2}$ and $\frac{1}{r}$

33. The displacement of a particle executing SHM is given by $x = 3\sin\left[2\pi t + \frac{\pi}{4}\right]$ where 'x' is in metres and 't' is in seconds. The amplitude and maximum speed of the particle is

(A) 3 m, 6π m s^{-1} **(B)** 3 m, 2π m s^{-1}

(C) 3 m, 8π m s^{-1} **(D)** 3 m, 4π m s^{-1}

34. Electrical as well as gravitational affects can be thought to be caused by fields. Which of the following is true for an electrical or gravitational field?

(A) Fields are useful for understanding forces acting through a distance.

(B) The field concept is often used to describe contact forces.

(C) There is no way to verify the existence of a force field since it is just a concept.

(D) Gravitational or Electric field does not exist in the space around an object.

35. Four charges $+q$, $+2q$, $+q$ and $-2q$ are placed at the corners of a square ABCD respectively. The force on a unit positive charge kept at the centre 'O' is

(A) along the diagonal AC

(B) zero

(C) perpendicular to AD

(D) along the diagonal BD

36. An electric dipole with dipole moment 4×10^{-9} C m is aligned at 30° with the direction of a uniform electric field of magnitude 5×10^4 N C^{-1}, the magnitude of the torque acting on the dipole is

(A) 10^{-5} N m **(B)** 10^{-4} N m

(C) 10×10^{-3} N m **(D)** $\sqrt{3} \times 10^{-4}$ N m

37. Ten identical cells each of potential 'E' and internal resistance 'r', are connected in series to form a closed circuit. An ideal voltmeter connected across three cells, will read

(A) $13E$ **(B)** $10E$ **(C)** $7E$ **(D)** $3E$

38. In an atom electrons revolve around the nucleus along a path of radius 0.72 Å making 9.4×10^{18} revolutions per second. The equivalent current is – [given $e = 1.6 \times 10^{-19}$ C]

(A) 1.4 A **(B)** 1.2 A **(C)** 1.8 A **(D)** 1.5 A

39. When a metal conductor connected to left gap of a meter bridge is heated, the balancing point

(A) remains unchanged

(B) shifts towards right

(C) shifts to the center

(D) shifts towards left

40. Two tiny spheres carrying charges 1.8 μC and 2.8 μC are located at 40 cm apart. The potential at the mid-point of the line joining the two charges is

(A) 4.3×10^4 V **(B)** 3.8×10^4 V

(C) 3.6×10^5 V **(D)** 2.1×10^5 V

41. A parallel plate capacitor is charged by connecting a 2 V battery across it. It is then disconnected from the battery and a glass slab is introduced between plates. Which of the following pairs of quantities decrease?

(A) Energy stored and capacitance.

(B) Charge and potential difference.

(C) Capacitance and charge.

(D) Potential difference and energy stored.

42. A charged particle is moving in an electric field of 3×10^{-10} V m^{-1} with mobility 2.5×10^6 m^2/V/s, its drift velocity is

(A) 2.5×10^4 m/s **(B)** 7.5×10^{-4} m/s

(C) 1.2×10^{-4} m/s **(D)** 8.33×10^{-4} m/s

43. Wire bound resistors are made by
(A) winding the wires of an alloy of Ge, Au, Ga
(B) winding the wires of an alloy of Cu, Al, Ag
(C) winding the wires of an alloy of manganin, constantan, nichrome
(D) winding the wires of an alloy of Si, Tu, Fe

44. A galvanometer of resistance 50 Ω is connected to a battery of 3 V along with a resistance 2950 Ω in series. A full scale deflection of 30 divisions is obtained in the galvanometer. In order to reduce this deflection to 20 divisions, the resistance in series should be
(A) 5050 Ω **(B)** 6050 Ω
(C) 4450 Ω **(D)** 5550 Ω

45. A circular coil of wire of radius 'r' has 'n' turns and carries a current 'I'. The magnetic induction 'B' at a point on the axis of the coil at a distance $\sqrt{3}\,r$ from its centre is
(A) $\frac{\mu_0 nI}{16r}$ **(B)** $\frac{\mu_0 nI}{32r}$ **(C)** $\frac{\mu_0 nI}{4r}$ **(D)** $\frac{\mu_0 nI}{8r}$

46. If voltage across a bulb rated 220 V, 100 W drops by 2.5% of its rated value, the percentage of the rated value by which the power would decrease is
(A) 5% **(B)** 20% **(C)** 10% **(D)** 2.5%

47. A wire of a certain material is stretched slowly by 10%. Its new resistance and specific resistance becomes respectively
(A) 1.21 times, same
(B) 1.1 times, 1.1 times
(C) both remains the same
(D) 1.2 times, 1.1 times

48. A proton moves with a velocity of $5 \times 10^6\,\hat{j}\,\text{m s}^{-1}$ through the uniform electric field, $\vec{E} = 4 \times 10^6\,[2\hat{i} + 0.2\hat{j} + 0.1\hat{k}]\,\text{V m}^{-1}$ and the uniform magnetic field $\vec{B} = 0.2\,[\hat{i} + 0.2\hat{j} + \hat{k}]$ T. The approximate net force acting on the proton is
(A) 2.2×10^{-13} N **(B)** 5×10^{-13} N
(C) 20×10^{-13} N **(D)** 25×10^{-13} N

49. A solenoid of length 50 cm having 100 turns carries a current of 2.5 A. The magnetic field at one end of the solenoid is
(A) 1.57×10^{-4} T **(B)** 3.14×10^{-4} T
(C) 9.42×10^{-4} T **(D)** 6.28×10^{-4} T

50. An alternating current is given by $i = i_1 \sin\omega t + i_2 \cos\omega t$. The r.m.s. current is given by
(A) $\sqrt{\frac{i_1^2 + i_2^2}{2}}$ **(B)** $\frac{i_1 + i_2}{\sqrt{2}}$
(C) $\sqrt{\frac{i_1^2 + i_2^2}{\sqrt{2}}}$ **(D)** $\frac{i_1 - i_2}{\sqrt{2}}$

51. Which of the following statements proves that Earth has a magnetic field?
(A) Earth is surrounded by ionosphere.
(B) The intensity of cosmic rays stream of charged particles is more at the poles than at the equator.
(C) A large quantity of iron-ore is found in the Earth.
(D) Earth is a planet rotating about the North South axis.

52. A long solenoid has 500 turns, when a current of 2 A is passed through it, the resulting magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb, then self induction of the solenoid is
(A) 2.0 henry **(B)** 4.0 henry
(C) 1.0 henry **(D)** 2.5 henry

53. A fully charged capacitor 'C' with initial charge 'q_0' is connected to a coil of self inductance 'L' at $t = 0$. The time at which the energy is stored equally between the electric and the magnetic field is
(A) $\pi\sqrt{LC}$ **(B)** $2\pi\sqrt{LC}$
(C) $\frac{\pi}{4}\sqrt{LC}$ **(D)** $\sqrt{LC}$

54. A magnetic field of flux density $1.0\,\text{Wb m}^{-2}$ acts normal to a 80 turn coil of $0.01\,\text{m}^2$ area. If this coil is removed from the field in 0.2 second, the emf induced in it is
(A) 0.8 V **(B)** 4 V **(C)** 5 V **(D)** 8 V

55. A ray of light passes through an equilateral glass prism in such a manner that the angle of incidence is equal to the angle of emergence and each of these angles is equal to $\frac{3}{4}$ of the angle of the prism. The angle of deviation is
(A) 20° **(B)** 45° **(C)** 30° **(D)** 39°

56. A convex lens of focal length 'f' is placed somewhere in between an object and a screen. The distance between the object and the screen is 'x'. If the numerical value of the magnification produced by the lens is 'm', then the focal length of the lens is
(A) $\frac{(m+1)^2 x}{m}$ **(B)** $\frac{mx}{(m+1)^2}$
(C) $\frac{(m-1)^2 x}{m}$ **(D)** $\frac{mx}{(m-1)^2}$

57. A series resonant ac circuit contains a capacitance 10^{-6} F and an inductor of 10^{-4} H. The frequency of electrical oscillations will be
(A) $\frac{10^5}{2\pi}$ Hz **(B)** 10^5 Hz
(C) $\frac{10}{2\pi}$ Hz **(D)** 10 Hz

58. In a series LCR circuit $R = 300\,\Omega$, $L = 0.9\,H$, $C = 2.0\,\mu F$ and $\omega = 1000$ rad/sec, then impedance of the circuit is

(A) 500 Ω **(B)** 1300 Ω
(C) 400 Ω **(D)** 900 Ω

59. Which of the following radiations of electromagnetic waves has the highest wavelength?

(A) IR-rays **(B)** X-rays
(C) Microwaves **(D)** UV-rays

60. The power of a equi-concave lens is −4.5 D and is made of a material of R.I. 1.6, the radii of curvature of the lens is

(A) −2.66 cm **(B)** −26.6 cm
(C) 115.44 cm **(D)** +36.6 cm

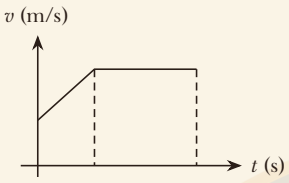
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KCET 2021

Physics Question Paper

Version A1 · 60 Questions · 60 Marks · 80 Minutes

1. For a body moving along a straight line, the following v-t graph is obtained.



According to the graph, the displacement during

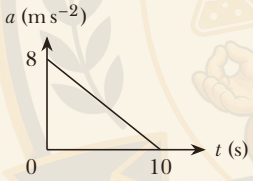
(A) Uniform acceleration is greater than that during uniform motion.

(B) Uniform acceleration is less than that during uniform motion.

(C) Uniform acceleration is equal to that during uniform motion.

(D) Uniform motion is zero.

2. A particle starts from rest. Its acceleration 'a' versus time 't' is shown in the figure. The maximum speed of the particle will be:



(A) $80\ \text{m s}^{-1}$ **(B)** $40\ \text{m s}^{-1}$

(C) $18\ \text{m s}^{-1}$ **(D)** $2\ \text{m s}^{-1}$

3. The maximum range of a gun on horizontal plane is 16 km. If $g = 10\ \text{m s}^{-2}$, then muzzle velocity of a shell is

(A) $160\ \text{m s}^{-1}$ **(B)** $200\sqrt{2}\ \text{m s}^{-1}$

(C) $400\ \text{m s}^{-1}$ **(D)** $800\ \text{m s}^{-1}$

4. The trajectory of a projectile is

(A) semicircle

(B) an ellipse

(C) a parabola always

(D) a parabola in the absence of air resistance

5. For a projectile motion, the angle between the velocity and acceleration is minimum and acute at

(A) only one point **(B)** two points

(C) three points **(D)** four points

6. A particle starts from the origin at $t = 0$ s with a velocity of $10\hat{j}\ \text{m s}^{-1}$ and moves in the x-y plane with a constant acceleration of $(8\hat{i} + 2\hat{j})\ \text{m s}^{-2}$. At an instant when the x-coordinate of the particle is 16 m, y-coordinate of the particle is:

(A) 16 m **(B)** 28 m **(C)** 36 m **(D)** 24 m

7. A coin placed on a rotating turn table just slips if it is placed at a distance of 4 cm from the centre. If the angular velocity of the turn table is doubled it will just slip at a distance of:

(A) 1 cm **(B)** 2 cm **(C)** 4 cm **(D)** 8 cm

8. A 1 kg ball moving at $12\ \text{m s}^{-1}$ collides with a 2 kg ball moving in opposite direction at $24\ \text{m s}^{-1}$. If the coefficient of restitution is $\frac{2}{3}$, then their velocities after the collision are

(A) $-4\ \text{m s}^{-1}$, $-28\ \text{m s}^{-1}$

(B) $-28\ \text{m s}^{-1}$, $-4\ \text{m s}^{-1}$

(C) $4\ \text{m s}^{-1}$, $28\ \text{m s}^{-1}$

(D) $28\ \text{m s}^{-1}$, $4\ \text{m s}^{-1}$

9. A ball hits the floor and rebounds after an inelastic collision. In this case

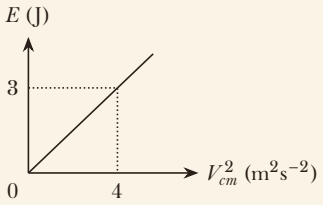
(A) the momentum of the ball is conserved

(B) the mechanical energy of the ball is conserved

(C) the total momentum of the ball and the earth is conserved

(D) the total mechanical energy of the ball and the earth is conserved

10. In figure E and V_{cm} represent the total energy and speed of centre of mass of an object of mass 1 kg in pure rolling. The object is:



(A) sphere **(B)** ring

(C) disc **(D)** Hollow Cylinder

11. Two bodies of masses 8 kg are placed at the vertices A and B of an equilateral triangle ABC. A third body of mass 2 kg is placed at the centroid G of the triangle. If $AG = BG = CG = 1$ m, where should a fourth body of mass 4 kg be placed so that the resultant force on the 2 kg body is zero?

(A) at C

(B) at a point P on the line CG such that $PG = \frac{1}{\sqrt{2}}$ m

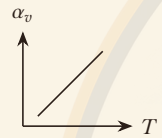
(C) at a point P on the line CG such that $PG = 0.5$ m

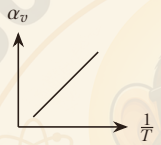
(D) at a point P on the line CG such that $PG = 2$ m

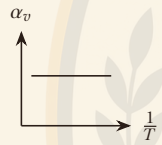
12. Two capillary tubes P and Q are dipped vertically in water. The height of water level in capillary tube P is $\frac{2}{3}$rd of the height in capillary tube Q. The ratio of their diameter is _____.

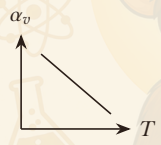
(A) 2:3 **(B)** 3:2 **(C)** 3:4 **(D)** 4:3

13. Which of the following curves represent the variation of coefficient of volume expansion of an ideal gas at constant pressure?

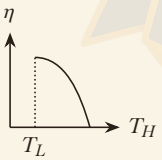
(A) 

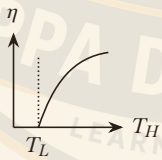
(B) 

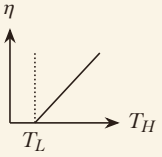
(C) 

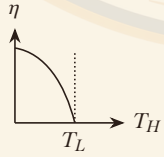
(D) 

14. A number of Carnot engines are operated at identical cold reservoir temperatures (T_L). However, their hot reservoir temperatures are kept different. A graph of the efficiency of the engines versus hot reservoir temperature (T_H) is plotted. The correct graphical representation is

(A) 

(B) 

(C) 

(D) 

15. A gas mixture contains monoatomic and diatomic molecules of 2 moles each. The mixture has a total internal energy of (symbols have usual meanings)

(A) $3RT$ **(B)** $5RT$ **(C)** $8RT$ **(D)** $9RT$

16. A pendulum oscillates simple harmonically if and only if

(I) the size of the bob of pendulum is negligible in comparison with the length of the pendulum.

(II) the angular amplitude is less than 10°.

(A) Both (I) and (II) are correct

(B) Both (I) and (II) are incorrect

(C) Only (I) is correct

(D) Only (II) is correct

17. To propagate both longitudinal and transverse waves, a material must have

(A) Bulk and shear moduli

(B) Only bulk modulus

(C) Only shear modulus

(D) Young's and Bulk modulus

18. A copper rod AB of length l is rotated about end A with a constant angular velocity ω. The electric field at a distance x from the axis of rotation is

(A) $\frac{m\omega^2 x}{e}$ **(B)** $\frac{m\omega x}{el}$ **(C)** $\frac{mx}{\omega^2 l}$ **(D)** $\frac{me}{\omega^2 x}$

19. Electric field due to infinite, straight uniformly charged wire varies with distance 'r' as

(A) r **(B)** $\frac{1}{r}$ **(C)** $\frac{1}{r^2}$ **(D)** r^2

20. A 2-gram object, located in a region of uniform electric field $\vec{E} = (300\ \text{N C}^{-1})\,\hat{i}$ carries a charge Q. The object released from rest at $x = 0$, has a kinetic energy of 0.12 J at $x = 0.5$ m. Then Q is

(A) 400 μC **(B)** −400 μC

(C) 800 μC **(D)** −800 μC

21. If a slab of insulating material (conceptual) 4×10^{-3} m thick is introduced between the plates of a parallel plate capacitor, the separation between the plates has to be increased by 3.5×10^{-3} m to restore the capacity to original value. The dielectric constant of the material will be

(A) 6 **(B)** 8 **(C)** 10 **(D)** 12

22. Eight drops of mercury of equal radii combine to form a big drop. The capacitance of a bigger drop as compared to each smaller drop is

(A) 2 times **(B)** 8 times

(C) 4 times **(D)** 16 times

23. Which of the statements is false in the case of polar molecules?

(A) Centers of positive and negative charges are separated in the absence of external electric field.

(B) Centers of positive and negative charges are separated in the presence of external electric field.

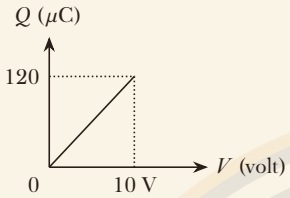
(C) Do not possess permanent dipole moments.

(D) Ionic molecule HCl is the example of polar molecule.

24. An electrician requires a capacitance of 6 μF in a circuit across a potential difference of 1.5 kV. A large number of 2 μF capacitors which can withstand a potential difference of not more than 500 V are available. The minimum number of capacitors required for the purpose is

(A) 3 **(B)** 9 **(C)** 6 **(D)** 27

25. In figure, charge on the capacitor is plotted against potential difference across the capacitor. The capacitance and energy stored in the capacitor are respectively.

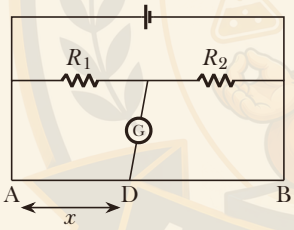


(A) 12 μF, 1200 μJ **(B)** 12 μF, 600 μJ
(C) 24 μF, 600 μJ **(D)** 24 μF, 1200 μJ

26. A wire of resistance 3 Ω is stretched to twice its original length. The resistance of the new wire will be

(A) 1.5 Ω **(B)** 3 Ω **(C)** 6 Ω **(D)** 12 Ω

27. In the given arrangement of experiment on metre bridge, if AD corresponding to null deflection of the galvanometer is X, what would be its value if the radius of the wire AB is doubled?



(A) X **(B)** $\frac{X}{4}$ **(C)** 4X **(D)** 2X

28. A copper wire of length 1 m and uniform cross-sectional area 5×10^{-7} m^2 carries a current of 1 A. Assuming that there are 8×10^{28} free electrons per m^3 in copper, how long will an electron take to drift from one end of the wire to the other?

(A) 0.8×10^3 s **(B)** 1.6×10^3 s
(C) 3.2×10^3 s **(D)** 6.4×10^3 s

29. Consider an electrical conductor connected across a potential difference V. Let Δq be a small charge moving through it in time Δt. If I is the electric current through it,

(I) the kinetic energy of the charge increases by $IV\Delta t$.
(II) the electric potential energy of the charge decreases by $IV\Delta t$.
(III) the thermal energy of the conductor increases by $IV\Delta t$.

Then the correct statement/s is / are

(A) (I) **(B)** (I), (II)
(C) (I) and (III) **(D)** (II), (III)

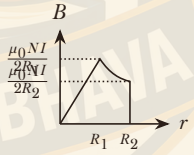
30. A strong magnetic field is applied on a stationary electron. Then the electron

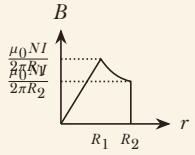
(A) Moves in the direction of the field
(B) Moves in an opposite direction of the field
(C) Remains stationary
(D) Starts spinning

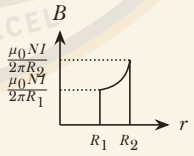
31. Two parallel wires in free space are 10 cm apart and each carries a current of 10 A in the same direction. The force exerted by one wire on the other [per unit length] is

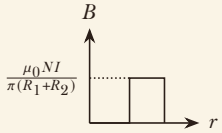
(A) 2×10^{-4} N m^{-1} [attractive]
(B) 2×10^{-7} N m^{-1} [attractive]
(C) 2×10^{-4} N m^{-1} [repulsive]
(D) 2×10^{-7} N m^{-1} [repulsive]

32. A toroid with thick windings of N turns has inner and outer radii R_1 and R_2 respectively. If it carries certain steady current I, the variation of the magnetic field due to the toroid with radial distance is correctly graphed in

(A)


(B)


(C)


(D)


33. A tightly wound long solenoid has 'n' turns per unit length, a radius 'r' and carries a current I. A particle having charge 'q' and mass 'm' is projected from a point on the axis in a direction perpendicular to the axis. The maximum speed of the particle for which the particle does not strike the solenoid is

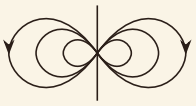
(A) $\frac{\mu_0 nIqr}{m}$ **(B)** $\frac{\mu_0 nIqr}{2m}$
(C) $\frac{\mu_0 nIqr}{4m}$ **(D)** $\frac{\mu_0 nIqr}{8m}$

34. Earth's magnetic field always has a horizontal component except at

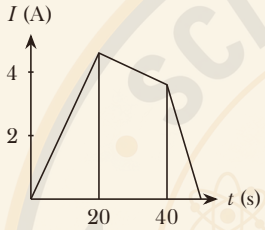
(A) equator **(B)** magnetic poles
(C) a latitude of 60° **(D)** an altitude of 60°

35. Which of the field pattern given below is valid for electric field as well as for magnetic field?

(A)  **(B)** 

(C)  **(D)** 

36. The current flowing through an inductance coil of self inductance 6 mH at different time instants is as shown. The emf induced between $t = 20$ s and $t = 40$ s is nearly

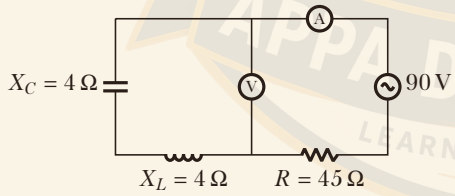


(A) 2×10^{-2} V **(B)** 3×10^{-2} V
(C) 4×10^{-3} V **(D)** 30×10^{2} V

37. The physical quantity which is measured in the unit of Wb A^{-1} is

(A) Self inductance **(B)** Mutual inductance
(C) Magnetic flux **(D)** Both (A) and (B)

38. What will be the reading in the voltmeter and ammeter of the circuit shown?



(A) 90 V, 2 A **(B)** 0 V, 2 A
(C) 90 V, 1 A **(D)** 0 V, 1 A

39. LC oscillations are similar and analogous to the mechanical oscillations of a block attached to a spring. The electrical equivalent of the force constant of the spring is

(A) reciprocal of capacitive reactance
(B) capacitive reactance
(C) reciprocal of capacitance
(D) capacitance

40. In an oscillating LC circuit, $L = 3.00$ mH and $C = 2.70\ \mu$F. At $t = 0$ the charge on the capacitor is zero and the current is 2.00 A. The maximum charge that will appear on the capacitor will be:

(A) 1.8×10^{-5} C **(B)** 18×10^{-5} C
(C) 9×10^{-5} C **(D)** 90×10^{-5} C

41. Suppose that the electric field amplitude of electromagnetic wave is $E_0 = 120$ N C^{-1} and its frequency is $f = 50$ MHz. Then which of the following value is incorrectly computed?

(A) Magnetic field amplitude is 400 nT.
(B) Angular frequency of EM wave is $\pi \times 10^8$ rad/s
(C) Propagation constant (angular wave number) is 2.1 rad/m
(D) Wavelength of the EM wave is 6 m.

42. The source of electromagnetic waves can be a charge.

(A) Moving with a constant velocity
(B) Moving in a circular orbit.
(C) At rest
(D) Moving parallel to the magnetic field

43. In refraction, light waves are bent on passing from one medium to second medium because, in the second medium.

(A) frequency is different
(B) speed is different
(C) coefficient of elasticity is different
(D) amplitude is smaller.

44. If the refractive index from air to glass is $\frac{3}{2}$ and that from air to water is $\frac{4}{3}$, then the ratio of focal lengths of a glass lens in water and in air is

(A) 1:2 **(B)** 2:1 **(C)** 1:4 **(D)** 4:1

45. Two thin biconvex lenses have focal lengths f_1 and f_2. A third thin biconcave lens has focal length of f_3. If the two biconvex lenses are in contact, the total power of the lenses is P_1. If the first convex lens is in contact with the third lens, the total power is P_2. If the second lens is in contact with the third lens, the total power is P_3 then

(A) $P_1 = \dfrac{f_1 f_2}{f_1 - f_2}$, $P_2 = \dfrac{f_1 f_3}{f_3 - f_1}$ and $P_3 = \dfrac{f_2 f_3}{f_3 - f_2}$

(B) $P_1 = \dfrac{f_1 - f_2}{f_1 f_2}$, $P_2 = \dfrac{f_3 - f_1}{f_3 + f_1}$ and $P_3 = \dfrac{f_3 - f_2}{f_2 f_3}$

(C) $P_1 = \dfrac{f_1 - f_2}{f_1 f_2}$, $P_2 = \dfrac{f_3 - f_1}{f_1 f_3}$ and $P_3 = \dfrac{f_3 - f_2}{f_2 f_3}$

(D) $P_1 = \dfrac{f_1 + f_2}{f_1 f_2}$, $P_2 = \dfrac{f_3 - f_1}{f_1 f_3}$ and $P_3 = \dfrac{f_3 - f_2}{f_2 f_3}$

46. The size of the image of an object, which is at infinity, as formed by a convex lens of focal length 30 cm is 2 cm. If a concave lens of focal length 20 cm is placed between the convex lens and the image at a distance of 26 cm from the lens, the new size of the image is:

(A) 1.25 cm **(B)** 2.5 cm
(C) 1.05 cm **(D)** 2 cm

47. A slit of width 'a' is illuminated by red light of wavelength 6500 Å. If the first diffraction minimum falls at 30°, then the value of 'a' is _____

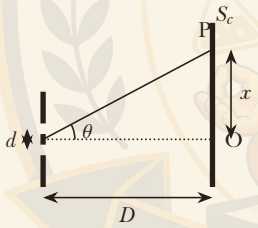
(A) 6.5×10^{-4} mm **(B)** 1.3 micron
(C) 3250 Å **(D)** 2.6×10^{-4} cm

48. Which of the statements are correct with reference to single slit diffraction pattern?

(i) Fringes are of unequal width
(ii) Fringes are of equal width
(iii) Light energy is conserved
(iv) Intensities of all bright fringes are equal

(A) (i) and (iii) **(B)** (i) and (iv)
(C) (ii) and (iv) **(D)** (ii) and (iii)

49. In the Young's double slit experiment a monochromatic source of wavelength λ is used. The intensity of light passing through each slit is I_0. The intensity of light reaching the screen S_c at a point P, a distance x from O is given by (Take $d \ll D$).



(A) $I_0 \cos^2\left(\frac{\pi D}{\lambda d}x\right)$ **(B)** $4I_0 \cos^2\left(\frac{\pi d}{\lambda D}x\right)$
(C) $I_0 \sin^2\left(\frac{\pi d}{2\lambda D}x\right)$ **(D)** $4I_0 \cos\left(\frac{\pi d}{2\lambda D}x\right)$

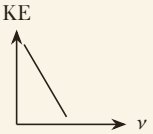
50. The work function of a metal is 1 eV. Light of wavelength 3000 Å is incident on this metal surface. The velocity of emitted photoelectrons will be

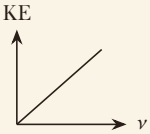
(A) $10\ \text{m s}^{-1}$ **(B)** $1 \times 10^3\ \text{m s}^{-1}$
(C) $1 \times 10^4\ \text{m s}^{-1}$ **(D)** $1 \times 10^6\ \text{m s}^{-1}$

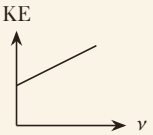
51. A proton moving with a momentum P_1 has a kinetic energy $\frac{1}{8}$th of its rest mass energy. Another light photon having energy equal to the kinetic energy of the proton possesses a momentum P_2. Then the ratio $\frac{P_1 - P_2}{P_1}$ is equal to

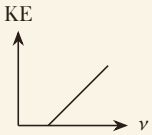
(A) 1 **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{2}$ **(D)** $\frac{3}{4}$

52. According to Einstein's photoelectric equation the graph between kinetic energy of photoelectrons ejected and the frequency of incident radiation is

(A) 

(B) 

(C) 

(D) 

53. Energy of an electron in the second orbit of hydrogen atom is E_2. The energy of electron in the third orbit of He^+ will be

(A) $\frac{9}{16}E_2$ **(B)** $\frac{16}{9}E_2$ **(C)** $\frac{3}{16}E_2$ **(D)** $\frac{16}{3}E_2$

54. The figure shows standing de Broglie waves due to the revolution of electron in a certain orbit of hydrogen atom. Then the expression for the orbit radius is (all notations have their usual meanings)

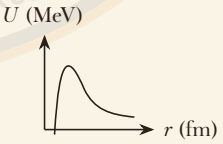


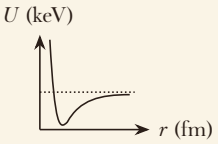
(A) $\frac{h^2\epsilon_0}{\pi me^2}$ **(B)** $\frac{4h^2\epsilon_0}{\pi me^2}$
(C) $\frac{9h^2\epsilon_0}{\pi me^2}$ **(D)** $\frac{16h^2\epsilon_0}{\pi me^2}$

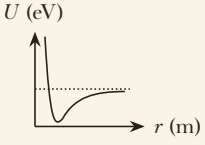
55. An electron in an excited state of Li^{2+} ion has angular momentum $\frac{3h}{2\pi}$. The de Broglie wavelength of electron in this state is $P\pi a_0$ (where a_0 = Bohr radius). The value of P is

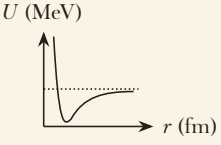
(A) 3 **(B)** 2 **(C)** 1 **(D)** 4

56. Which graph in the following diagrams correctly represents the potential energy of a pair of nucleons as a function of their separation?

(A) 

(B) 

(C) 

(D) 

57. In a nuclear reactor heavy nuclei is not used as moderators because

(A) They will break up

(B) Elastic collision of neutrons with heavy nuclei will not slow them down.

(C) The net weight of the reactor would be unbearably high

(D) Substances with heavy nuclei do not occur in liquid or gaseous state at room temperature.

58. The circuit given represents which of the logic operations?



(A) OR **(B)** AND **(C)** NOT **(D)** NOR

59. Identify the incorrect statement:

(A) when a P-N junction diode is forward biased, the width of the depletion region decreases.

(B) when a P-N junction diode is reverse biased, the barrier potential increases.

(C) a photo diode is operated in the reverse bias.

(D) an LED is a lightly doped P-N junction diode which emits spontaneous radiation on forward biasing.

60. Three photodiodes D_1, D_2 and D_3 are made of semiconductors having band gaps of 2.5 eV, 2 eV and 3 eV respectively. Which one will be able to detect light of wavelength 600 nm?

(A) D_1 only

(B) Both D_1 and D_3

(C) D_2 only

(D) All the three diodes

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KCET 2020
Physics Question Paper

Version A-1 · 60 Questions · 60 Marks · 80 Minutes

1. The value of acceleration due to gravity at a height of 10 km from the surface of earth is x. At what depth inside the earth is the value of the acceleration due to gravity has the same value x?

(A) 5 km **(B)** 20 km **(C)** 10 km **(D)** 15 km

2. Young's modulus of a perfect rigid body is

(A) zero
(B) unity
(C) infinity
(D) between zero and unity

3. A wheel starting from rest gains an angular velocity of 10 rad/s after uniformly accelerated for 5 sec. The total angle through which it has turned is

(A) 25 rad
(B) 100 rad
(C) 25π rad
(D) 50π rad about a vertical axis

4. Iceberg floats in water with part of it submerged. What is the fraction of the volume of iceberg submerged if the density of ice is $\rho_i = 0.917\ \text{g cm}^{-3}$?

(A) 0.917 **(B)** 1 **(C)** 0.458 **(D)** 0

5. A sphere, a cube and a thin circular plate all of same material and same mass initially heated to same high temperature are allowed to cool down under similar conditions. Then the

(A) plate will cool the fastest and cube the slowest.
(B) sphere will cool the fastest and cube the slowest.
(C) plate will cool the fastest and sphere the slowest.
(D) cube will cool the fastest and plate the slowest.

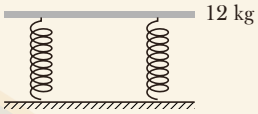
6. In an adiabatic expansion of an ideal gas the product of pressure and volume.

(A) Decreases
(B) Increases
(C) Remains constant
(D) At first increases and then decreases

7. A certain amount of heat energy is supplied to a monoatomic ideal gas which expands at constant pressure. What fraction of the heat energy is converted into work?

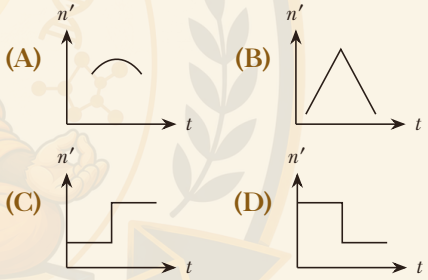
(A) 1 **(B)** $\frac{2}{3}$ **(C)** $\frac{2}{5}$ **(D)** $\frac{5}{7}$

8. A tray of mass 12 kg is supported by two identical springs as shown in figure. When the tray is pressed down slightly and then released, it executes SHM with a time period of 1.5 s. The spring constant of each spring is

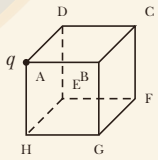


(A) $50\ \text{N m}^{-1}$ **(B)** 0
(C) $105\ \text{N m}^{-1}$ **(D)** ∞

9. A train whistling at constant frequency 'n' is moving towards a station at a constant speed V. The train goes past a stationary observer on the station. The frequency 'n' of the sound as heard by the observer is plotted as a function of time 't'. Identify the correct curve



10. A point charge 'q' is placed at the corner of a cube of side 'a' as shown in the figure. What is the electric flux through the face ABCD?



(A) 0 **(B)** $\frac{q}{24\varepsilon_0}$ **(C)** $\frac{q}{6\varepsilon_0}$ **(D)** $\frac{q}{72\varepsilon_0}$

11. The electric field lines on the left have twice the separation on those on the right as shown in figure. If the magnitude of the field at A is $40\ \text{V m}^{-1}$, what is the force on 20 μC charge kept at B?



(A) $4 \times 10^{-4}\ \text{V m}^{-1}$ **(B)** $8 \times 10^{-4}\ \text{V m}^{-1}$
(C) $16 \times 10^{-4}\ \text{V m}^{-1}$ **(D)** $1 \times 10^{-4}\ \text{V m}^{-1}$

12. An infinitely long thin straight wire has uniform charge density of $\frac{1}{4}\times10^{-2}\,\text{C m}^{-1}$. What is the magnitude of electric field at a distance 20 cm from the axis of the wire?

(A) $1.12\times10^{8}\,\text{N C}^{-1}$ **(B)** $4.5\times10^{8}\,\text{N C}^{-1}$
(C) $2.25\times10^{8}\,\text{N C}^{-1}$ **(D)** $9\times10^{8}\,\text{N C}^{-1}$
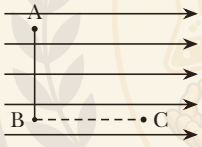
13. A dipole of dipole moment 'P' and moment of inertia I is placed in a uniform electric field $\vec{E}$. If it is displaced slightly from its stable equilibrium position, the period of oscillation of dipole is

(A) $\sqrt{\frac{PE}{I}}$ **(B)** $2\pi\sqrt{\frac{I}{PE}}$
(C) $\frac{1}{2\pi}\sqrt{\frac{PE}{I}}$ **(D)** $\pi\sqrt{\frac{I}{PE}}$

14. The difference between equivalent capacitances of two identical capacitors connected in parallel to that in series is 6 μF. The value of capacitance of each capacitor is

(A) 2 μF **(B)** 3 μF **(C)** 4 μF **(D)** 6 μF

15. Figure shows three points A, B and C in a region of uniform electric field $\vec{E}$. The line AB is perpendicular and BC is parallel to the field lines. Then which of the following holds good? (V_A, V_B and V_C represent the electric potential at points A, B and C respectively)



(A) $V_A = V_B = V_C$ **(B)** $V_A = V_B > V_C$
(C) $V_A = V_B < V_C$ **(D)** $V_A > V_B = V_C$

16. When a soap bubble is charged?

(A) Its radius increases.
(B) Its radius decreases.
(C) The radius remains the same.
(D) Its radius may increase or decrease.

17. A hot filament liberates an electron with zero initial velocity. The anode potential is 1200 V. The speed of the electron when it strikes the anode is

(A) $1.5\times10^{5}\,\text{m s}^{-1}$ **(B)** $2.5\times10^{6}\,\text{m s}^{-1}$
(C) $2.1\times10^{7}\,\text{m s}^{-1}$ **(D)** $2.5\times10^{8}\,\text{m s}^{-1}$

18. A metal rod of length 10 cm and a rectangular cross-section of 1 cm $\times$ $\frac{1}{2}$ cm is connected to a battery across opposite faces. The resistance will be

(A) maximum when the battery is connected across 1 cm $\times$ $\frac{1}{2}$ cm faces.
(B) maximum when the battery is connected across 10 cm $\times$ $\frac{1}{2}$ cm faces.
(C) maximum when the battery is connected across 10 cm $\times$ 1 cm faces.
(D) same irrespective of the three faces.

19. A car has a fresh storage battery of e.m.f. 12 V and internal resistance $2\times10^{-2}\,\Omega$. If the starter motor draws a current of 80 A. Then the terminal voltage when the starter is on is

(A) 12 V **(B)** 8.4 V **(C)** 10.4 V **(D)** 9.3 V

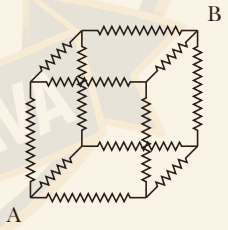
20. A potentiometer has a uniform wire of length 5 m. A battery of emf 10 V and negligible internal resistance is connected between its ends. A secondary cell connected to the circuit gives balancing length at 200 cm. The emf of the secondary cell is

(A) 4 V **(B)** 6 V **(C)** 2 V **(D)** 8 V

21. The colour code for a carbon resistor of resistance 0.28 kΩ ± 10% is

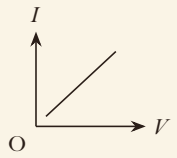
(A) Red, Grey, Brown, Silver
(B) Red, Green, Brown, Silver
(C) Red, Grey, Silver, Silver
(D) Red, Green, Silver

22. Each resistance in the given cubical network has resistance of 1 Ω and equivalent resistance between A and B is



(A) $\frac{5}{6}\,\Omega$ **(B)** $\frac{6}{5}\,\Omega$ **(C)** $\frac{5}{12}\,\Omega$ **(D)** $\frac{12}{5}\,\Omega$

23. I-V characteristic of a copper wire of length L and area of cross-section A is shown in figure. The slope of the curve becomes



(A) More if experiment is performed at higher temperature.
(B) More if a wire of steel of same dimension is used.
(C) Less if the area of the wire is increased.
(D) Less if the length of the wire is increased.

24. In the given figure, the magnetic field at 'O'.

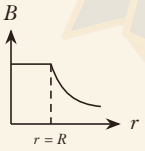


(A) $\frac{3}{4}\frac{\mu_0 I}{r}+\frac{\mu_0 I}{4\pi r}$ **(B)** $\frac{3}{10}\frac{\mu_0 I}{r}-\frac{\mu_0 I}{4\pi r}$
(C) $\frac{3}{8}\frac{\mu_0 I}{r}+\frac{\mu_0 I}{4\pi r}$ **(D)** $\frac{3}{8}\frac{\mu_0 I}{r}-\frac{\mu_0 I}{4\pi r}$

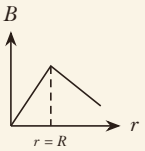
25. The magnetic field at the origin due to a current element $i\,\vec{dl}$ placed at a point with vector position $\vec{r}$ is

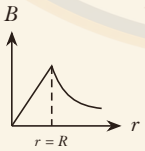
(A) $\frac{\mu_0 i}{4\pi}\frac{\vec{dl}\times\vec{r}}{r^3}$ **(B)** $\frac{\mu_0 i}{4\pi}\frac{\vec{r}\times\vec{dl}}{r^3}$
(C) $\frac{\mu_0 i}{4\pi}\frac{\vec{dl}\times\vec{r}}{r^2}$ **(D)** $\frac{\mu_0 i}{4\pi}\frac{\vec{r}\times\vec{dl}}{r^2}$

26. A long cylindrical wire of radius R carries a uniform current I flowing through it. The variation of magnetic field with distance 'r' from the axis of the wire is shown by

(A)


(B)


(C)


(D)


27. A cyclotron is used to accelerate protons ($^1_1$H), Deuterons ($^2_1$H) and α-particles ($^4_2$He). While exiting under similar conditions, the minimum K.E. is gained by

(A) α-particle **(B)** proton
(C) deuteron **(D)** same for all

28. A paramagnetic sample shows a net magnetization of $8\,A\,m^{-1}$ when placed in an external magnetic field of 0.6 T at a temperature of 4 K. When the same sample is placed in an external magnetic field of 0.2 T at a temperature of 16 K. The magnetization will be

(A) $\frac{32}{3}\,A\,m^{-1}$ **(B)** $\frac{2}{3}\,A\,m^{-1}$
(C) $6\,A\,m^{-1}$ **(D)** $2.4\,A\,m^{-1}$

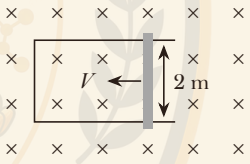
29. The ratio of magnetic field at the centre of a current carrying circular coil to its magnetic moment is 'x'. If the current and the radius both are doubled. The new ratio will become

(A) $2x$ **(B)** $4x$ **(C)** $\frac{x}{4}$ **(D)** $\frac{x}{8}$

30. In a permanent magnet at room temperature

(A) Magnetic moment of each molecule is zero.
(B) The individual molecules have non-zero magnetic moment which are all perfectly aligned.
(C) Domains are partially aligned.
(D) Domains are all perfectly aligned.

31. A rod of length 2 m slides with a speed of $5\,m\,s^{-1}$ on a rectangular conducting frame as shown in figure. There exists a uniform magnetic field of 0.04 T perpendicular to the plane of the figure. If the resistance of the rod is 3 Ω. The current through the rod is

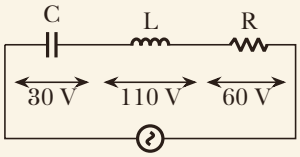


(A) 75 mA **(B)** 133 mA
(C) 0.75 A **(D)** 1.33 A

32. The current in a coil of inductance 0.2 H changes from 5 A to 2 A in 0.5 sec. The magnitude of the average induced emf in the coil is

(A) 0.6 V **(B)** 1.2 V **(C)** 30 V **(D)** 0.3 V

33. In the given circuit the peak voltages across C, L and R are 30 V, 110 V and 60 V respectively. The rms value of the applied voltage is



(A) 100 V **(B)** 200 V **(C)** 70.7 V **(D)** 141 V

34. The power factor of R-L circuit is $\frac{1}{\sqrt{3}}$. If the inductive reactance is 2 Ω. The value of resistance is

(A) 2 Ω **(B)** $\sqrt{2}$ Ω **(C)** 0.5 Ω **(D)** $\frac{1}{\sqrt{2}}$ Ω

35. In the given circuit, the resonant frequency is



(A) 15.92 Hz **(B)** 159.2 Hz
(C) 1592 Hz **(D)** 15910 Hz

36. A light beam of intensity $20\ W/cm^2$ is incident normally on a perfectly reflecting surface of sides 25 cm × 15 cm. The momentum imparted to the surface by the light per second is

(A) $2 \times 10^{-5}\ kg\ m\ s^{-1}$
(B) $1 \times 10^{-5}\ kg\ m\ s^{-1}$
(C) $5 \times 10^{-5}\ kg\ m\ s^{-1}$
(D) $1.2 \times 10^{-5}\ kg\ m\ s^{-1}$

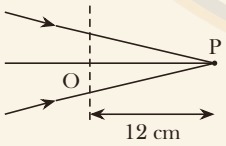
37. An object approaches a convergent lens from the left of the lens with a uniform speed 5 m/s and stops at the focus, the image

(A) moves away from the lens with an uniform speed 5 m/s.
(B) moves away from the lens with an uniform acceleration.
(C) moves away from the lens with a non-uniform acceleration.
(D) moves towards the lens with a non-uniform acceleration.

38. The refracting angle of a prism is A and refractive index of material of prism is $\cot\frac{A}{2}$. The angle of minimum deviation is

(A) $180° - 3A$ **(B)** $180° + 2A$
(C) $90° - A$ **(D)** $180° - 2A$

39. The following figure shows a beam of light converging at point P. When a concave lens of focal length 16 cm is introduced in the path of the beam at a place shown by dotted line such that OP becomes the axis of the lens, the beam converges at a distance x from the lens. The value of x will be equal to



(A) 12 cm **(B)** 24 cm **(C)** 36 cm **(D)** 48 cm

40. Three polaroid sheets P_1, P_2 and P_3 are kept parallel to each other such that the angle between pass axes of P_1 and P_2 is 45° and that between P_2 and P_3 is 45°. If unpolarised beam of light of intensity $128\ W\ m^{-2}$ is incident on P_1. What is the intensity of light coming out of P_3?

(A) $128\ W\ m^{-2}$ **(B)** 0
(C) $16\ W\ m^{-2}$ **(D)** $64\ W\ m^{-2}$

41. Two poles are separated by a distance of 3.14 m. The resolving power of human eye is 1 minute of an arc. The maximum distance from which he can identify the two poles distinctly is

(A) 10.8 km **(B)** 5.4 km
(C) 188 m **(D)** 376 m

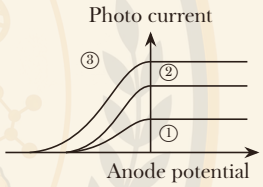
42. In Young's Double Slit Experiment, the distance between the slits and the screen is 1.2 m and the distance between the two slits is 2.4 mm. If a thin transparent mica sheet of thickness 1 μm and R.I. 1.5 is introduced between one of the interfering beams, the shift in the position of central bright fringe is

(A) 2 mm **(B)** 0.5 mm
(C) 0.125 mm **(D)** 0.25 mm

43. The de-Broglie wavelength associated with electron of hydrogen atom in this ground state is

(A) 0.3 Å **(B)** 3.3 Å **(C)** 6.26 Å **(D)** 10 Å

44. The following graph represents the variation of photo current with anode potential for a metal surface. Here I_1, I_2 and I_3 represents intensities and γ_1, γ_2, γ_3 represent frequency for curves 1, 2 and 3 respectively, then



(A) $\gamma_1 = \gamma_2$ and $I_1 \neq I_2$ **(B)** $\gamma_1 = \gamma_3$ and $I_1 = I_3$
(C) $\gamma_1 = \gamma_2$ and $I_1 = I_2$ **(D)** $\gamma_2 = \gamma_3$ and $I_1 = I_3$

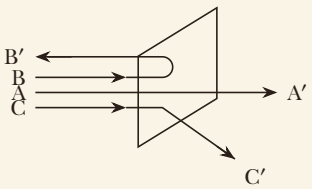
45. The period of revolution of an electron revolving in n^{th} orbit of H-atom is proportional to

(A) n^2 **(B)** $\frac{1}{n}$
(C) n^3 **(D)** Independent of n

46. Angular momentum of an electron in hydrogen atom is $\frac{3h}{2\pi}$ (h is the Planck's constant). The K.E. of the electron is

(A) 4.35 eV **(B)** 1.51 eV
(C) 3.4 eV **(D)** 6.8 eV

47. A beam of fast moving alpha particles were directed towards a thin film of gold. The parts A, B and C of the transmitted and reflected beams corresponding to the incident parts A, B and C of the beam are shown in the adjoining diagram. The number of alpha particles in



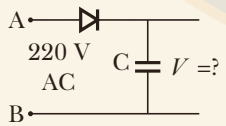
(A) B′ will be minimum and in C′ maximum
(B) A′ will be maximum and in C′ minimum
(C) A′ will be minimum and in B′ maximum
(D) C′ will be minimum and in B′ maximum

48. Two protons are kept at a separation of 10 nm. Let F_n and F_e be the nuclear force and the electromagnetic force between them
(A) $F_e = F_n$
(B) $F_e \gg F_n$
(C) $F_e \ll F_n$
(D) F_e and F_n differ only slightly

49. During a β^- decay
(A) an atomic electron is ejected.
(B) an electron which is already present within the nucleus is ejected.
(C) A neutron in the nucleus decays emitting an electron.
(D) A proton in the nucleus decays emitting an electron.

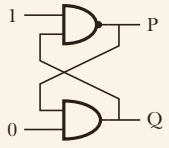
50. A radio-active element has half-life of 15 years. What is the fraction that will decay in 30 years?
(A) 0.25 **(B)** 0.5 **(C)** 0.75 **(D)** 0.85

51. A 220 V A.C. supply is connected between points A and B as shown in figure. What will be the potential difference V across the capacitor?



(A) 220 V **(B)** 110 V
(C) 0 **(D)** $220\sqrt{2}$ V

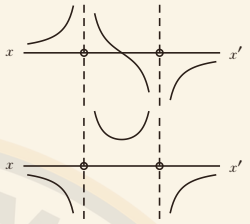
52. In the following circuit what are P and Q:

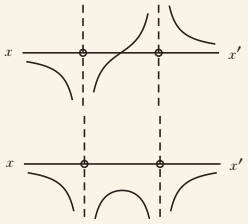


(A) $P = 1, Q = 0$ **(B)** $P = 0, Q = 1$
(C) $P = 0, Q = 0$ **(D)** $P = 1, Q = 1$

53. A positive hole in a semiconductor is
(A) an anti-particle of electron.
(B) a vacancy created when an electron leaves a co-valent bond.
(C) absence of free electrons.
(D) an artificially created particle.

54. Two long straight parallel wires are a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper. The variation of magnetic field B along the line xx' is given by

(A) 

(B) 

(C)

(D)

55. A cylindrical wire has a mass (0.3 ± 0.003) g, radius (0.5 ± 0.005) mm and length (6 ± 0.06) cm. The maximum percentage error in the measurement of its density is
(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

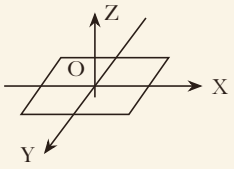
56. At a metro station, a girl walks up a stationary escalator in 20 sec. If she remains stationary on the escalator, then the escalator take her up in 30 sec. The time taken by her to walk up on the moving escalator will be
(A) 25 sec **(B)** 60 sec **(C)** 12 sec **(D)** 10 sec

57. Rain is falling vertically with a speed of $12\ \text{m s}^{-1}$. A woman rides a bicycles with a speed of $12\ \text{m s}^{-1}$ in east to west direction. What is the direction in which she should hold her umbrella?
(A) 30° towards East **(B)** 45° towards East
(C) 30° towards West **(D)** 45° towards West

58. One end of a string of length 'l' is connected to a particle of mass 'm' and the other to a small peg on a smooth horizontal table. If the particle moves in a circle with speed 'v', the net force on the particle (directed towards the centre) is: (T is the tension in the string)
(A) T **(B)** $T - \frac{mv^2}{l}$
(C) $T + \frac{mv^2}{l}$ **(D)** 0

59. A body is initially at rest. It undergoes one-dimensional motion with constant acceleration. The power delivered to it at time 't' is proportional to
(A) $t^{1/2}$ **(B)** t **(C)** $t^{3/2}$ **(D)** t^2

60. A thin uniform rectangular plate of mass 2 kg is placed in X-Y plane as shown in figure. The moment of inertia about x-axis is $I_x = 0.2\,\text{kg m}^2$ and the moment of inertia about y-axis is $I_y = 0.3\,\text{kg m}^2$. The radius of gyration of the plate about the axis passing through O and perpendicular to the plane of the plate is



(A) 50 cm **(B)** 5 cm
(C) 38.7 cm **(D)** 31.6 cm

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KCET 2019
Physics Question Paper

Version A-1 · 60 Questions · 60 Marks · 80 Minutes

1. In a cyclotron a charged particle
(A) undergoes acceleration all the time
(B) speeds up between the dees because of the magnetic field.
(C) speeds up in dee
(D) slows down within a dee and speeds up between dees

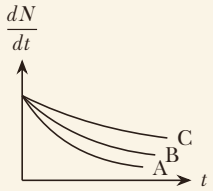
2. The number of turns in a coil of Galvanometer is tripled, then
(A) Voltage sensitivity increases 3 times and current sensitivity remains constant
(B) Voltage sensitivity remains constant and current sensitivity increases 3 times
(C) Both voltage and current sensitivity remains constant
(D) Both voltage and current sensitivity decreases by 33%

3. A circular current loop of magnetic moment M is in an arbitrary orientation in an external uniform magnetic field $\vec{B}$. The work done to rotate the loop by 30° about an axis perpendicular to its plane is
(A) MB **(B)** $\sqrt{3}\frac{MB}{2}$
(C) $\frac{MB}{2}$ **(D)** Zero

4. In a permanent magnet at room temperature
(A) magnetic moment of each molecule is zero.
(B) the individual molecules have non zero magnetic moment which are all perfectly aligned.
(C) domains are partially aligned.
(D) domains are all perfectly aligned.

5. Coersivity of a magnet where the ferromagnet gets completely demagnetized is $3 \times 10^3\ \text{A m}^{-1}$. The minimum current required to be passed in a solenoid having 1000 turns per metre, so that the magnet gets completely demagnetized when placed inside the solenoid is
(A) 30 mA **(B)** 60 mA **(C)** 3 A **(D)** 6 A

6. Which one of the following nuclei has shorter mean life?

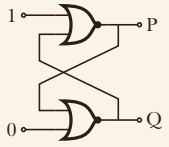


(A) A **(B)** B
(C) C **(D)** Same for all

7. The conductivity of semiconductor increases with increase in temperature because
(A) number density of charge carriers increases
(B) relaxation time increases
(C) both number density of charge carriers and relaxation time increase
(D) number density of current carriers increases, relaxation time decreases but effect of decrease in relaxation time is much less than increase in number density

8. For a transistor amplifier, the voltage gain
(A) remains constant for all frequencies
(B) is high at high and low frequencies and constant in the middle frequency range
(C) is low at high and low frequencies and constant at mid frequencies
(D) constant at high frequencies and low at low frequencies

9. In the following circuit, what are P and Q?



(A) $P = 0, Q = 0$ **(B)** $P = 1, Q = 0$
(C) $P = 0, Q = 1$ **(D)** $P = 1, Q = 1$

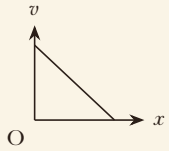
10. An antenna uses electromagnetic waves of frequency 5 MHz. For proper working, the size of the antenna should be
(A) 15 m **(B)** 300 m **(C)** 15 km **(D)** 3 km

11. If P, Q and R are physical quantities having different dimensions, which of the following combinations can never be a meaningful quantity?

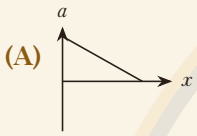
(A) $\frac{P-Q}{R}$ **(B)** $PQ-R$

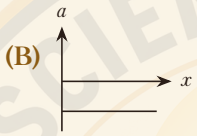
(C) $\frac{PQ}{R}$ **(D)** $\frac{PR-Q^2}{R}$

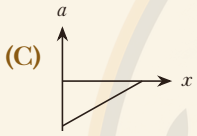
12. The given graph shows the variation of velocity (v) with position (x) for a particle moving along a straight line

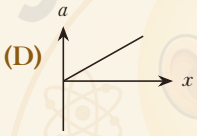


Which of the following graph shows the variation of acceleration (a) with position (x)?

(A) 

(B) 

(C) 

(D) 

13. The trajectory of a projectile projected from origin is given by the equation $y = x - \frac{2x^2}{5}$. The initial velocity of the projectile is

(A) $\frac{2}{5}\,\text{m s}^{-1}$ **(B)** $5\,\text{m s}^{-1}$

(C) $25\,\text{m s}^{-1}$ **(D)** $\frac{5}{2}\,\text{m s}^{-1}$

14. An object with mass 5 kg is acted upon by a force, $\vec{F} = (-3\hat{i} + 4\hat{j})$ N. If its initial velocity at $t = 0$ is $\vec{v} = (6\hat{i} - 12\hat{j})\,\text{m s}^{-1}$, the time at which it will just have a velocity along y-axis is

(A) 5 s **(B)** 10 s **(C)** 2 s **(D)** 15 s

15. During inelastic collision between two objects, which of the following quantity always remains conserved?

(A) Total kinetic energy
(B) Total mechanical energy
(C) Total linear momentum
(D) Speed of each body

16. In Rutherford experiment, for head-on collision of α-particles with a gold nucleus, the impact parameter is

(A) zero
(B) of the order of 10^{-14} m
(C) of the order of 10^{-10} m
(D) of the order of 10^{-6} m

17. Frequency of revolution of an electron revolving in n^{th} orbit of H-atom is proportional to

(A) $\frac{1}{n^2}$ **(B)** n

(C) n independent of n **(D)** $\frac{1}{n^3}$

18. A hydrogen atom in ground state absorbs 10.2 eV of energy. The orbital angular momentum of the electron is increased by

(A) 1.05×10^{-34} J s **(B)** 2.11×10^{-34} J s

(C) 3.16×10^{-34} J s **(D)** 4.22×10^{-34} J s

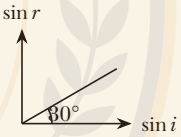
19. The end product of decay of ${}_{90}\text{Th}^{232}$ is ${}_{82}\text{Pb}^{208}$. The number of α and β particles emitted are respectively

(A) 3, 3 **(B)** 6, 4 **(C)** 6, 0 **(D)** 4, 6

20. Two protons are kept at a separation of 10 nm. Let F_n and F_e be the nuclear force and the electromagnetic force between them

(A) $F_e = F_n$
(B) $F_e \gg F_n$
(C) $F_e \ll F_n$
(D) F_e and F_n differ only slightly

21. A transparent medium shows relation between i and r as shown. If the speed of light in vacuum is c the Brewster angle for the medium is



(A) 30° **(B)** 45° **(C)** 60° **(D)** 90°

22. In Young's double slit experiment, using monochromatic light of wavelength λ, the intensity of light at a point on the screen where path difference is λ is K units. The intensity of light at a point where path difference is $\frac{\lambda}{3}$ is

(A) K **(B)** $\frac{K}{4}$ **(C)** 4K **(D)** 2K

23. Due to Doppler's effect the shift in wavelength observed is 0.1 Å for a star producing wavelength 6000 Å. Velocity of recession of the star will be

(A) 25 km/s **(B)** 10 km/s
(C) 5 km/s **(D)** 20 km/s

24. An electron is moving with an initial velocity $\vec{V} = V_0\hat{i}$ and is in a uniform magnetic field $\vec{B} = B_0\hat{j}$. Then its de Broglie wavelength

(A) remains constant
(B) increases with time
(C) decreases with time
(D) increase and decreases periodically

25. Light of certain frequency and intensity incident on a photosensitive material causes photoelectric effect. If both the frequency and intensity are doubled, the photoelectric saturation current becomes

(A) quadrupled **(B)** doubled
(C) halved **(D)** unchanged

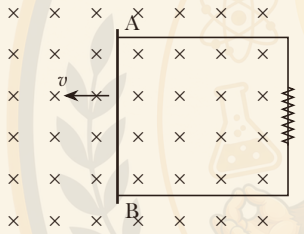
26. A magnetic needle has a magnetic moment of 5×10^{-2} A m^2 and moment of inertia 8×10^{-6} kg m^2. It has a period of oscillation of 2 s in a magnetic field $\vec{B}$. The magnitude of magnetic field is approximately

(A) 1.6×10^{-4} T **(B)** 0.4×10^{-4} T
(C) 3.2×10^{-4} T **(D)** 0.8×10^{-4} T

27. A toroid has 500 turns per metre length. If it carries a current of 2 A, the magnetic energy density inside the toroid is

(A) 0.628 J/m^3 **(B)** 0.314 J/m^3
(C) 6.28 J/m^3 **(D)** 3.14 J/m^3

28. Consider the situation given in figure. The wire AB is slid on the fixed rails with a constant velocity. If the wire AB is replaced by a semicircular wire, the magnitude of the induced current will

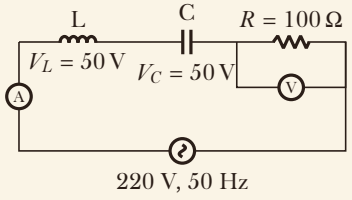


(A) increase
(B) remain same
(C) decrease
(D) increase or decrease depending on whether the semicircle bulges towards the resistance or away from it

29. The frequency of an alternating current is 50 Hz. What is the minimum time taken by current to reach its peak value from rms value?

(A) 5×10^{-3} s **(B)** 2.5×10^{-3} s
(C) 0.02 s **(D)** 10×10^{-3} s

30. The readings of ammeter and voltmeter in the following circuit are respectively

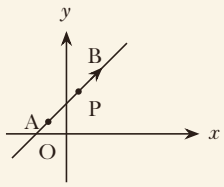


(A) 1.2 A, 120 V **(B)** 1.5 A, 100 V
(C) 2.7 A, 220 V **(D)** 2.2 A, 220 V

31. Two particles which are initially at rest move towards each other under the action of their mutual attraction. If their speeds are v and $2v$ at any instant, then the speed of center of mass of the system is,

(A) $2v$ **(B)** Zero **(C)** $1.5v$ **(D)** v

32. A particle is moving uniformly along a straight line as shown in the figure. During the motion of the particle from A to B, the angular momentum of the particle about 'O'



(A) increases
(B) decreases
(C) remains constant
(D) first increases then decreases

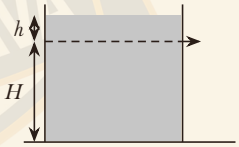
33. A satellite is orbiting close to the earth and has a kinetic energy K. The minimum extra kinetic energy required by it to just overcome the gravitation pull of the earth is

(A) K **(B)** 2K **(C)** $\sqrt{3}$K **(D)** $2\sqrt{2}$K

34. A wire is stretched such that its volume remains constant. The Poission's ratio of the material of the wire is

(A) 0.50 **(B)** −0.50 **(C)** 0.25 **(D)** −0.25

35. A cylindrical container containing water has a small hole at height of H = 8 cm from the bottom and at a depth of 2 cm from the top surface of the liquid. The maximum horizontal distance travelled by the water before it hits the ground (x) is



(A) 8 cm **(B)** $4\sqrt{2}$ cm
(C) 4 cm **(D)** 6 cm

36. An inductor of inductance L and resistor R are joined together in series and connected by a source of frequency ω. The power dissipated in the circuit is

(A) $\dfrac{R^2+\omega^2L^2}{V}$ **(B)** $\dfrac{V^2R}{R^2+\omega^2L^2}$

(C) $\dfrac{V}{R^2+\omega^2L^2}$ **(D)** $\dfrac{V^2R}{\sqrt{R^2+\omega^2L^2}}$

37. An electromagnetic wave is travelling in x-direction with electric field vector given by, $\vec{E}_y = E_0 \sin(kx - \omega t)\,\hat{j}$. The correct expression for magnetic field vector is

(A) $\vec{B}_y = E_0 C \sin(kx - \omega t)\,\hat{j}$

(B) $\vec{B}_z = E_0 C \sin(kx - \omega t)\,\hat{k}$

(C) $\vec{B}_y = \dfrac{E_0}{C} \sin(kx - \omega t)\,\hat{j}$

(D) $\vec{B}_z = \dfrac{E_0}{C} \sin(kx - \omega t)\,\hat{k}$

38. The phenomenon involved in the reflection of radiowaves by ionosphere is similar to

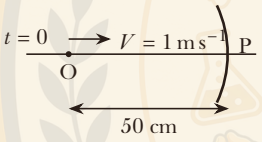
(A) reflection of light by plane mirror

(B) total internal reflection of light in air during a mirage

(C) dispersion of light by water molecules during the formation of a rainbow

(D) scattering of light by air particles

39. A point object is moving uniformly towards the pole of a concave mirror of focal length 25 cm along its axis as shown below. The speed of the object is $1\ \text{m s}^{-1}$. At $t = 0$, the distance of the object from the mirror is 50 cm. The average velocity of the image formed by the mirror between time $t = 0$ and $t = 0.25$ s is:



(A) $40\ \text{cm s}^{-1}$ **(B)** $20\ \text{cm s}^{-1}$

(C) Zero **(D)** Infinity

40. A certain prism is found to produce a minimum deviation of 38°. It produces a deviation of 44° when the angle of incidence is either 42° or 62°. What is the angle of incidence when it is undergoing minimum deviation?

(A) 30° **(B)** 40° **(C)** 49° **(D)** 60°

41. An aluminium sphere is dipped into water. Which of the following is true?

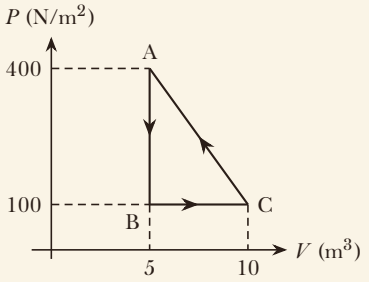
(A) Buoyancy will be less in water at 0°C than that in water at 4°C

(B) Buoyancy will be more in water at 0°C than that in water at 4°C

(C) Buoyancy in water at 0°C will be same as that in water at 4°C

(D) Buoyancy may be more or less in water at 4°C depending on the radius of the sphere

42. A thermodynamic system undergoes a cyclic process ABC as shown in the diagram. The work done by the system per cycle is



(A) 750 J **(B)** −1250 J

(C) −750 J **(D)** 1250 J

43. One mole of O_2 gas is heated at constant pressure starting at 27°C. How much energy must be added to the gas as heat to double its volume?

(A) Zero **(B)** 450 R

(C) 750 R **(D)** 1050 R

44. A piston is performing S.H.M. in the vertical direction with a frequency of 0.5 Hz. A block of 10 kg is placed on the piston. The maximum amplitude of the system such that the block remains in contact with the piston is

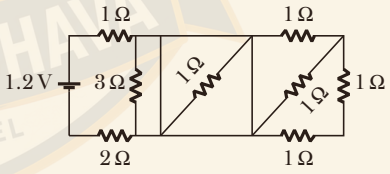
(A) 1 m **(B)** 0.5 m **(C)** 1.5 m **(D)** 0.1 m

45. The equation of a stationary wave is $y = 2 \sin\left(\dfrac{\pi x}{15}\right) \cos(48\pi t)$. The distance between a node and its next antinode is

(A) 7.5 units **(B)** 1.5 units

(C) 22.5 units **(D)** 30 units

46. In the given circuit, the current through 2 Ω resistor is



(A) 0.2 A **(B)** 0.3 A **(C)** 0.4 A **(D)** 0.1 A

47. Kirchhoff's junction rule is a reflection of

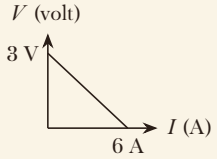
(A) Conservation of current density vector

(B) Conservation of energy

(C) Conservation of momentum

(D) Conservation of charges

48. The variation of terminal potential difference (V) with current flowing through a cell is as shown



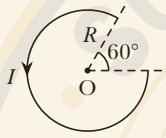
The emf and internal resistance of the cell are

(A) 3 V, 2 Ω **(B)** 3 V, 0.5 Ω
(C) 6 V, 2 Ω **(D)** 6 V, 0.5 Ω

49. In a potentiometer experiment, the balancing point with a cell is at a length 240 cm. On shunting the cell with a resistance of 2 Ω, the balancing length becomes 120 cm. The internal resistance of the cell is

(A) 4 Ω **(B)** 2 Ω **(C)** 1 Ω **(D)** 0.5 Ω

50. The magnetic field at the centre 'O' in the given figure is

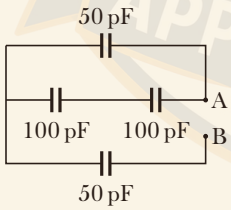


(A) $\frac{7}{14}\frac{\mu_0 I}{R}$ **(B)** $\frac{5}{12}\frac{\mu_0 I}{R}$
(C) $\frac{3}{10}\frac{\mu_0 I}{R}$ **(D)** $\frac{\mu_0 I}{12R}$

51. Two metal plates are separated by 2 cm. The potentials of the plates are –10 V and +30 V. The electric field between the two plates is

(A) 500 V/m **(B)** 1000 V/m
(C) 2000 V/m **(D)** 3000 V/m

52. The equivalent capacitance between A and B is,



(A) 50 pF **(B)** $\frac{100}{3}$ pF
(C) 150 pF **(D)** 300 pF

53. A capacitor of capacitance C charged by an amount Q is connected in parallel with an uncharged capacitor of capacitance 2C. The final charges on the capacitors are

(A) $\frac{Q}{2}, \frac{Q}{2}$ **(B)** $\frac{Q}{4}, \frac{3Q}{4}$
(C) $\frac{Q}{3}, \frac{2Q}{3}$ **(D)** $\frac{Q}{5}, \frac{4Q}{5}$

54. Though the electron drift velocity is small and electron charge is very small, a conductor can carry an appreciably large current because

(A) electron number density is very large
(B) drift velocity of electron is very large
(C) electron number density depends on temperature
(D) relaxation time is small

55. Masses of three wires of copper are in the ratio 1:3:5 and their lengths are in the ratio 5:3:1. The ratio of their electrical resistance are

(A) 1:3:5 **(B)** 5:3:1
(C) 1:15:125 **(D)** 125:15:1

56. A certain charge 2Q is divided at first into two parts q_1 and q_2. Later the charges are placed at a certain distance. If the force of interaction between two charges is maximum then $\frac{Q}{q_1}$ = _____.

(A) 4 **(B)** 2 **(C)** 1 **(D)** 0.5

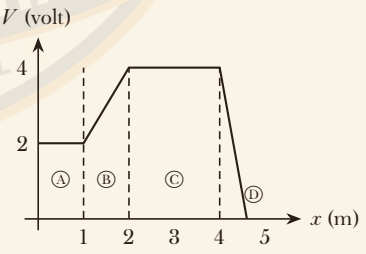
57. A particle of mass m and charge q is placed at rest in uniform electric field E and then released. The kinetic energy attained by the particle after moving a distance y is

(A) qEy^2 **(B)** qE^2y **(C)** qEy **(D)** q^2Ey

58. An electric dipole is kept in non-uniform electric field. It generally experiences

(A) A force and torque
(B) A force but not a torque
(C) A torque but not a force
(D) Neither a force nor a torque

59. The figure gives the electric potential V as a function of distance through four regions on x-axis. Which of the following is true for the magnitude of the electric field E in these regions?



(A) $E_A > E_B > E_C > E_D$
(B) $E_A = E_C$ and $E_B < E_D$
(C) $E_B = E_D$ and $E_A < E_C$
(D) $E_A < E_B < E_C < E_D$

60. A system of two charges separated by a certain distance apart stores electrical potential energy. If the distance between them is increased, the potential energy of the system,

(A) increases in any case

(B) decreases in any case

(C) may increase or decrease

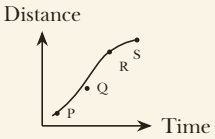
(D) remains the same

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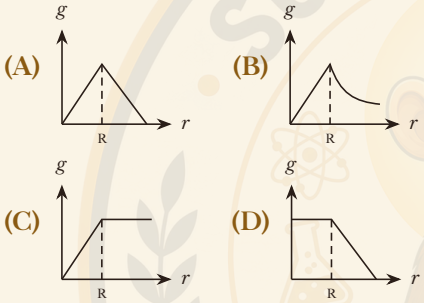
Version A · 60 Questions · 60 Marks · 80 Minutes

1. A particle shows distance-time curve as shown in the figure. The maximum instantaneous velocity of the particle is around the point



(A) P **(B)** S **(C)** R **(D)** Q

2. Which of the following graphs correctly represents the variation of 'g' on the Earth?



3. A cup of tea cools from 65.5°C to 62.5°C in 1 minute in a room at 22.5°C. How long will it take to cool from 46.5°C to 40.5°C in the same room?

(A) 4 minutes **(B)** 2 minutes
(C) 1 minute **(D)** 3 minutes

4. The dimensions of the ratio of magnetic flux (ϕ) and permeability (μ) are

(A) $[M^0L^1T^0A^1]$ **(B)** $[M^0L^{-3}T^0A^1]$
(C) $[M^0L^1T^1A^{-1}]$ **(D)** $[M^0L^2T^0A^1]$

5. A mass 'm' on the surface of the Earth is shifted to a target equal to the radius of the Earth. If 'R' is the radius and 'M' is the mass of the Earth, then work done in this process is

(A) $\frac{mgR}{2}$ **(B)** mgR **(C)** $2mgR$ **(D)** $\frac{mgR}{4}$

6. First overtone frequency of a closed pipe of length 'l_1' is equal to the 2nd harmonic frequency of an open pipe of length 'l_2'. The ratio $\frac{l_1}{l_2}$ =

(A) $\frac{3}{4}$ **(B)** $\frac{4}{3}$ **(C)** $\frac{3}{2}$ **(D)** $\frac{2}{3}$

7. The resistance $R = \frac{V}{I}$ where $V = (100 \pm 5)$ V and $I = (10 \pm 0.2)$ A. The percentage error in R is

(A) 5.2% **(B)** 4.8% **(C)** 7% **(D)** 3%

8. A block rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.8. If the frictional force on the block is 10 N, the mass of the block is ($g = 10\,m\,s^{-2}$)

(A) 1 kg **(B)** 2 kg **(C)** 3 kg **(D)** 4 kg

9. Two particles of masses m_1 and m_2 have equal kinetic energies. The ratio of their momenta is

(A) $m_1 : m_2$ **(B)** $m_2 : m_1$
(C) $\sqrt{m_1} : \sqrt{m_2}$ **(D)** $m_1^2 : m_2^2$

10. The pressure at the bottom of a liquid tank is ***not*** proportional to the

(A) Acceleration due to gravity
(B) Density of the liquid
(C) Height of the liquid
(D) Area of the liquid surface

11. A Carnot engine takes 300 calories of heat from a source at 500 K and rejects 150 calories of heat to the sink. The temperature of the sink is

(A) 125 K **(B)** 250 K
(C) 750 K **(D)** 1000 K

12. Pressure of an ideal gas is increased by keeping temperature constant. The kinetic energy of molecules

(A) Decreases
(B) Increases
(C) Remains same
(D) Increases or decreases depending on the nature of gas

13. A man weighing 60 kg is in a lift moving down with an acceleration of $1.8\,m\,s^{-2}$. The force exerted by the floor on him is

(A) 588 N **(B)** 480 N **(C)** Zero **(D)** 696 N

14. Moment of inertia of a body about two perpendicular axes X and Y in the plane of lamina are 20 kg m^2 and 25 kg m^2 respectively. Its moment of inertia about an axis perpendicular to the plane of the lamina and passing through the point of intersection of X and Y axes is

(A) 5 kg m^2 **(B)** 45 kg m^2
(C) 12.5 kg m^2 **(D)** 500 kg m^2

15. Two wires A and B are stretched by the same load. If the area of cross-section of wire 'A' is double that of 'B', then the stress on 'B' is

(A) Equal to that on A
(B) Twice that on A
(C) Half that on A
(D) Four times that on A

16. The magnitude of point charge due to which the electric field 30 cm away has the magnitude 2 N C^{-1} will be

(A) 2×10^{-11} C **(B)** 3×10^{-11} C
(C) 5×10^{-11} C **(D)** 9×10^{-11} C

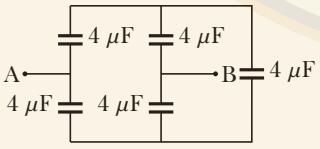
17. A mass of 1 kg carrying a charge of 2 C is accelerated through a potential of 1 V. The velocity acquired by it is

(A) $\sqrt{2}$ m s^{-1} **(B)** 2 m s^{-1}
(C) $\frac{1}{\sqrt{2}}$ m s^{-1} **(D)** $\frac{1}{2}$ m s^{-1}

18. The force of repulsion between two identical positive charges when kept with a separation 'r' in air is 'F'. Half the gap between the two charges is filled by a dielectric slab of dielectric constant = 4. Then the new force of repulsion between those two charges becomes

(A) $\frac{F}{3}$ **(B)** $\frac{F}{2}$ **(C)** $\frac{F}{4}$ **(D)** $\frac{4F}{9}$

19. For the arrangement of capacitors as shown in the circuit, the effective capacitance between the points A and B is (capacitance of each capacitor is 4 μF)



(A) 4 μF **(B)** 2 μF **(C)** 1 μF **(D)** 8 μF

20. The work done to move a charge on an equipotential surface is

(A) Infinity **(B)** Less than 1
(C) Greater than 1 **(D)** Zero

21. Two capacitors of 3 μF and 6 μF are connected in series and a potential difference of 900 V is applied across the combination. They are then disconnected and reconnected in parallel. The potential difference across the combination is

(A) Zero **(B)** 100 V **(C)** 200 V **(D)** 400 V

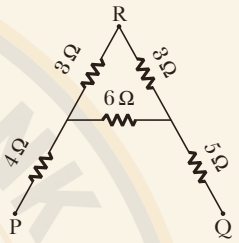
22. Ohm's Law is applicable to

(A) Diode **(B)** Transistor
(C) Electrolyte **(D)** Conductor

23. If the last band on the carbon resistor is absent, then the tolerance is

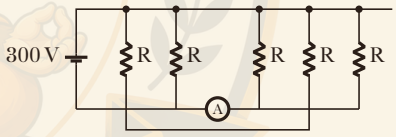
(A) 5% **(B)** 20% **(C)** 10% **(D)** 15%

24. The effective resistance between P and Q for the following network is



(A) $\frac{1}{12}\ \Omega$ **(B)** 21 Ω **(C)** 12 Ω **(D)** $\frac{1}{21}\ \Omega$

25. Five identical resistors each of resistance R = 1500 Ω are connected to a 300 V battery as shown in the circuit. The reading of the ideal ammeter A is

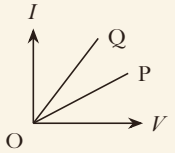


(A) $\frac{1}{5}$ A **(B)** $\frac{3}{5}$ A **(C)** $\frac{2}{5}$ A **(D)** $\frac{4}{5}$ A

26. Two cells of internal resistances r_1 and r_2 and of same emf are connected in series, across a resistor of resistance R. If the terminal potential difference across the cells of internal resistance r_1 is zero, then the value of R is

(A) $R = 2(r_1 + r_2)$ **(B)** $R = r_2 - r_1$
(C) $R = r_1 - r_2$ **(D)** $R = 2(r_1 - r_2)$

27. The I-V graphs for two different electrical appliances P and Q are shown in the diagram. If R_P and R_Q be the resistances of the devices, then



(A) $R_P = R_Q$ **(B)** $R_P > R_Q$
(C) $R_P < R_Q$ **(D)** $R_P = \frac{R_Q}{2}$

28. The correct Biot-Savart law in vector form is

(A) $d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(\vec{dl}\times\vec{r})}{r^2}$ **(B)** $d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(\vec{dl}\times\vec{r})}{r^3}$

(C) $d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,\vec{dl}}{r^2}$ **(D)** $d\vec{B} = \frac{\mu_0}{4\pi}\cdot\frac{I\,\vec{dl}}{r^3}$

29. An electron is moving in a circle of radius r in a uniform magnetic field B. Suddenly the field is reduced to $\frac{B}{2}$. The radius of the circular path now becomes

(A) $\frac{r}{2}$ **(B)** $2r$ **(C)** $\frac{r}{4}$ **(D)** $4r$

30. A charge q is accelerated through a potential difference V. It is then passed normally through a uniform magnetic field, where it moves in a circle of radius r. The potential difference required to move it in a circle of radius $2r$ is

(A) 2 V **(B)** 4 V **(C)** 1 V **(D)** 3 V

31. A cyclotron's oscillator frequency is 10 MHz and the operating magnetic field is 0.66 T. If the radius of its dees is 60 cm, then the kinetic energy of the proton beam produced by the accelerator is

(A) 9 MeV **(B)** 10 MeV
(C) 7 MeV **(D)** 11 MeV

32. Needles N_1, N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will

(A) Attract all three of them
(B) Attract N_1 strongly, N_2 weakly and repel N_3 weakly
(C) Attract N_1 strongly but repel N_2 and N_3 weakly
(D) Attract N_1 and N_2 strongly but repel N_3

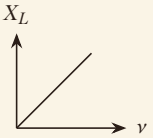
33. The strength of the Earth's magnetic field is

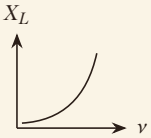
(A) Constant everywhere
(B) Zero everywhere
(C) Having very high value
(D) Varying from place to place on the Earth's surface

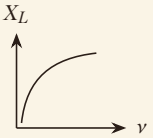
34. A jet plane having a wing-span of 25 m is travelling horizontally towards east with a speed of 3600 km/hour. If the Earth's magnetic field at the location is 4×10^{-4} T and the angle of dip is 30°, then, the potential difference between the ends of the wing is

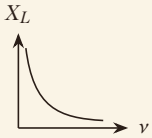
(A) 4 V **(B)** 5 V **(C)** 2 V **(D)** 2.5 V

35. Which of the following, represents the variation of inductive reactance (X_L) with the frequency of voltage source (ν)?

(A) 

(B) 

(C) 

(D) 

36. The magnetic flux linked with a coil varies as $\phi = 3t^2 + 4t + 9$. The magnitude of the emf induced at $t = 2$ seconds is

(A) 8 V **(B)** 16 V **(C)** 32 V **(D)** 64 V

37. A 100 W bulb is connected to an AC source of 220 V, 50 Hz. Then the current flowing through the bulb is

(A) $\frac{5}{11}$ A **(B)** $\frac{1}{2}$ A **(C)** 2 A **(D)** $\frac{3}{4}$ A

38. In the series LCR circuit, the power dissipation is through

(A) R **(B)** L
(C) C **(D)** Both L and C

39. In Karnataka, the normal domestic power supply AC is 220 V, 50 Hz. Here 220 V and 50 Hz refer to

(A) Peak value of voltage and frequency
(B) rms value of voltage and frequency
(C) Mean value of voltage and frequency
(D) Peak value of voltage and angular frequency

40. A step-up transformer operates on a 230 V line and a load current of 2 A. The ratio of primary and secondary windings is 1:25. Then the current in the primary is

(A) 25 A **(B)** 50 A **(C)** 15 A **(D)** 12.5 A

41. The number of photons falling per second on a completely darkened plate to produce a force of 6.62×10^{-5} N is 'n'. If the wavelength of the light falling is 5×10^{-7} m, then $n =$ _____ $\times 10^{22}$. ($h = 6.62 \times 10^{-34}$ J s)

(A) 1 **(B)** 5 **(C)** 0.2 **(D)** 3.3

42. An object is placed at the principal focus of a convex mirror. The image will be at

(A) Centre of curvature
(B) Principal focus
(C) Infinity
(D) No image will be formed

43. An object is placed at a distance of 20 cm from the pole of a concave mirror of focal length 10 cm. The distance of the image formed is

(A) +20 cm **(B)** +10 cm
(C) −20 cm **(D)** −10 cm

44. A candle placed 25 cm from a lens forms an image on a screen placed 75 cm on the other side of the lens. The focal length and type of the lens should be

(A) +18.75 cm and convex lens
(B) −18.75 cm and concave lens
(C) +20.25 cm and convex lens
(D) −20.25 cm and concave lens

45. A plane wavefront of wavelength λ is incident on a single slit of width a. The angular width of principal maximum is

(A) $\frac{\lambda}{a}$ **(B)** $\frac{2\lambda}{a}$ **(C)** $\frac{a}{\lambda}$ **(D)** $\frac{a}{2\lambda}$

46. In a Fraunhofer diffraction at a single slit, if yellow light illuminating the slit is replaced by blue light, then diffraction bands

(A) Remain unchanged **(B)** Become wider
(C) Disappear **(D)** Become narrower

47. In Young's double slit experiment, two wavelengths $\lambda_1 = 780$ nm and $\lambda_2 = 520$ nm are used to obtain interference fringes. If the nth bright band due to λ_1 coincides with $(n+1)^{th}$ bright band due to λ_2, then the value of n is

(A) 4 **(B)** 3 **(C)** 2 **(D)** 6

48. In Young's double slit experiment, slits are separated by 2 mm and the screen is placed at a distance of 1.2 m from the slits. Light consisting of two wavelengths 6500 Å and 5200 Å are used to obtain interference fringes. Then the separation between the fourth bright fringes of two different patterns produced by the two wavelengths is

(A) 0.312 mm **(B)** 0.123 mm
(C) 0.213 mm **(D)** 0.412 mm

49. The maximum kinetic energy of emitted photoelectrons depends on

(A) Intensity of incident radiation
(B) Frequency of incident radiation
(C) Speed of incident radiation
(D) Number of photons in the incident radiation

50. A proton and an α particle are accelerated through the same potential difference V. The ratio of their de Broglie wavelengths is

(A) $\sqrt{2}$ **(B)** $2\sqrt{2}$ **(C)** $\sqrt{3}$ **(D)** $2\sqrt{3}$

51. The total energy of an electron revolving in the second orbit of hydrogen atom is

(A) −13.6 eV **(B)** −1.51 eV
(C) −3.4 eV **(D)** Zero

52. The period of revolution of an electron in the ground state of hydrogen atom is T. The period of revolution of the electron in the first excited state is

(A) 2T **(B)** 4T **(C)** 6T **(D)** 8T

53. The energy equivalent to a substance of mass 1 g is

(A) 18×10^{13} J **(B)** 9×10^{13} J
(C) 18×10^{6} J **(D)** 9×10^{6} J

54. The half-life of tritium is 12.5 years. What mass of tritium of initial mass 64 mg will remain undecayed after 50 years?

(A) 32 mg **(B)** 8 mg **(C)** 16 mg **(D)** 4 mg

55. In a CE amplifier, the input ac signal to be amplified is applied across

(A) Forward biased emitter-base junction
(B) Reverse biased collector-base junction
(C) Reverse biased emitter-base junction
(D) Forward biased collector-base junction

56. If $A = 1$ and $B = 0$, then in terms of Boolean algebra, $A + \bar{B} =$

(A) B **(B)** $\bar{B}$ **(C)** A **(D)** $\bar{A}$

57. The density of an electron-hole pair in a pure germanium is 3×10^{16} m^{-3} at room temperature. On doping with aluminium, the hole density increases to 4.5×10^{22} m^{-3}. Now the electron density (in m^{-3}) in doped germanium will be

(A) 1×10^{10} **(B)** 2×10^{10}
(C) 0.5×10^{10} **(D)** 4×10^{10}

58. The dc common emitter current gain of a n-p-n transistor is 50. The potential difference applied across the collector and emitter of a transistor used in CE configuration is, $V_{CE} = 2$ V. If the collector resistance, $R_C = 4$ kΩ, the base current (I_B) and the collector current (I_C) are

(A) $I_B = 10\ \mu$A, $I_C = 0.5$ mA
(B) $I_B = 0.5\ \mu$A, $I_C = 10$ mA
(C) $I_B = 5\ \mu$A, $I_C = 1$ mA
(D) $I_B = 1\ \mu$A, $I_C = 0.5$ mA

59. The radius of the Earth is 6400 km. If the height of an antenna is 500 m, then its range is

(A) 800 km **(B)** 100 km
(C) 80 km **(D)** 10 km

60. A space station is at a height equal to the radius of the Earth. If 'v_E' is the escape velocity on the surface of the Earth, the same on the space station is _____ times v_E.

(A) $\frac{1}{2}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{\sqrt{2}}$ **(D)** $\frac{1}{\sqrt{3}}$

KCET 2017
Physics Question Paper

60 Questions · 60 Marks · 80 Minutes

1. A substance of mass 49.53 g occupies 1.5 cm^3 of volume. The density of the substance (in g cm^{-3}) with correct number of significant figures is
(A) 3.302 **(B)** 3.300 **(C)** 3.3 **(D)** 3.30

2. A car moving with a velocity of 20 m s^{-1} is stopped in a distance of 40 m. If the same car is travelling at double the velocity, the distance travelled by it for same retardation is
(A) 640 m **(B)** 320 m
(C) 1280 m **(D)** 160 m

3. The angle between velocity and acceleration of a particle describing uniform circular motion is
(A) 45° **(B)** 60° **(C)** 90° **(D)** 180°

4. If $\vec{A} = 2\hat{i}+3\hat{j}+8\hat{k}$ is perpendicular to $\vec{B} = 4\hat{j}-4\hat{i}+\alpha\hat{k}$, then the value of '$\alpha$' is
(A) $\frac{1}{2}$ **(B)** $-\frac{1}{2}$ **(C)** 1 **(D)** −1

5. A body of mass 50 kg is suspended using a spring balance inside a lift at rest. If the lift starts falling freely, the reading of the spring balance is
(A) = 50 kg **(B)** > 50 kg
(C) < 50 kg **(D)** = 0

6. A motor pump lifts 6 tonnes of water from a well of depth 25 m to the first floor of height 35 m from the ground floor in 20 minutes. The power of the pump (in kW) is [g = 10 m s^{-2}]
(A) 3 **(B)** 6 **(C)** 1.5 **(D)** 12

7. Two balls are thrown simultaneously in air. The acceleration of the centre of mass of the two balls when in air,
(A) depends on the masses of the two balls
(B) depends on the speeds of the two balls
(C) is equal to g (Acceleration due to gravity)
(D) depends on the direction of motion of the two balls.

8. The value of acceleration due to gravity at a depth of 1600 km is equal to [Radius of earth = 6400 km]
(A) 9.8 m s^{-2} **(B)** 4.9 m s^{-2}
(C) 19.6 m s^{-2} **(D)** 7.35 m s^{-2}

9. 'Young's modulus' is defined as the ratio of
(A) tensile stress and longitudinal strain
(B) hydraulic stress and hydraulic strain
(C) shearing stress and shearing strain
(D) bulk stress and longitudinal strain

10. 'Hydraulic lift' works on the basis of
(A) Stoke's law **(B)** Toricelli's law
(C) Pascal's Law **(D)** Bernoulli's Law

11. The S.I. unit of specific heat capacity is
(A) J $mol^{-1}K^{-1}$ **(B)** J $kg^{-1}K^{-1}$
(C) J K^{-1} **(D)** J kg^{-1}

12. For which combination of working temperatures, the efficiency of 'Carnot's engine' is the least?
(A) 60 K, 40 K **(B)** 40 K, 20 K
(C) 80 K, 60 K **(D)** 100 K, 80 K

13. The mean energy of a molecule of an ideal gas is
(A) $2KT$ **(B)** $\frac{3}{2}KT$ **(C)** KT **(D)** $\frac{1}{2}KT$

14. Two simple pendulums A and B are made to oscillate simultaneously and it is found that A completes 10 oscillations in 20 sec and B completes 8 oscillations in 10 sec. The ratio of the lengths of A and B is
(A) $\frac{8}{5}$ **(B)** $\frac{64}{25}$ **(C)** $\frac{5}{4}$ **(D)** $\frac{25}{64}$

15. The waves set up in a closed pipe are
(A) Transverse and Progressive
(B) Longitudinal and Stationary
(C) Transverse and Stationary
(D) Longitudinal and Progressive

16. Two spheres of electric charges +2 nC and −8 nC are placed at a distance 'd' apart. If they are allowed to touch each other, what is the new distance between them to get a repulsive force of same magnitude as before?
(A) $\frac{4d}{3}$ **(B)** $\frac{3d}{4}$ **(C)** d **(D)** $\frac{d}{2}$

17. Three point charges of $+2q$, $+2q$ and $-4q$ are placed at the corners A, B and C of an equilateral triangle ABC of side 'x'. The magnitude of the electric dipole moment of this system is
(A) $2qx$ **(B)** $2\sqrt{3}\,qx$ **(C)** $3\sqrt{2}\,qx$ **(D)** $3qx$

18. 4×10^{10} electrons are removed from a neutral metal sphere of diameter 20 cm placed in air. The magnitude of the electric field (in $N\,C^{-1}$) at a distance of 20 cm from its center is

(A) 5760 **(B)** 1440 **(C)** 640 **(D)** Zero

19. Two point charges $A = +3$ nC and $B = +1$ nC are placed 5 cm apart in air. The work done to move charge B towards A by 1 cm is

(A) 1.35×10^{-7} J **(B)** 2.7×10^{-7} J
(C) 2.0×10^{-7} J **(D)** 12.1×10^{-7} J

20. A system of 2 capacitors of capacitance 2 μF and 4 μF is connected in series across a potential difference of 6 V. The electric charge and energy stored in the system are

(A) 10 μC and 30 μJ **(B)** 36 μC and 108 μJ
(C) 8 μC and 24 μJ **(D)** 1 μC and 3 μJ

21. The minimum value of effective capacitance that can be obtained by combining 3 capacitors of capacitances 1 pF, 2 pF and 4 pF is

(A) $\frac{4}{7}$ pF **(B)** 1 pF **(C)** $\frac{7}{4}$ pF **(D)** 2 pF

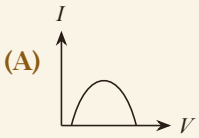
22. A cylindrical conductor of diameter 0.1 mm carries a current of 90 mA. The current density (in $A\,m^{-2}$) is ($\pi \simeq 3$)

(A) 1.2×10^7 **(B)** 2.4×10^7
(C) 3×10^6 **(D)** 6×10^6

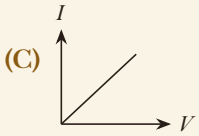
23. A piece of copper is to be shaped into a conducting wire of maximum resistance. The suitable length and diameter are _____ and _____ respectively.

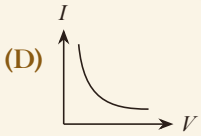
(A) L and d **(B)** 2L and d
(C) L/2 and 2d **(D)** 2L and d/2

24. Of the following graphs, the one that correctly represents the I-V characteristics of a 'Ohmic device' is

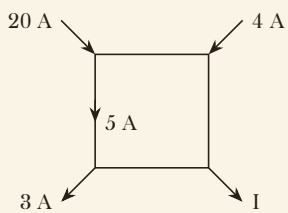
(A) 

(B) 

(C) 

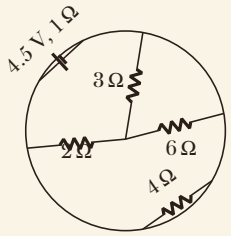
(D) 

25. The value of I in the figure shown below is



(A) 8 A **(B)** 21 A **(C)** 19 A **(D)** 4 A

26. The power dissipated in 3 Ω resistance in the following circuit is



(A) 0.75 W **(B)** 0.25 W
(C) 1 W **(D)** 0.5 W

27. In metre bridge experiment, with a standard resistance in the right gap and a resistance coil dipped in water (in a beaker) in the left gap, the balancing length obtained is 'l'. If the temperature of water is increased, the new balancing length is

(A) $> l$ **(B)** $< l$ **(C)** $= l$ **(D)** $= 0$

28. A proton, a deuteron and an α-particle are projected perpendicular to the direction of a uniform magnetic field with same kinetic energy. The ratio of the radii of the circular paths described by them is

(A) $1 : \sqrt{2} : 1$ **(B)** $1 : \sqrt{2} : \sqrt{2}$
(C) $\sqrt{2} : 1 : 1$ **(D)** $\sqrt{2} : \sqrt{2} : 1$

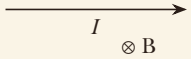
29. A galvanometer of resistance 50 Ω is connected to a battery of 3 V along with a resistance of 2950 Ω in series shows full-scale deflection of 30 divisions. The additional series resistance required to reduce the deflection to 20 divisions is

(A) 1500 Ω **(B)** 4440 Ω
(C) 7400 Ω **(D)** 2950 Ω

30. The magnetic field at the center of a current carrying loop of radius 0.1 m is $5\sqrt{5}$ times that at a point along its axis. The distance of this point from the centre of the loop is

(A) 0.2 m **(B)** 0.1 m
(C) 0.05 m **(D)** 0.25 m

31. A straight wire of length 50 cm carrying a current of 2.5 A is suspended in mid-air by a uniform magnetic field of 0.5 T (as shown in figure). The mass of the wire is ($g = 10\,m\,s^{-2}$)



(A) 62.5 g **(B)** 250 g **(C)** 125 g **(D)** 100 g

32. Which of the following properties is 'False' for a bar magnet?

(A) Its poles cannot be separated.
(B) It points in North-South direction when suspended.
(C) Its like poles repel and unlike poles attract.
(D) It doesn't produce magnetic field.

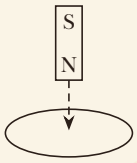
33. A magnetic dipole of magnetic moment 6×10^{-2} A m^2 and moment of inertia 12×10^{-6} kg m^2 performs oscillations in a magnetic field of 2×10^{-2} T. The time taken by the dipole to complete 20 oscillations is ($\pi \simeq 3$)

(A) 36 s **(B)** 6 s **(C)** 12 s **(D)** 18 s

34. The susceptibility of a ferromagnetic substance is

(A) $\gg 1$ **(B)** > 1 **(C)** < 1 **(D)** Zero

35. A bar magnet is allowed to fall vertically through a copper coil placed in a horizontal plane. The magnet falls with a net acceleration



(A) $= g$ **(B)** $> g$ **(C)** $< g$ **(D)** Zero

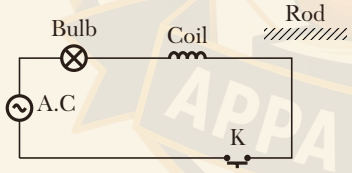
36. The working of magnetic braking of trains is based on

(A) Alternating current **(B)** Eddy current
(C) Steady current **(D)** Pulsating current

37. A jet plane of wing span 20 m is travelling towards west at a speed of 400 m s^{-1}. If the earth's total magnetic field is 4×10^{-4} T and the dip angle is 30°, at that place, the voltage difference developed across the ends of the wing is

(A) 1.6 V **(B)** 3.2 V **(C)** 0.8 V **(D)** 6.4 V

38. In the A.C. circuit shown, keeping 'K' pressed, if an iron rod is inserted into the coil, the bulb in the circuit,



(A) glows more brightly
(B) glows less brightly
(C) glows with same brightness (as before the rod is inserted)
(D) gets damaged

39. The output of a step down transformer is measured to be 48 V when connected to a 12 W bulb. The value of peak current is

(A) $\frac{1}{\sqrt{2}}$ A **(B)** $\sqrt{2}$ A **(C)** $\frac{1}{2\sqrt{2}}$ A **(D)** $\frac{1}{4}$ A

40. A coil of inductive reactance $1/\sqrt{3}\ \Omega$ and resistance $1\ \Omega$ is connected to a 200 V, 50 Hz A.C. supply. The time lag between maximum voltage and current is

(A) $\frac{1}{300}$ s **(B)** $\frac{1}{600}$ s **(C)** $\frac{1}{500}$ s **(D)** $\frac{1}{200}$ s

41. If $\vec{E}$ and $\vec{B}$ represent electric and magnetic field vectors of an electromagnetic wave, the direction of propagation of the wave is along

(A) $\vec{E}$ **(B)** $\vec{B}$ **(C)** $\vec{E} \times \vec{B}$ **(D)** $\vec{B} \times \vec{E}$

42. According to Cartesian sign convention, in ray optics

(A) all distances are taken positive
(B) all distances are taken negative
(C) all distances in the direction of incident ray are taken positive
(D) all distances in the direction of incident ray are taken negative

43. A linear object of height 10 cm is kept in front of a concave mirror of radius of curvature 15 cm, at a distance of 10 cm. The image formed is

(A) magnified and erect
(B) magnified and inverted
(C) diminished and erect
(D) diminished and inverted

44. During scattering of light, the amount of scattering is inversely proportional to _____ of wavelength of light.

(A) cube **(B)** square
(C) fourth power **(D)** half

45. In Young's double-slit experiment if yellow light is replaced by blue light, the interference fringes become

(A) wider **(B)** narrower
(C) brighter **(D)** darker

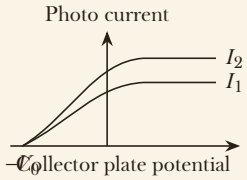
46. According to Huygens' principle, during refraction of light from air to a denser medium

(A) Wavelength and speed decrease
(B) Wavelength and speed increase
(C) Wavelength increases but speed decreases
(D) Wavelength decreases but speed increases

47. In a system of two crossed polarisers, it is found that the intensity of light from the second polariser is half from that of first polariser. The angle between their pass axes is

(A) 45° **(B)** 60° **(C)** 30° **(D)** 0°

48. From the following graph of photo current against collector plate potential, for two different intensities of light I_1 and I_2, one can conclude



(A) $I_1 = I_2$
(B) $I_1 > I_2$
(C) $I_1 < I_2$
(D) Comparison is not possible.

49. A particle is dropped from a height 'H'. The de Broglie wavelength of the particle depends on height as
(A) H **(B)** H^0 **(C)** $H^{1/2}$ **(D)** $H^{-1/2}$

50. The scientist who is credited with the discovery of 'nucleus' in an atom is
(A) J.J. Thomson **(B)** Rutherford
(C) Niels Bohr **(D)** Balmer

51. The energy (in eV) required to excite an electron from $n = 2$ to $n = 4$ state in hydrogen atom is
(A) +2.55 **(B)** −3.4 **(C)** −0.85 **(D)** +4.25

52. In a nuclear reactor the function of the Moderator is to decrease
(A) number of neutrons
(B) speed of neutrons
(C) escape of neutrons
(D) temperature of the reactor

53. The particles emitted in the decay of $^{238}_{92}U$ to $^{234}_{92}U$
(A) 1 α and 2 β **(B)** 1 α only
(C) 1 α and 1 β **(D)** 2 α and 2 β

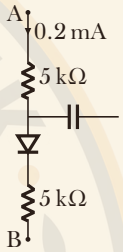
54. The mass defect of ^4_2He is 0.03 u. The binding energy per nucleon of helium (in MeV) is
(A) 27.93 **(B)** 6.9825
(C) 2.793 **(D)** 69.825

55. The energy gap in case of which of the following is less than 3 eV?
(A) Copper **(B)** Iron
(C) Aluminium **(D)** Germanium

56. Which of the following semi-conducting devices is used as voltage regulator?
(A) Photo diode **(B)** LASER diode
(C) Zener diode **(D)** Solar cell

57. In the three parts of a transistor, 'Emitter' is of
(A) moderate size and heavily doped
(B) large size and lightly doped
(C) thin size and heavily doped
(D) large size and moderately doped

58. In the figure shown, if the diode forward voltage drop is 0.2 V, the voltage difference between A and B is



(A) 1.3 V **(B)** 2.2 V **(C)** 0 **(D)** 0.5 V

59. Which of the following logic gates is considered as 'universal'?

(A)  **(B)** 

(C)  **(D)** 

60. A basic communication system consists of
(a) Transmitter
(b) Information source
(c) User of information
(d) Channel
(e) Receiver
The correct sequence of the arrangement is
(A) a, b, c, d and e **(B)** b, a, d, e and c
(C) b, d, a, c and e **(D)** b, e, a, d and c

KCET 2016
Physics Question Paper

Version A-1 · 60 Questions · 60 Marks · 80 Minutes

1. A body falls freely for 10 sec. Its average velocity during this journey (take $g = 10\,m\,s^{-2}$)

(1) $100\,m\,s^{-1}$ **(2)** $10\,m\,s^{-1}$
(3) $50\,m\,s^{-1}$ **(4)** $5\,m\,s^{-1}$

2. Three projectiles A, B and C are projected at an angle of 30°, 45°, 60° respectively. If R_A, R_B and R_C are ranges of A, B and C respectively, then (velocity of projection is same for A, B & C)

(1) $R_A = R_B = R_C$ **(2)** $R_A = R_C > R_B$
(3) $R_A < R_B < R_C$ **(4)** $R_A = R_C < R_B$

3. The component of a vector $\vec{r}$ along x-axis will have a maximum value if

(1) $\vec{r}$ is along +ve x-axis
(2) $\vec{r}$ is along +ve y-axis
(3) $\vec{r}$ is along −ve y-axis
(4) $\vec{r}$ makes an angle of 45° with the x-axis

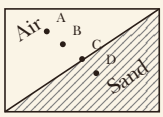
4. Maximum acceleration of the train in which a 50 kg box lying on its floor will remain stationary (Given: Co-efficient of static friction between the box and the train's floor is 0.3 and $g = 10\,m\,s^{-2}$)

(1) $5.0\,m\,s^{-2}$ **(2)** $3.0\,m\,s^{-2}$
(3) $1.5\,m\,s^{-2}$ **(4)** $15\,m\,s^{-2}$

5. A 12 kg bomb at rest explodes into two pieces of 4 kg and 8 kg. If the momentum of 4 kg piece is 20 N s, the kinetic energy of the 8 kg piece is

(1) 25 J **(2)** 20 J **(3)** 50 J **(4)** 40 J

6. Which of the points is likely position of the centre of mass of the system shown in the figure?

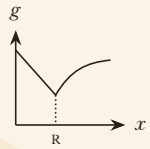


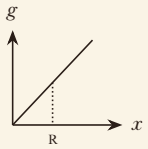
(1) A **(2)** D **(3)** B **(4)** C

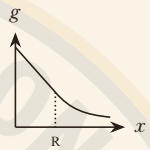
7. Three bodies a ring (R), a solid cylinder (C) and a solid sphere (S) having same mass and same radius roll down the inclined plane without slipping. They start from rest, if v_R, v_C and v_S are velocities of respective bodies on reaching the bottom of the plane, then

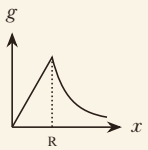
(1) $v_R = v_C = v_S$ **(2)** $v_R > v_C > v_S$
(3) $v_R < v_C < v_S$ **(4)** $v_R = v_C > v_S$

8. Variation of acceleration due to gravity (g) with distance x from the centre of the earth is best represented by ($R \rightarrow$ Radius of the earth)

(1)


(2)


(3)


(4)


9. A spring is stretched by applying a load to its free end. The strain produced in the spring is

(1) Volumetric
(2) Shear
(3) Longitudinal & Shear
(4) Longitudinal

10. An ideal fluid flows through a pipe of circular cross section with diameters 5 cm and 10 cm as shown. The ratio of velocities of fluid at A and B is



(1) 4:1 **(2)** 1:4 **(3)** 2:1 **(4)** 1:2

11. A pan filled with hot food cools from 94°C to 86°C in 2 minutes. When the room temperature is 20°C. How long will it cool from 74°C to 66°C?

(1) 2 minutes **(2)** 2.8 minutes
(3) 2.5 minutes **(4)** 1.8 minutes

12. Four rods with different radii r and length l are used to connect two heat reservoirs at different temperature. Which one will conduct most heat?

(1) $r = 1$ cm, $l = 1$ m **(2)** $r = 1$ cm, $l = \frac{1}{2}$ m
(3) $r = 2$ cm, $l = 2$ m **(4)** $r = 2$ cm, $l = \frac{1}{2}$ m

13. A Carnot engine working between 300 K and 400 K has 800 J of useful work. The amount of heat energy supplied to the engine from the source is

(1) 2400 J **(2)** 3200 J **(3)** 1200 J **(4)** 3600 J

14. A particle executing SHM has a maximum speed of $0.5\ m\ s^{-1}$ and maximum acceleration of $1.0\ m\ s^{-2}$. The angular frequency of oscillation is

(1) $2\ rad\ s^{-1}$ **(2)** $0.5\ rad\ s^{-1}$

(3) $2\pi\ rad\ s^{-1}$ **(4)** $0.5\pi\ rad\ s^{-1}$

15. A source of sound is moving with a velocity of $50\ m\ s^{-1}$ towards a stationary observer. The observer measures the frequency of sound as 500 Hz. The apparent frequency of sound as heard by the observer when source is moving away from him with the same speed is (Speed of sound at room temperature $350\ m\ s^{-1}$)

(1) 400 Hz **(2)** 666 Hz

(3) 375 Hz **(4)** 177.5 Hz

16. If there are only one type of charge in the universe, then ($\vec{E}$ → Electric field, $\vec{ds}$ → Area vector)

(1) $\oint \vec{E} \cdot \vec{ds} \neq 0$ on any surface

(2) $\oint \vec{E} \cdot \vec{ds}$ could not be defined

(3) $\oint \vec{E} \cdot \vec{ds} = \infty$ if charge is inside

(4) $\oint \vec{E} \cdot \vec{ds} = 0$ if charge is outside, $= \frac{q}{\varepsilon_0}$ if charge is inside

17. An electron of mass m, charge e falls through a distance h meter in a uniform electric field E. Then time of fall

(1) $t = \sqrt{\frac{2hm}{eE}}$ **(2)** $t = \frac{2hm}{eE}$

(3) $t = \sqrt{\frac{2eE}{hm}}$ **(4)** $t = \frac{2eE}{hm}$

18. If $\vec{E}_{ax}$ and $\vec{E}_{eq}$ represents electric field at a point on the axial and equatorial line of a dipole. If points are at a distance r from the centre of the dipole, for $r \gg a$

(1) $\vec{E}_{ax} = \vec{E}_{eq}$ **(2)** $\vec{E}_{ax} = -\vec{E}_{eq}$

(3) $\vec{E}_{ax} = -2\vec{E}_{eq}$ **(4)** $\vec{E}_{eq} = 2\vec{E}_{ax}$

19. Nature of equipotential surface for a point charge is

(1) Ellipsoid with charge at foci.

(2) Sphere with charge at the centre of the sphere.

(3) Sphere with charge on the surface of the sphere.

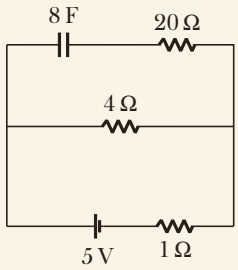
(4) Plane with charge on the surface.

20. A particle of mass 1 g and charge 1 μC is held at rest on a frictionless horizontal surface at distance 1 m from the fixed charge 2 mC. If the particle is released, it will be repelled. The speed of the particle when it is at a distance of 10 m from the fixed charge

(1) $60\ m\ s^{-1}$ **(2)** $100\ m\ s^{-1}$

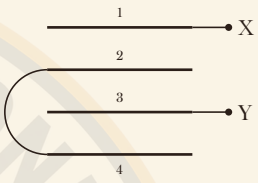
(3) $90\ m\ s^{-1}$ **(4)** $180\ m\ s^{-1}$

21. A capacitor of 8 F is connected as shown. Charge on the plates of the capacitor



(1) 32 C **(2)** 40 C **(3)** 0 C **(4)** 80 C

22. Four metal plates are arranged as shown. Capacitance between X and Y (A → Area of each plate, d → distance between the plates)



(1) $\frac{3}{2}\frac{\varepsilon_0 A}{d}$ **(2)** $\frac{2\varepsilon_0 A}{d}$ **(3)** $\frac{2}{3}\frac{\varepsilon_0 A}{d}$ **(4)** $\frac{3\varepsilon_0 A}{d}$

23. Mobility of free electrons in a conductor is

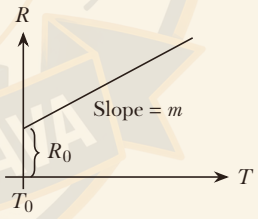
(1) directly proportional to electron density.

(2) directly proportional to relaxation time.

(3) inversely proportional to electron density.

(4) inversely proportional to relaxation time.

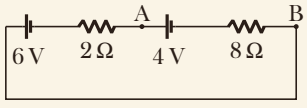
24. Variation of resistance of the conductor with temperature is as shown



The temperature co-efficient (α) of the conductor is

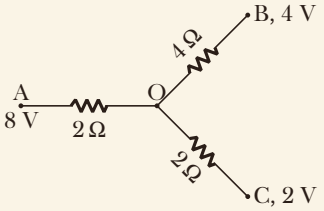
(1) $\frac{R_0}{m}$ **(2)** mR_0 **(3)** m^2R_0 **(4)** $\frac{m}{R_0}$

25. Potential difference between A and B in the following circuit



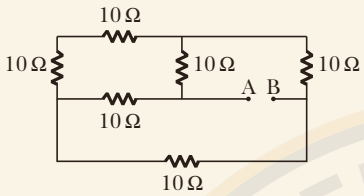
(1) 4 V **(2)** 5.6 V **(3)** 2.8 V **(4)** 6 V

26. In the following network potential at 'O'



(1) 4 V **(2)** 3 V **(3)** 6 V **(4)** 4.8 V

27. Effective resistance between A and B in the following circuit



(1) 10 Ω **(2)** 20 Ω **(3)** 5 Ω **(4)** $\frac{20}{3}$ Ω

28. Two heating coils of resistances 10 Ω and 20 Ω are connected in parallel and connected to a battery of emf 12 V and internal resistance 1 Ω. The power consumed by them are in the ratio

(1) 1:4 **(2)** 1:3 **(3)** 2:1 **(4)** 4:1

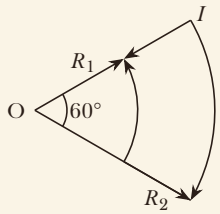
29. A proton is projected with a uniform velocity 'v' along the axis of a current carrying solenoid, then

(1) the proton will be accelerated along the axis.
(2) the proton path will be circular about the axis.
(3) the proton move along helical path.
(4) the proton will continue to move with velocity 'v' along the axis.

30. In the cyclotron, as radius of the circular path of the charged particle increases (ω = angular velocity, v = linear velocity)

(1) both ω and v increases
(2) ω only increases, v remains constant
(3) v increases, ω remains constant
(4) v increases, ω decreases

31. A conducting wire carrying current is arranged as shown. The magnetic field at 'O'



(1) $\frac{\mu_0 i}{12}\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$ **(2)** $\frac{\mu_0 i}{12}\left[\frac{1}{R_1}+\frac{1}{R_2}\right]$

(3) $\frac{\mu_0 i}{6}\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$ **(4)** $\frac{\mu_0 i}{6}\left[\frac{1}{R_1}+\frac{1}{R_2}\right]$

32. The quantity of a charge that will be transferred by a current flow of 20 A over 1 hour 30 minutes period is

(1) 10.8×10^3 C **(2)** 10.8×10^4 C
(3) 5.4×10^3 C **(4)** 1.8×10^4 C

33. A galvanometer coil has a resistance of 50 Ω and the meter shows full scale deflection for a current of 5 mA. This galvanometer is converted into voltmeter of range 0–20 V by connecting

(1) 3950 Ω in series with galvanometer
(2) 4050 Ω in series with galvanometer
(3) 3950 Ω in parallel with galvanometer
(4) 4050 Ω in parallel with galvanometer

34. χ_1 and χ_2 are susceptibility of a paramagnetic material at temperatures T_1 K and T_2 K respectively, then

(1) $\chi_1 = \chi_2$ **(2)** $\chi_1 T_1 = \chi_2 T_2$
(3) $\chi_1 T_2 = \chi_2 T_1$ **(4)** $\chi_1\sqrt{T_1} = \chi_2\sqrt{T_2}$

35. At certain place, the horizontal component of earth's magnetic field is 3.0 G and the angle dip at that place is 30°. The magnetic field of earth at that location

(1) 4.5 G **(2)** 5.1 G **(3)** 3.5 G **(4)** 6.0 G

36. The process of super imposing message signal on high frequency carrier wave is called

(1) Amplification **(2)** Demodulation
(3) Transmission **(4)** Modulation

37. A long solenoid with 40 turns per cm carries a current of 1 A. The magnetic energy stored per unit volume is _____ J/m^3.

(1) 3.2π **(2)** 32π **(3)** 1.6π **(4)** 6.4π

38. A wheel with 10 spokes each of length 'L' m is rotated with a uniform angular velocity 'ω' in a plane normal to the magnetic field 'B'. The emf induced between the axle and the rim of the wheel.

(1) $\frac{1}{2}N\omega BL^2$ **(2)** $\frac{1}{2}\omega BL^2$
(3) ωbL^2 **(4)** $N\omega BL^2$

39. The rms value of current in a 50 Hz AC circuit is 6 A. The average value of AC current over a cycle is

(1) $6\sqrt{2}$ **(2)** $\frac{3}{\pi\sqrt{2}}$ **(3)** Zero **(4)** $\frac{6}{\pi\sqrt{2}}$

40. A capacitor of capacitance 10 μF is connected to an AC source and an AC Ammeter. If the source voltage varies as $V = 50\sqrt{2}\sin 100t$, the reading of the ammeter is

(1) 50 mA **(2)** 70.7 mA
(3) 5.0 mA **(4)** 7.07 mA

41. In a series L.C.R circuit, the potential drop across L, C and R respectively are 40 V, 120 V and 60 V. Then the source voltage is

(1) 220 V **(2)** 160 V **(3)** 180 V **(4)** 100 V

42. In a series L.C.R. circuit an alternating emf (v) and current (i) are given by the equation $v = v_0 \sin \omega t$, $i = i_0 \sin\left(\omega t + \frac{\pi}{3}\right)$. The average power dissipated in the circuit over a cycle of AC is

(1) $\frac{v_0 i_0}{2}$ **(2)** $\frac{v_0 i_0}{4}$ **(3)** $\frac{\sqrt{3}}{2} v_0 i_0$ **(4)** Zero

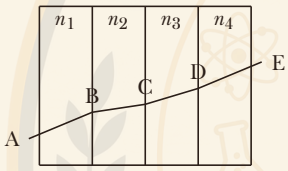
43. Electromagnetic radiation used to sterilise milk is

(1) X-ray **(2)** γ-ray
(3) UV rays **(4)** Radiowaves

44. A plane glass plate is placed over a various coloured letters (violet, green, yellow, red). The letter which appears to raised more

(1) Red **(2)** Yellow **(3)** Green **(4)** Violet

45. A ray of light passes through four transparent media with refractive index n_1, n_2, n_3 and n_4 as shown. The surfaces of all media are parallel



If the emergent ray DE is parallel to incident ray AB, then

(1) $n_1 = n_4$ **(2)** $n_2 = n_4$
(3) $n_3 = n_4$ **(4)** $n_1 = \frac{n_2 + n_3 + n_4}{3}$

46. Focal length of a convex lens is 20 cm and its RI is 1.5. It produces an erect, enlarged image if the distance of the object from the lens is

(1) 40 cm **(2)** 30 cm **(3)** 15 cm **(4)** 20 cm

47. A ray of light suffers a minimum deviation when incident on an equilateral prism of refractive index $\sqrt{2}$. The angle of incidence is

(1) 30° **(2)** 45° **(3)** 60° **(4)** 50°

48. In Young's double slit experiment the source is white light. One slit is covered with red filter and the other with blue filter. There shall be

(1) Alternate red & blue fringes
(2) Alternate dark & pink fringes
(3) Alternate dark & yellow fringes
(4) No interference

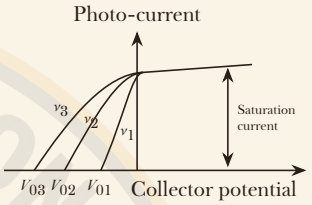
49. Light of wavelength 600 nm is incident normally on a slit of width 0.2 mm. The angular width of central maxima in the diffraction pattern is (measured from minimum to minimum)

(1) 6×10^{-3} rad **(2)** 4×10^{-3} rad
(3) 2.4×10^{-3} rad **(4)** 4.5×10^{-3} rad

50. For what distance is ray optics is good approximation when the aperture is 4 mm and the wavelength of light is 400 nm?

(1) 24 m **(2)** 40 m **(3)** 18 m **(4)** 30 m

51. The variation of photo-current with collector potential for different frequencies of incident radiation ν_1, ν_2 and ν_3 is as shown in the graph, then



(1) $\nu_1 = \nu_2 = \nu_3$ **(2)** $\nu_1 > \nu_2 > \nu_3$
(3) $\nu_1 < \nu_2 < \nu_3$ **(4)** $\nu_3 = \frac{\nu_1 + \nu_2}{2}$

52. The de Broglie wavelength of an electron accelerated to a potential of 400 V is approximately

(1) 0.03 nm **(2)** 0.04 nm
(3) 0.12 nm **(4)** 0.06 nm

53. Total energy of electron in an excited state of hydrogen atom is −3.4 eV. The kinetic and potential energy of electron in this state

(1) $K = -3.4$ eV, $U = -6.8$ eV
(2) $K = 3.4$ eV, $U = -6.8$ eV
(3) $K = -6.8$ eV, $U = +3.4$ eV
(4) $K = +10.2$ eV, $U = -13.6$ eV

54. When electron jumps from $n = 4$ level to $n = 1$ level, the angular momentum of electron changes by

(1) $\frac{h}{2\pi}$ **(2)** $\frac{2h}{2\pi}$ **(3)** $\frac{3h}{2\pi}$ **(4)** $\frac{4h}{2\pi}$

55. A radio-active sample of half-life 10 days contains $1000x$ nuclei. Number of original nuclei present after 5 days is

(1) $707x$ **(2)** $750x$ **(3)** $500x$ **(4)** $250x$

56. An element X decays into element Z by two-step process.

$$X \longrightarrow Y + {}^{4}_{2}He, \qquad Y \longrightarrow Z + 2\bar{e}$$

then

(1) X & Z are isobars. **(2)** X & Y are isotopes.
(3) X & Z are isotones. **(4)** X & Z are isotopes.

57. A nucleus of mass 20 u emits a γ photon of energy 6 MeV. If the emission assume to occur when nucleus is free and rest, then the nucleus will have kinetic energy nearest to (take 1 u = 1.6×10^{-27} kg)

(1) 10 keV **(2)** 1 keV
(3) 0.1 keV **(4)** 100 keV

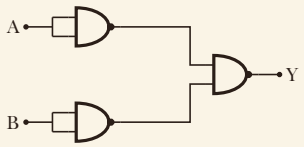
58. Constant DC voltage is required from a variable AC voltage. Which of the following is correct order of operation?

(1) Regulator, filter, rectifier
(2) Rectifier, regulator, filter
(3) Rectifier, filter, regulator
(4) Filter, regulator, rectifier

59. In a transistor, the collector current varies by 0.49 mA and emitter current varies by 0.50 mA. Current gain β measured is

(1) 49 **(2)** 150 **(3)** 99 **(4)** 100

60. Identify the logic operation carried out by the following circuit.



(1) AND **(2)** NAND **(3)** NOR **(4)** OR

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